

数学物理方程讲义

CSSE60003.01

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Abstract

This document collects notes for the course CSSE60003.01 (数学物理方程). Content is organized by topic into five parts: classical elliptic equations, heat and wave equations, Sobolev spaces, weak solutions and regularity, and numerical methods for PDEs.

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Part I

Classical elliptic equations

1 Introduction to PDEs

1.1 Course outline

- **Part I.** Classical elliptic equations
- **Part II.** Heat and wave equations
- **Part III.** Sobolev spaces
- **Part IV.** Weak solutions and regularity
- **Part V.** Numerical methods

1.2 Examples of linear PDEs

0. Transport equation: $u_t + b \cdot \nabla u = 0$

1. Laplace equation: $\Delta u = 0$

2. Heat equation: $u_t = \Delta u$

3. Wave equation: $u_{tt} = \Delta u$

Remark 1.1 (Linear operator / equation). An operator L is linear if

$$L(u + v) = L(u) + L(v), \quad L(cu) = cL(u).$$

A linear equation has the form $L(u) = f(x)$.

1.3 Examples of nonlinear PDEs

$ \nabla u = 1$	(Eikonal)
$\Delta u + u^2 = 0$	(Emden–Fowler)
$\operatorname{div}(\nabla u ^{p-2} \nabla u) = 0$	(p -Laplacian)
$u_t + u u_x = 0$	(Burgers)
$u_{tt} - \Delta u + \sin u = 0$	(Klein–Gordon)
$i u_t - \Delta u - u^3 = 0$	(NLS)
$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1 + \nabla u ^2}}\right) = 0$	(minimal surface)
$\begin{cases} u_t - \Delta u + u \cdot \nabla u + \nabla p = 0 \\ \operatorname{div} u = 0 \end{cases}$	(Navier–Stokes)

1.4 Guiding questions

- Existence & uniqueness
- What do we mean by a solution?
- Stability of IVP & BVP

- Other properties

Definition 1.2 (Classical / weak solutions). **Classical solutions:** all derivatives appearing in the equation are continuous and the equation holds pointwise.

Weak solutions: the equation holds in a weak sense (prerequisites: L^p spaces).

1.5 Transport equation

Consider

$$\begin{cases} \partial_t u + \vec{b} \cdot \nabla u = f(t, x), \\ u(0, x) = g(x), \end{cases} \quad u : \mathbb{R}^n \times [0, +\infty) \rightarrow \mathbb{R}. \quad (1)$$

1.5.1 Method of characteristics

Along characteristic curves,

$$\frac{dt}{1} = \frac{dx_j}{b_j}, \quad \dot{x}(t) = \vec{b}, \quad \frac{d}{dt}u(x(t), t) = \dot{x}(t) \cdot \nabla u + \partial_t u.$$

Parametrize initial data by $(x_0(\xi), t_0(\xi))$. With parameter s ,

$$\begin{cases} \frac{dt}{ds} = 1, & t(0) = 0, \\ \frac{dx_j}{ds} = b_j, & x_j(0) = \xi_j, \\ \frac{du}{ds} = f(x, t), & u(0) = g(\xi). \end{cases} \quad (2)$$

Hence $t = s$, $x_j = b_j s + \xi_j$ so $\xi = x - bt$, and

$$u(t, x) = g(x - \vec{b}t) + \int_0^t f(s, \vec{b}(s - t) + x) ds. \quad (3)$$

Existence and uniqueness hold provided $b \in C^1$.

Derivation via $u(t, x(t))$. Set $\dot{x}(t) = \vec{b}$, so $x(t) = \vec{b}t + \vec{c}$ with $x(0) = \vec{c}$. Then

$$\frac{d}{dt}u = f, \quad u(t, x(t)) = u(0) + \int_0^t f(s, x(s)) ds.$$

With $x(s) = \vec{b}s + \vec{c} = \vec{b}s + x - \vec{b}t$ and $u(0) = g(\vec{c}) = g(x - \vec{b}t)$, one recovers (3).

1.6 Wave equation via the method of characteristics

Consider

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0, & c > 0, \\ u(0, x) = u_0(x), \\ \partial_t u(0, x) = u_1(x). \end{cases} \quad (4)$$

Factorization:

$$u_{tt} - c^2 u_{xx} = (\partial_t + c\partial_x)(\partial_t u - c\partial_x u) = 0.$$

Set $v := \partial_t u - c\partial_x u$. Then

$$\begin{cases} \partial_t v + c\partial_x v = 0, \\ v(0, x) = u_1(x) - cu'_0(x). \end{cases} \quad (5)$$

By the homogeneous transport formula ($f = 0$),

$$v(t, x) = u_1(x - ct) - cu'_0(x - ct).$$

Hence

$$\partial_t u - c \partial_x u = v = u_1(x - ct) - cu'_0(x - ct).$$

Apply the transport formula again with forcing $f(t, x) = u_1(x - ct) - cu'_0(x - ct)$ and initial data u_0 :

$$u = u_0(x + ct) + \int_0^t f(s, x - c(s - t)) ds.$$

Change variables $y = x + ct - 2cs$ (so $ds = -\frac{1}{2c} dy$), yielding d'Alembert's formula

$$u(t, x) = \frac{1}{2}(u_0(x + ct) + u_0(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} u_1(y) dy. \quad (6)$$

1.7 Laplace and Poisson equations

$$-\Delta u = 0 \quad (\text{Laplace}), \quad -\Delta u = f \quad (\text{Poisson}).$$

Classical solution: $u \in C^2$, $f \in C^0$, equation holds pointwise.

Aim. Solution formula in a bounded domain Ω with suitable boundary conditions:

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases} \quad (7)$$

Existence & uniqueness in the class of continuous functions (to be discussed).

1.7.1 Laplace equation in $\mathbb{R}^n$; radial solutions

In spherical coordinates,

$$\Delta u = u_{rr} + \frac{n-1}{r} u_r + \frac{1}{r^2} \Delta_{S^{n-1}} u,$$

where $\Delta_{S^{n-1}}$ is the Laplace–Beltrami operator on S^{n-1} (for $n = 2$, $\Delta_{S^1} = \partial_{\theta\theta}$).

Radial solutions satisfy $u_{rr} + \frac{n-1}{r} u_r = 0$, hence

$$u = \begin{cases} C_1 r^{2-n} + C_2, & n > 2, \\ C_1 \log r + C_2, & n = 2. \end{cases} \quad (8)$$

1.7.2 Newtonian potential

For f suitably decaying / compactly supported,

$$V(x) = \begin{cases} \int_{\mathbb{R}^n} f(y) |x - y|^{2-n} dy = (f * |\cdot|^{2-n})(x), & n > 2, \\ \int_{\mathbb{R}^n} f(y) \log |x - y| dy, & n = 2. \end{cases} \quad (9)$$

Theorem 1.3. Let $f \in C_c^1(\mathbb{R}^n)$. Then $V \in C^2(\mathbb{R}^n)$ and

$$\begin{aligned} \Delta V &= -(n-2) \omega_n f(x), & x \in \mathbb{R}^n, \quad n \geq 3, \\ \Delta V &= 2\pi f(x), & x \in \mathbb{R}^n, \quad n = 2, \end{aligned}$$

where $\omega_n = |S^{n-1}|$ and $|B_1| = \omega_n/n$.

Sketch for $n = 3$. Differentiate under the integral (justified by cutting out a small ball):

$$\begin{aligned} V_{x_i} &= \int_{\mathbb{R}^n} f(y) \partial_{x_i} (|x - y|^{-1}) dy \\ &= \int_{\mathbb{R}^n \setminus B_\rho(x)} f(y) \partial_{x_i} (|x - y|^{-1}) dy + \int_{B_\rho(x)} f_{y_i}(y) |x - y|^{-1} dy - \int_{\partial B_\rho} f(y) |x - y|^{-1} n_i d\sigma_y, \end{aligned}$$

using $\partial_{x_i} |x - y|^{-1} = -\partial_{y_i} |x - y|^{-1}$ and integration by parts. The first and third integrals are C^1 ; the second is C^1 since $f \in C^1$. Hence

$$\begin{aligned} \Delta V &= \int_{\mathbb{R}^n \setminus B_\rho(x)} f(y) \Delta_x (|x - y|^{-1}) dy \\ &\quad + \int_{B_\rho} \sum_{i=1}^n f_{y_i}(y) \partial_{x_i} (|x - y|^{-1}) dy - \int_{\partial B_\rho} f(y) \sum_{i=1}^n \partial_{x_i} (|x - y|^{-1}) n_i d\sigma. \end{aligned}$$

Away from the diagonal, $\Delta_x |x - y|^{-1} = 0$, so $\Delta V = I_1 + I_2$ with

$$\begin{aligned} |I_1| &\leq C \int_{B_\rho(x)} \frac{1}{|x - y|^2} dy \leq C\rho \rightarrow 0 \quad (\rho \rightarrow 0), \\ I_2 &= - \int_{\partial B_\rho(x)} f(y) \rho^{-2} d\sigma = -\omega_n f(\bar{y}), \end{aligned}$$

where $f(\bar{y})$ is the spherical mean of f on $\partial B_\rho(x)$. Thus $\lim_{\rho \rightarrow 0} I_2 = -\omega_n f(x)$.

Other dimensions: exercise (same idea). □

Sketch for $n = 2$. With $V(x) = \int_{\mathbb{R}^2} f(y) \log |y - x| dy$,

$$\Delta V = \sum_{i=1}^2 \int_{\mathbb{R}^2} f(y) \partial_{x_i x_i} \log |y - x| dy.$$

Since $\Delta_x \log |y - x| = 0$ for $y \neq x$, cutting out $B_\varepsilon(x)$ yields

$$\Delta V = \sum_{i=1}^2 \int_{B_\varepsilon(x)} f(y) \partial_{x_i x_i} \log |y - x| dy.$$

Using $\partial_{x_i} \log |y - x| = -\partial_{y_i} \log |y - x|$ and integration by parts,

$$\Delta V = \sum_{i=1}^2 \left(\int_{B_\varepsilon(x)} \partial_{y_i} f(y) \partial_{x_i} \log |y - x| dy - \int_{\partial B_\varepsilon(x)} f(y) \partial_{x_i} \log |y - x| n_i d\sigma_y \right) =: I_1 + I_2.$$

Recall $\partial_{x_i} \log |y - x| = -(y_i - x_i)/|y - x|^2$. Then

$$|I_1| \leq C \|f\|_{C^1} \int_{B_\varepsilon(x)} \frac{1}{|y - x|} dy = C \|f\|_{C^1} \cdot 2\pi\varepsilon \rightarrow 0,$$

and $(y - x) \cdot \vec{n} = \varepsilon$ on $\partial B_\varepsilon(x)$ implies

$$I_2 = \int_{\partial B_\varepsilon(x)} f(y) \varepsilon^{-1} d\sigma_y.$$

Thus

$$\left| I_2 - 2\pi f(x) \right| = \left| \varepsilon^{-1} \int_{\partial B_\varepsilon(x)} (f(y) - f(x)) d\sigma_y \right| \leq C\varepsilon \rightarrow 0,$$

so $\lim_{\varepsilon \rightarrow 0} I_2 = 2\pi f(x)$. □

Remark 1.4. 1. The result holds for $f \in C_c^\alpha(\mathbb{R}^n)$.

2. It also holds under decay $|f(y)| \leq C(1 + |y|)^{-l}$ with $l > 2$.

Exercise 1.5.

$$\int \frac{1}{|x - y|^{n-2}} \frac{1}{(1 + |y|)^l} dy \lesssim \begin{cases} (1 + |x|)^{2-l}, & 2 < l < n, \\ (1 + |x|)^{2-n} \log |x|, & l = n, \\ (1 + |x|)^{2-n}, & l > n. \end{cases}$$

Next aim. General solution formula for

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega, \end{cases}$$

with $f, g \in C^0$; prove existence and uniqueness.

1.7.3 Properties of harmonic functions

Theorem 1.6 (Mean-value property). *If $u \in C^2(\Omega)$ and $\Delta u = 0$, then for every ball $B_\rho(x) \Subset \Omega$,*

$$u(x) = \frac{1}{|\partial B_\rho(x)|} \int_{\partial B_\rho(x)} u d\sigma = \frac{1}{|B_\rho(x)|} \int_{B_\rho(x)} u dy. \quad (10)$$

Proof. By the divergence theorem,

$$\begin{aligned} \int_{B_\rho} \Delta u &= \int_{\partial B_\rho} \frac{\partial u}{\partial \nu} d\sigma = \rho^{n-1} \int_{|\omega|=1} \partial_\rho u(x + \rho\omega) d\sigma \\ &= \rho^{n-1} \frac{\partial}{\partial \rho} \int_{|\omega|=1} u(x + \rho\omega) d\sigma. \end{aligned}$$

If $\Delta u = 0$, the spherical mean is independent of ρ . Integrating from 0 to r ,

$$\int_{|\omega|=1} u(x + r\omega) d\sigma = u(x) \omega_n,$$

hence

$$u(x) = \frac{1}{\omega_n r^{n-1}} \int_{\partial B_r(x)} u(y) d\sigma.$$

Integrating in ρ from 0 to r gives the volume form of the MVP:

$$u(x) \frac{\omega_n}{n} r^n = \int_{B_r(x)} u(y) dy \implies u(x) = \frac{1}{|B_r(x)|} \int_{B_r(x)} u(y) dy.$$

□

Remark 1.7. If $\Delta u \leq 0$, then

$$\frac{\partial}{\partial \rho} \int_{|\omega|=1} u(x + \rho\omega) d\sigma \leq 0,$$

so

$$\frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u d\sigma \leq u(x)$$

(sub-mean-value inequality). Moreover, the MVP characterizes harmonic functions.

Theorem 1.8 (MVP $\implies$ harmonic). *If $u \in C^0(\Omega)$ satisfies the MVP, then u is smooth and harmonic in Ω .*

Proof. Choose a radial mollifier $\varphi \in C_c^\infty(B_1(0))$ with $\int_{B_1} \varphi = 1$, and set $\varphi_\varepsilon(z) = \varepsilon^{-n} \varphi(z/\varepsilon)$. Then for $x \in \Omega_\varepsilon := \{y \in \Omega : d(y, \partial\Omega) > \varepsilon\}$,

$$\begin{aligned} (\varphi_\varepsilon * u)(x) &= \int u(x+y) \varphi_\varepsilon(y) dy = \int_{|y|<1} u(x+\varepsilon y) \varphi(y) dy \\ &= \int_0^1 \varphi(r) r^{n-1} dr \int_{|\omega|=1} u(x+\varepsilon r\omega) d\sigma = u(x), \end{aligned}$$

by the MVP (and normalization of φ). Thus $u = \varphi_\varepsilon * u$ on Ω_ε , so $u \in C^\infty(\Omega)$ and $\partial^\alpha u = \partial^\alpha \varphi_\varepsilon * u$. Moreover, for every $B_r(x) \Subset \Omega$,

$$\int_{B_r(x)} \Delta u = r^{n-1} \frac{\partial}{\partial r} (\omega_n u(x)) = 0,$$

hence $\Delta u = 0$ in Ω . □

1.7.4 Consequences of the MVP

Theorem 1.9 (Maximum principle). *If $u \in C^0(\overline{\Omega})$ satisfies the MVP, then u attains its maximum and minimum only on $\partial\Omega$, unless u is constant.*

Proof. Let $M = \max_{\overline{\Omega}} u$ and $\Sigma := \{x \in \Omega : u(x) = M\}$. Continuity implies Σ is relatively closed in Ω . If $x_0 \in \Sigma$ and $\overline{B}_r(x_0) \subset \Omega$, the MVP gives

$$M = u(x_0) = \frac{1}{|B_r(x_0)|} \int_{B_r(x_0)} u \leq M,$$

so $u \equiv M$ on $B_r(x_0)$, i.e. Σ is open. Thus $\Sigma = \emptyset$ or $\Sigma = \Omega$: either the maximum is attained only on $\partial\Omega$, or $u \equiv M$. □

Corollary 1.10 (Uniqueness for the Dirichlet problem). *Solutions of*

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega \end{cases}$$

are unique in $C^0(\overline{\Omega}) \cap C^2(\Omega)$.

Theorem 1.11 (Gradient estimate). *Let $u \in C(\overline{B}_R)$ be harmonic in $B_R(x_0)$. Then*

$$|Du(x_0)| \leq \frac{n}{R} \max_{\overline{B}_R} |u|.$$

Proof. Harmonicity implies $u \in C^\infty(B_R)$ and $\Delta(D_{x_i} u) = 0$. By the volume MVP and the divergence theorem,

$$D_{x_i} u(x_0) = \frac{n}{\omega_n R^n} \int_{B_R(x_0)} D_{x_i} u = \frac{n}{\omega_n R^n} \int_{\partial B_R} u \nu_i d\sigma,$$

which yields the stated bound. □

2 Gradient estimates, Harnack inequality, and Green's functions

2.1 Gradient estimates

The gradient estimate for harmonic functions yields

$$|D_{x_i} u(x_0)| \leq \frac{n}{\omega_n R^n} \max |u| \cdot \omega_n R^{n-1} \leq \frac{n}{R} \max |u|.$$

Lemma 2.1. *If $u \geq 0$ and $\Delta u = 0$, then*

$$|Du(x_0)| \leq \frac{n}{R} u(x_0).$$

Corollary 2.2. *If $\Delta u = 0$ in $\mathbb{R}^n$ and $u \leq M$ for some constant M , then u is constant.*

Corollary 2.3. *If $\Delta u = 0$ in $\mathbb{R}^n$ and $|u| \leq (1 + |y|)^\ell$ with $\ell < 1$, then $u \equiv C$.*

Theorem 2.4 (Higher-order derivative estimates). *Let $u \in C(\overline{B_R})$ with $\Delta u = 0$ in B_R . Then*

$$|D^m u(x_0)| \leq \frac{n^m e^{m-1} m!}{R^m} \max_{B_R} |u|.$$

Equivalently, for a multi-index $\alpha = (\alpha_1, \dots, \alpha_n)$ with $|\alpha| = m$,

$$|D^\alpha u(x_0)| \leq \frac{n^m e^{m-1} m!}{R^m} \max_{B_R} |u|.$$

Proof of Lemma 2.1 / Corollary 1. Since $\Delta \partial_{x_i} u = 0$, the volume MVP and the divergence theorem give

$$\begin{aligned} \partial_{x_i} u(x_0) &= \frac{n}{\omega_n R^n} \int_{B_R(x_0)} \partial_{x_i} u \, dy = \frac{n}{\omega_n R^n} \int_{\partial B_R(x_0)} u n_i \, d\sigma_y \\ &\leq \frac{n}{R} \cdot \frac{1}{\omega_n R^{n-1}} \int_{\partial B_R(x_0)} u \, d\sigma_y \leq \frac{n}{R} u(x_0), \end{aligned}$$

where the last step uses $u \geq 0$ and the surface MVP. $\square$

Proof of Corollary 2.2. Apply Lemma 2.1 to the nonnegative harmonic function $v := M - u$:

$$|Dv(x_0)| \leq \frac{n}{R} v(x_0) \leq \frac{n}{R} (M - \inf_{\mathbb{R}^n} u) \quad \forall R > 0.$$

(If u is not bounded from below one first replaces M by a local bound on $B_R(x_0)$ and lets $R \rightarrow \infty$ after estimating on each ball.) Letting $R \rightarrow \infty$ yields $Dv \equiv 0$, hence u is constant. $\square$

Proof of Corollary 2.3. For $\ell < 1$,

$$\begin{aligned} |D_k u(x_0)| &\leq \frac{n}{R} \max_{x \in B_R(x_0)} |u| \leq \frac{n}{R} \max_{x \in B_R(x_0)} (1 + |x|)^\ell \\ &\leq \frac{C_n}{R} (1 + R)^\ell \leq C_n R^{\ell-1}. \end{aligned}$$

As $R \rightarrow \infty$ we obtain $D_k u(x_0) = 0$, so $u \equiv C$. $\square$

Proof of Theorem 2.4. We proceed by induction on m . The case $m = 1$ is the gradient estimate.

Assume the claim holds for m , and consider $m + 1$. Fix $\theta \in (0, 1)$ and set $r = (1 - \theta)R$. Then $B_{R-r}(y) \subset B_R(x_0)$ for every $y \in \overline{B_r}(x_0)$, and

$$|D^{m+1} u(x_0)| \leq \frac{n}{r} \max_{\overline{B_r}} |D^m u|.$$

By the maximum principle, $\max_{B_r(x_0)} |D^m u| = \max_{\partial B_r(x_0)} |D^m u|$. Applying the induction hypothesis on balls of radius $R - r = \theta R$ gives

$$\max_{\overline{B_r}} |D^m u| \leq \frac{n^m e^{m-1} m!}{(R - r)^m} \max_{B_R} |u|.$$

Hence

$$\begin{aligned} |D^{m+1}u(x_0)| &\leq \frac{n}{r} \cdot \frac{n^m e^{m-1} m!}{(R-r)^m} \max_{B_R} |u| \\ &= \frac{n^{m+1} e^{m-1} m!}{R^{m+1} \theta^m (1-\theta)} \max_{B_R} |u|, \end{aligned}$$

where we used $(R-r)^m r = (\theta R)^m \cdot (1-\theta)R = R^{m+1} \theta^m (1-\theta)$.

Choose $\theta = \frac{m}{m+1}$. Then

$$\frac{1}{\theta^m (1-\theta)} = \left(1 + \frac{1}{m}\right)^m (m+1) < e(m+1),$$

and therefore

$$|D^{m+1}u(x_0)| \leq \frac{n^{m+1} (m+1)! e^m}{R^{m+1}} \max_{B_R(x_0)} |u|.$$

This completes the induction. $\square$

2.2 Analyticity of harmonic functions

Theorem 2.5. *Harmonic functions are (real) analytic.*

Proof. Write the Taylor expansion

$$u(x+h) = u(x) + \sum_{i=1}^{m-1} \frac{1}{i!} [(h_1 \partial_1 + \cdots + h_n \partial_n)^i u](x) + R_m(h),$$

with remainder

$$R_m(h) = \frac{1}{m!} [(h_1 \partial_1 + \cdots + h_n \partial_n)^m u](x_1 + \theta h_1, \dots, x_n + \theta h_n)$$

for some $\theta \in (0, 1)$. Using Theorem 2.4,

$$\begin{aligned} |R_m(h)| &\leq \frac{1}{m!} |h|^m n^m \cdot \frac{n^m e^{m-1} m!}{R^m} \max_{B_R} |u| \\ &\leq \left(\frac{|h| n^2 e}{R}\right)^m \max |u|. \end{aligned}$$

Whenever $|h| n^2 e < R/2$, we have $R_m(h) \rightarrow 0$ as $m \rightarrow +\infty$. Thus the Taylor series converges to u in a neighborhood of x . $\square$

2.3 Harnack inequality

Theorem 2.6 (Harnack inequality). *Let $\Delta u = 0$ in Ω with $u \geq 0$. Then for every compact $K \subset \Omega$ there exists $C = C(n, K, \Omega)$ such that*

$$\frac{1}{C} u(y) \leq u(x) \leq C u(y) \quad \forall x, y \in K.$$

Proof. Let $r = \frac{1}{4} \text{dist}(K, \partial\Omega)$. For all $x, y \in K$ with $|x - y| \leq r$ one has $B_r(y) \subseteq B_{2r}(x) \subseteq \Omega$. By the volume MVP,

$$\begin{aligned} u(x) &= \frac{n}{\omega_n (2r)^n} \int_{B_{2r}(x)} u(z) dz \geq \frac{n}{\omega_n (2r)^n} \int_{B_r(y)} u(z) dz \\ &= \frac{1}{2^n} \cdot \frac{n}{\omega_n r^n} \int_{B_r(y)} u(z) dz = \frac{1}{2^n} u(y). \end{aligned}$$

Similarly $u(y) \geq \frac{1}{2^n}u(x)$, so

$$\frac{1}{2^n} u(y) \leq u(x) \leq 2^n u(y) \quad \forall x, y \in K \text{ with } |x - y| \leq r.$$

Since K is compact, finitely many balls of radius $r/2$ cover K . Chaining the local comparison along a cover of length N yields

$$\frac{1}{2^{nN}} u(y) \leq u(x) \leq 2^{nN} u(y) \quad \forall x, y \in K.$$

□

Remark 2.7. In particular, if $B_{4R}(x_0) \subset \Omega$ and $x, y \in B_R(x_0)$, a direct MVP comparison on balls of radii comparable to R gives a constant $C = C(n)$ (e.g. $C = 3^n$ in a standard configuration). For a general compact K one covers K by balls $B_R(x_i)$ with $4R < \text{dist}(K, \partial\Omega)$ and takes $C = c^N$.

Remark 2.8. Harnack's inequality implies

$$\sup_K u \leq C \inf_K u.$$

Theorem 2.9 (Harnack convergence theorem). *Let $\{u_j\}$ be a monotone increasing sequence of harmonic functions in Ω . Suppose that at some point $y \in \Omega$ the sequence $\{u_j(y)\}$ is bounded. Then $\{u_j\}$ converges uniformly on every compact $K \Subset \Omega$.*

2.4 Green's function

2.4.1 Introduction

Recall the fundamental solution

$$\Gamma(a, x) = \begin{cases} \frac{1}{2\pi} \log |x - a|, & n = 2, \\ -\frac{1}{(n-2)\omega_n} |x - a|^{2-n}, & n \geq 3. \end{cases}$$

We have $\Delta_x \Gamma(a, x) = 0$ for $x \neq a$. Moreover,

$$\int_{\partial B_r(a)} \nabla_x \Gamma(a, x) \cdot \vec{n} \, d\sigma = 1 \quad \forall r > 0,$$

since $\frac{\partial \Gamma}{\partial n} = \nabla \Gamma \cdot n$.

Case $n = 2$.

$$\int_{\partial B_r(a)} \nabla_x \Gamma \cdot \vec{n} \, d\sigma = \frac{1}{2\pi} \int_{\partial B_r(a)} \frac{x-a}{|x-a|^2} \cdot \frac{x-a}{|x-a|} \, d\sigma = \frac{1}{2\pi} \int_{\partial B_r(a)} \frac{1}{|x-a|} \, d\sigma = 1.$$

Case $n \geq 3$.

$$\begin{aligned} \int_{\partial B_r(a)} \nabla_x \Gamma \cdot \vec{n} \, d\sigma &= \frac{n-2}{(n-2)\omega_n} \int_{\partial B_r(a)} \frac{1}{|x-a|^{n-1}} \frac{x-a}{|x-a|} \cdot \frac{x-a}{|x-a|} \, d\sigma \\ &= \frac{1}{\omega_n} \int_{\partial B_r(a)} r^{1-n} \, d\sigma = 1. \end{aligned}$$

Lemma 2.10 (Representation formula). *Suppose Ω is bounded and $u \in C^1(\overline{\Omega}) \cap C^2(\Omega)$. Then for every $a \in \Omega$,*

$$u(a) = \int_{\Omega} \Gamma(a, y) \Delta u(y) \, dy - \int_{\partial\Omega} \left(\Gamma(a, y) \frac{\partial u}{\partial n}(y) - u(y) \frac{\partial \Gamma}{\partial n}(a, y) \right) d\sigma_y.$$

Proof. We treat the case $n \geq 3$. Apply Green's formula on $\Omega \setminus B_\varepsilon(a)$:

$$\begin{aligned} & \int_{\Omega \setminus B_\varepsilon(a)} (\Gamma \Delta u - u \Delta \Gamma) dy \\ &= \int_{\partial \Omega} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) d\sigma_y - \int_{\partial B_\varepsilon(a)} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) d\sigma_y. \end{aligned}$$

Since $\Delta \Gamma \equiv 0$ in $\Omega \setminus B_\varepsilon(a)$ for every $\varepsilon > 0$,

$$\begin{aligned} & \lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(a)} \underbrace{\left(\Gamma \frac{\partial u}{\partial n} \right)}_{I_1} - \underbrace{\left(u \frac{\partial \Gamma}{\partial n} \right)}_{I_2} d\sigma_y \\ &= \int_{\Omega} \Gamma \Delta u dy - \int_{\partial \Omega} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) d\sigma_y. \end{aligned}$$

Estimate of I_1 .

$$\left| \int_{\partial B_\varepsilon(a)} \Gamma \frac{\partial u}{\partial n} d\sigma_y \right| \leq \frac{1}{(n-2)\omega_n} \max |\nabla u| \int_{\partial B_\varepsilon(a)} \varepsilon^{2-n} d\sigma_y = \frac{\max |\nabla u|}{n-2} \varepsilon \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0.$$

Estimate of I_2 .

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(a)} u \frac{\partial \Gamma}{\partial n} d\sigma_y &= \lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(a)} \left(\frac{1}{\omega_n} u(y) \frac{y-a}{|y-a|^n} \cdot \frac{y-a}{|y-a|} \right) d\sigma_y \\ &= \lim_{\varepsilon \rightarrow 0} \frac{1}{\omega_n \varepsilon^{n-1}} \int_{\partial B_\varepsilon(a)} u(y) d\sigma_y = \lim_{\varepsilon \rightarrow 0} \frac{1}{|\partial B_\varepsilon(a)|} \int_{\partial B_\varepsilon(a)} u(y) d\sigma_y = u(a), \end{aligned}$$

as long as $u \in C^1(\Omega)$. Combining the limits yields the claimed formula. $\square$

Remark 2.11. (i) If $a \notin \overline{\Omega}$, the right-hand side of the representation formula vanishes. Indeed $\Delta \Gamma(a, \cdot) \equiv 0$ in Ω (no singularity), so

$$0 \equiv \int_{\Omega} u \Delta \Gamma dy = \int_{\Omega} \Gamma \Delta u dy - \int_{\partial \Omega} \left(\Gamma \frac{\partial u}{\partial n} - u \frac{\partial \Gamma}{\partial n} \right) d\sigma_y.$$

(ii) For every $a \in \Omega$, $\Gamma(a, \cdot)$ is integrable on Ω (despite the singularity at $y = a$).

(iii) Taking $u \equiv 1$ gives

$$\int_{\partial \Omega} \frac{\partial \Gamma}{\partial n} d\sigma = 1.$$

Derivation of the Green's function. Let $\Omega \subseteq \mathbb{R}^n$ be bounded and $u \in C^1(\overline{\Omega}) \cap C^2(\Omega)$. For $x \in \Omega$,

$$u(x) = \int_{\Omega} \Gamma(x, y) \Delta u(y) dy - \int_{\partial \Omega} \left(\Gamma(x, y) \frac{\partial u}{\partial n}(y) - u(y) \frac{\partial \Gamma}{\partial n}(x, y) \right) d\sigma_y. \quad (11)$$

Assume u solves the Dirichlet problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial \Omega. \end{cases}$$

Set $G(x, y) = \Gamma(x, y) + \Phi(x, y)$, where $\Phi(x, \cdot) \in C^2(\overline{\Omega})$ satisfies $\Delta_y \Phi(x, y) = 0$ for $y \in \Omega$. Then

$$0 = \int_{\Omega} \Phi(x, y) \Delta u(y) dy - \int_{\partial \Omega} \left(\Phi(x, y) \frac{\partial u}{\partial n}(y) - u(y) \frac{\partial \Phi}{\partial n}(x, y) \right) d\sigma_y. \quad (12)$$

Adding (11) and (12) yields

$$u(x) = \int_{\Omega} G(x, y) \Delta u(y) dy - \int_{\partial\Omega} \left(G(x, y) \frac{\partial u}{\partial n}(y) - u(y) \frac{\partial G}{\partial n}(x, y) \right) d\sigma_y.$$

Choosing the corrector so that

$$\begin{cases} \Delta_y \Phi(x, y) = 0 & y \in \Omega, \\ \Phi(x, y) = -\Gamma(x, y) & y \in \partial\Omega, \end{cases}$$

we obtain $G(x, y) = 0$ for $y \in \partial\Omega$, and therefore

$$u(x) = - \int_{\Omega} G(x, y) f(y) dy + \int_{\partial\Omega} g(y) \frac{\partial G}{\partial n}(x, y) d\sigma_y.$$

In other words, the Green's function satisfies

$$\begin{cases} \Delta G = \delta_x & \text{in } \Omega, \\ G = 0 & \text{on } \partial\Omega. \end{cases}$$

It remains to solve for Φ (equivalently G) on concrete domains Ω .

2.4.2 Properties of the Green's function

Theorem 2.12 (Symmetry of the Green's function). $G(x_1, x_2) = G(x_2, x_1)$ for all $x_1 \neq x_2$ in Ω . (The diagonal $x_1 = x_2$ is singular.)

Proof. Pick $x_1, x_2 \in \Omega$ and $\varepsilon > 0$ small enough that $B_\varepsilon(x_1) \cap B_\varepsilon(x_2) = \emptyset$. Write $G_1(y) = G(x_1, y)$ and $G_2(y) = G(x_2, y)$. Green's second identity on $\Omega \setminus (B_\varepsilon(x_1) \cup B_\varepsilon(x_2))$ gives

$$\begin{aligned} & \int_{\Omega \setminus (B_\varepsilon(x_1) \cup B_\varepsilon(x_2))} (G_1 \Delta G_2 - G_2 \Delta G_1) dy \\ &= \int_{\partial\Omega} \left(G_1 \frac{\partial G_2}{\partial n} - G_2 \frac{\partial G_1}{\partial n} \right) d\sigma \\ & \quad - \int_{\partial B_\varepsilon(x_1)} \left(G_1 \frac{\partial G_2}{\partial n} - G_2 \frac{\partial G_1}{\partial n} \right) d\sigma \\ & \quad - \int_{\partial B_\varepsilon(x_2)} \left(G_1 \frac{\partial G_2}{\partial n} - G_2 \frac{\partial G_1}{\partial n} \right) d\sigma. \end{aligned}$$

Note that

$$\begin{cases} \Delta_y G_1 = \Delta_y G_2 = 0 & \text{in } \Omega \setminus (B_\varepsilon(x_1) \cup B_\varepsilon(x_2)), \\ G_1 = G_2 = 0 & \text{on } \partial\Omega. \end{cases}$$

Hence

$$\begin{aligned} 0 &= \int_{\partial B_\varepsilon(x_1)} \left(G_1 \frac{\partial G_2}{\partial n} - G_2 \frac{\partial G_1}{\partial n} \right) d\sigma_y \\ & \quad + \int_{\partial B_\varepsilon(x_2)} \left(G_1 \frac{\partial G_2}{\partial n} - G_2 \frac{\partial G_1}{\partial n} \right) d\sigma_y. \end{aligned} \tag{*}$$

We check the case $n \geq 3$. First,

$$\int_{\partial B_\varepsilon(x_1)} G_1 \frac{\partial G_2}{\partial n} d\sigma_y = \int_{\partial B_\varepsilon(x_1)} (\Gamma(x_1, y) + \Phi(x_1, y)) \frac{\partial G_2}{\partial n} d\sigma_y.$$

Since $\Gamma(x_1, y) = -\frac{1}{(n-2)\omega_n}|x_1 - y|^{2-n}$,

$$\begin{aligned} \int_{\partial B_\varepsilon(x_1)} \frac{1}{(n-2)\omega_n}|x_1 - y|^{2-n} \left| \frac{\partial G_2}{\partial n} \right| d\sigma_y &\leq C \varepsilon^{n-1} \varepsilon^{2-n} \max |DG_2| = C\varepsilon \rightarrow 0, \\ \int_{\partial B_\varepsilon(x_1)} |\Phi(x_1, y)| \left| \frac{\partial G_2}{\partial n} \right| d\sigma_y &\leq C\varepsilon^{n-1} \rightarrow 0. \end{aligned}$$

Thus

$$\lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(x_1)} G_1 \frac{\partial G_2}{\partial n} d\sigma_y = 0,$$

and similarly

$$\lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(x_2)} G_2 \frac{\partial G_1}{\partial n} d\sigma_y = 0.$$

Next,

$$\int_{\partial B_\varepsilon(x_1)} G_2 \frac{\partial G_1}{\partial n} d\sigma_y = \int_{\partial B_\varepsilon(x_1)} G_2(y) \nabla_y (\Gamma(x_1, y) + \Phi(x_1, y)) \cdot \vec{n} d\sigma_y.$$

A direct computation gives

$$\nabla_y \Gamma(x_1, y) \cdot \vec{n} = -\frac{(2-n)}{\omega_n(n-2)} |y - x_1|^{1-n} \frac{y - x_1}{|y - x_1|} \cdot \frac{y - x_1}{|y - x_1|} = \frac{1}{\omega_n} |y - x_1|^{1-n}.$$

It is then not hard to see that

$$\lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(x_1)} G_2(y) \nabla_y \Gamma(x_1, y) \cdot \vec{n} d\sigma = G_2(x_1),$$

while

$$\left| \int_{\partial B_\varepsilon(x_1)} G_2(y) \nabla_y \Phi \cdot \vec{n} d\sigma_y \right| \leq \varepsilon^{n-1} \max |G_2| \max |\nabla_y \Phi| \rightarrow 0.$$

Therefore

$$\lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(x_1)} G_2(y) \frac{\partial G_1}{\partial n} d\sigma_y = G_2(x_1).$$

Similarly,

$$\lim_{\varepsilon \rightarrow 0} \int_{\partial B_\varepsilon(x_2)} G_1(y) \frac{\partial G_2}{\partial n} d\sigma_y = G_1(x_2).$$

Passing to the limit in (*) yields $G_1(x_2) = G_2(x_1)$, i.e. $G(x_1, x_2) = G(x_2, x_1)$. $\square$

Remark 2.13. (i) $G(x, y) \in C_y^\infty(\Omega \setminus \{x\})$.

(ii) $G(x, y) < 0$ for $y \in \Omega \setminus \{x\}$. Indeed, with the convention $\Delta \Gamma = \delta_x$, one has $\Gamma \rightarrow -\infty$ as $y \rightarrow x$, while $\Phi = -\Gamma$ on $\partial\Omega$ is harmonic in Ω . Away from the pole, G is harmonic, $G = 0$ on $\partial\Omega$, and $G \rightarrow -\infty$ at $y = x$, hence the maximum principle yields $G < 0$ in $\Omega \setminus \{x\}$.

Recall the representation

$$u(x) = - \int_{\Omega} G(x, y) f(y) dy + \int_{\partial\Omega} g(y) \frac{\partial G}{\partial n} d\sigma_y.$$

Writing $u = u_1 + u_2$, we have the decomposition

$$u_1 \text{ solves } \begin{cases} -\Delta u_1 = f & \text{in } \Omega, \\ u_1 = 0 & \text{on } \partial\Omega, \end{cases} \quad u_2 \text{ solves } \begin{cases} -\Delta u_2 = 0 & \text{in } \Omega, \\ u_2 = g & \text{on } \partial\Omega, \end{cases}$$

and therefore u solves

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases}$$

Remark 2.14 (Task). Find $G(x, y) / \Phi(x, y)$ for concrete domains Ω .

2.5 Solving for the Green's function

3 Green's functions on special domains; Perron's method

3.1 Green's function for the ball

3.1.1 Ball $\Omega = B_R(0)$

Definition 3.1 (Inversion). The inversion of $x \in \mathbb{R}^n \setminus \{0\}$ with respect to $\partial B_R(0)$ is

$$x^* = \frac{R^2}{|x|^2} x.$$

This map sends $0 \mapsto \infty$, $\infty \mapsto 0$, fixes $\partial B_R(0)$, and interchanges the interior and exterior of the ball.

Idea. Invert the singularity from $x \in B_R(0)$ to $x^* \notin B_R(0)$. Recall $\Delta_y \Gamma(z, y) = 0$ for all $y \neq z$. Hence, for $n = 3$ (and similarly in general),

$$\Delta_y \Gamma\left(\frac{|x|}{R} x^*, \frac{|x|}{R} y\right) = 0 \quad \text{for } y \neq x^*.$$

Define the image term

$$\Phi(x, y) := -\Gamma\left(\frac{|x|}{R} x^*, \frac{|x|}{R} y\right)$$

(so that $G = \Gamma + \Phi$), and obtain the explicit formula

$$G(x, y) = \begin{cases} \frac{1}{2\pi} \left(\log|x - y| - \log\left|\frac{R}{|x|}x - \frac{|x|}{R}y\right| \right), & n = 2, \\ -\frac{1}{(n-2)\omega_n} \left(|x - y|^{2-n} - \left|\frac{R}{|x|}x - \frac{|x|}{R}y\right|^{2-n} \right), & n \geq 3. \end{cases} \quad (13)$$

Verification. We need to check:

1. $\Delta_y \Phi(x, y) = 0$ for $y \in B_R(0)$. Since $x \in B_R(0)$ implies $x^* \notin \overline{B_R(0)}$, one has $y \neq x^*$ for $y \in B_R(0)$.
2. $\Phi(x, y) = -\Gamma(x, y)$ for $y \in \partial B_R(0)$.

Proof of the boundary condition. Let $y \in \partial B_R(0)$ and $x \neq 0$. Then $|y| = R$ and

$$\begin{aligned} \left| \frac{|x|}{R} x^* - \frac{|x|}{R} y \right|^2 &= \frac{|x|^2}{R^2} |x^* - y|^2 = \frac{|x|^2}{R^2} (|x^*|^2 + |y|^2 - 2x^* \cdot y) \\ &= \frac{|x|^2}{R^2} \left(\left| \frac{R^2}{|x|^2} x \right|^2 + |y|^2 - 2 \frac{R^2}{|x|^2} x \cdot y \right) \\ &= R^2 + |x|^2 - 2x \cdot y = |x - y|^2. \end{aligned}$$

Hence

$$\Gamma\left(\frac{|x|}{R} x^*, \frac{|x|}{R} y\right) = \Gamma(x, y) \quad \text{for } y \in \partial B_R(0),$$

so $\Phi = -\Gamma$ on $\partial B_R(0)$. □

Lemma 3.2 (Normal derivative on $\partial B_R(0)$). Suppose G is given by (13). Then for all $x \in B_R(0)$ and $y \in \partial B_R(0)$,

$$\frac{\partial G}{\partial n}(x, y) = \frac{R^2 - |x|^2}{\omega_n R |x - y|^n}.$$

Proof. Consider the case $n \geq 3$. From (13),

$$G(x, y) = -\frac{1}{(n-2)\omega_n} \left(|x-y|^{2-n} - \left| \frac{R}{|x|}x - \frac{|x|}{R}y \right|^{2-n} \right).$$

Differentiating in y ,

$$\nabla_y G = \frac{1}{\omega_n} \left(|x-y|^{1-n} \frac{y-x}{|y-x|} - \left| \frac{R}{|x|}x - \frac{|x|}{R}y \right|^{1-n} \frac{\frac{|x|}{R}y - \frac{R}{|x|}x}{\left| \frac{R}{|x|}x - \frac{|x|}{R}y \right|} \cdot \frac{|x|}{R} \right).$$

On $\partial B_R(0)$ one has $\left| \frac{R}{|x|}x - \frac{|x|}{R}y \right| = |x-y|$, whence

$$\nabla_y G = \frac{1}{\omega_n} |x-y|^{-n} \left((y-x) - \left(\frac{|x|^2}{R^2} y - x \right) \right) = \frac{1}{\omega_n} |x-y|^{-n} \left(y - \frac{|x|^2}{R^2} y \right).$$

Therefore, with outward unit normal $n = y/R$,

$$\frac{\partial G}{\partial n}(x, y) = D_y G \cdot \frac{y}{R} = \frac{1}{\omega_n} |x-y|^{-n} R \left(1 - \frac{|x|^2}{R^2} \right) = \frac{1}{\omega_n} |x-y|^{-n} \frac{R^2 - |x|^2}{R}.$$

□

Definition 3.3 (Poisson kernel). The function

$$K(x, y) := \frac{\partial G}{\partial n}(x, y)$$

is called the Poisson kernel (for the ball).

Properties.

1. $K(x, y)$ is smooth for $x \neq y$.
2. $K(x, y) > 0$ for all $|x| < R$.
3. $\int_{|y|=R} K(x, y) d\sigma_y = 1$ for all $|x| < R$.

Indeed, recalling Green's representation and taking $u \equiv 1$ (so $f = 0$ and $G|_{\partial\Omega} = 0$),

$$1 = \int_{\partial\Omega} \frac{\partial G}{\partial n}(x, y) d\sigma_y.$$

Theorem 3.4 (Poisson integral formula). For every $\varphi \in C(\partial B_R(0))$, define

$$u(x) = \begin{cases} \int_{\partial B_R(0)} K(x, y) \varphi(y) d\sigma_y, & x \in B_R(0), \\ \varphi(x), & x \in \partial B_R(0). \end{cases}$$

Then u solves

$$\begin{cases} \Delta u = 0 & \text{in } B_R(0), \\ u = \varphi & \text{on } \partial B_R(0). \end{cases}$$

Proof. For $x \in B_R(0)$,

$$\Delta u(x) = \int_{\partial B_R(0)} \Delta_x K(x, y) \varphi(y) d\sigma_y.$$

Since $\Delta_x K(x, y) = 0$ for $x \in B_R(0)$ and $y \in \partial B_R(0)$ (the singularity of G lies at $y = x$, which is interior), and φ is bounded,

$$\int_{\partial B_R(0)} \Delta_x K(x, y) \varphi(y) d\sigma_y = 0.$$

It remains to check the boundary values. □

Theorem 3.5. For every $x^0 \in \partial B_R(0)$,

$$\lim_{B_R(0) \ni x \rightarrow x^0} u(x) = \varphi(x^0).$$

Proof. Fix $x^0 \in \partial B_R(0)$. Given $\varepsilon > 0$, choose $\delta > 0$ small enough that

$$|\varphi(y) - \varphi(x^0)| < \varepsilon \quad \text{whenever } y \in \partial B_R(0) \text{ and } |y - x^0| < \delta.$$

Then, for $|x - x^0| < \delta/2$,

$$u(x) - \varphi(x^0) = \int_{\partial B_R(0)} K(x, y) \varphi(y) d\sigma_y - \varphi(x^0) = \int_{\partial B_R(0)} K(x, y) (\varphi(y) - \varphi(x^0)) d\sigma_y,$$

where we used $\int K = 1$. Splitting the integral,

$$\begin{aligned} |u(x) - \varphi(x^0)| &\leq \int_{\partial B_R(0) \cap \{|y - x^0| < \delta\}} K(x, y) |\varphi(y) - \varphi(x^0)| d\sigma_y =: I_1 \\ &\quad + \int_{\partial B_R(0) \cap \{|y - x^0| \geq \delta\}} K(x, y) |\varphi(y) - \varphi(x^0)| d\sigma_y =: I_2. \end{aligned}$$

For the near region, $|I_1| \leq \varepsilon$.

For the far region, $|x^0 - y| \geq \delta$ and $|x - x^0| < \delta/2$ imply

$$|y - x^0| \leq |y - x| + \frac{\delta}{2} \leq |y - x| + \frac{1}{2}|y - x^0|,$$

hence $|y - x^0| \leq 2|y - x|$ (so y is not close to x or x^0). Thus

$$\begin{aligned} |I_2| &\leq 2 \max |\varphi| \int_{\partial B_R(0) \cap \{|y - x^0| \geq \delta\}} K(x, y) d\sigma_y \\ &\leq C \int_{\partial B_R(0) \cap \{|y - x^0| \geq \delta\}} \frac{R^2 - |x|^2}{R} |x - y|^{-n} d\sigma_y \\ &\leq C(R^2 - |x|^2) \int_{\partial B_R(0) \cap \{|y - x^0| \geq \delta\}} |x^0 - y|^{-n} d\sigma_y \\ &\leq C(R^2 - |x|^2) \delta^{-n}. \end{aligned}$$

As $x \rightarrow x^0$ one has $R^2 - |x|^2 \rightarrow 0$, so $I_2 \rightarrow 0$. Combining with $|I_1| \leq \varepsilon$ yields the claim. $\square$

Remark 3.6. In the Poisson integral formula, taking $x = 0$,

$$K(0, y) = \frac{1}{\omega_n} |y|^{-n} \frac{R^2}{R} = \frac{R}{\omega_n} |y|^{-n},$$

hence

$$u(0) = \int_{\partial B_R(0)} \frac{R}{\omega_n} |y|^{-n} \varphi(y) d\sigma_y = \frac{1}{R^{n-1} \omega_n} \int_{\partial B_R(0)} \varphi(y) d\sigma_y,$$

which is the mean value property.

Lemma 3.7 (Harnack's inequality in a ball). Suppose u is harmonic in $B_R(x_0)$ and $u \geq 0$. Then for $r = |x - x_0| < R$,

$$\left(\frac{R}{R+r}\right)^{n-2} \frac{R-r}{R+r} u(x_0) \leq u(x) \leq \left(\frac{R}{R-r}\right)^{n-2} \frac{R+r}{R-r} u(x_0).$$

Proof. Without loss of generality take $x_0 = 0$. The Poisson formula gives

$$u(x) = \frac{1}{\omega_n R} \int_{\partial B_R} \frac{R^2 - |x|^2}{|x - y|^n} u(y) d\sigma_y.$$

For $|y| = R$ one has $R - |x| \leq |y - x| \leq R + |x|$, and therefore

$$\begin{aligned} \frac{1}{\omega_n R} \frac{R - |x|}{R + |x|} \left(\frac{1}{R + |x|} \right)^{n-2} \int_{\partial B_R} u(y) d\sigma_y &\leq u(x) \\ &\leq \frac{1}{\omega_n R} \frac{R + |x|}{R - |x|} \left(\frac{1}{R - |x|} \right)^{n-2} \int_{\partial B_R} u(y) d\sigma_y. \end{aligned}$$

By the mean value property,

$$\frac{1}{\omega_n R^{n-1}} \int_{\partial B_R} u(y) d\sigma_y = u(0),$$

whence

$$\frac{R - |x|}{R + |x|} \left(\frac{R}{R + |x|} \right)^{n-2} u(0) \leq u(x) \leq \frac{R + |x|}{R - |x|} \left(\frac{R}{R - |x|} \right)^{n-2} u(0).$$

□

Corollary 3.8 (Liouville). *If $\Delta u = 0$ in $\mathbb{R}^n$ and u is bounded, then u is constant.*

Second approach. From Harnack's inequality,

$$\left(\frac{R}{R + |x|} \right)^{n-2} \frac{R - |x|}{R + |x|} u(0) \leq u(x) \leq \left(\frac{R}{R - |x|} \right)^{n-2} \frac{R + |x|}{R - |x|} u(0).$$

Letting $R \rightarrow \infty$ yields $u(0) \leq u(x) \leq u(0)$, so $u \equiv C$.

□

3.2 Green's function for a half-space

3.2.1 Half-space

Let

$$\mathbb{R}_+^n = \{x = (x_1, \dots, x_n) : x_n > 0\}.$$

Definition 3.9 (Reflection). For $x \in \mathbb{R}_+^n$, write

$$x^* = (x_1, \dots, x_{n-1}, -x_n) \in \mathbb{R}_-^n.$$

Idea. Take $\Phi(x, y) = -\Gamma(x^*, y)$. One needs

$$\begin{cases} \Delta_y \Phi(x, y) = 0, & y \in \mathbb{R}_+^n, \\ \Phi(x, y) = -\Gamma(x, y), & y \in \partial \mathbb{R}_+^n. \end{cases}$$

The first holds because the only singularity is at $y = x^* \in \mathbb{R}_-^n$, while $y \in \mathbb{R}_+^n$. For the second: if $y \in \partial \mathbb{R}_+^n$, then $y = y^*$ and $|x - y| = |x^* - y|$, so

$$\Gamma(x^*, y) = \Gamma(x, y).$$

Thus $G(x, y) = \Gamma(x, y) - \Gamma(x^*, y)$ vanishes on the boundary (equivalently $G = \Gamma + \Phi$ with $\Phi = -\Gamma(x^*, \cdot)$).

Theorem 3.10 (Green's function for the half-space).

$$G(x, y) = \begin{cases} \frac{1}{2\pi} (\log |x - y| - \log |x^* - y|), & n = 2, \\ -\frac{1}{(n-2)\omega_n} (|x - y|^{2-n} - |x^* - y|^{2-n}), & n \geq 3. \end{cases}$$

Lemma 3.11. For $x \in \mathbb{R}_+^n$ and $y \in \partial\mathbb{R}_+^n$,

$$\frac{\partial G}{\partial n}(x, y) = \frac{2x_n}{\omega_n} \frac{1}{|x - y|^n}.$$

Proof. Only the case $n \geq 3$ is written. One has $\partial G / \partial n = \nabla_y G \cdot \vec{n}$, and

$$\nabla_y G = \frac{1}{\omega_n} \left(|y - x|^{1-n} \frac{y - x}{|y - x|} - |y - x^*|^{1-n} \frac{y - x^*}{|y - x^*|} \right).$$

On the boundary $|y - x| = |y - x^*|$, so

$$\nabla_y G = \frac{1}{\omega_n} |y - x|^{-n} (y - x - y + x^*) = \frac{1}{\omega_n} |x - y|^{-n} (-x + x^*).$$

Here $-x + x^* = (0, \dots, -2x_n)$. Taking the outward unit normal $\vec{n} = (0, \dots, 0, -1)$ gives

$$\nabla_y G \cdot \vec{n} = \frac{2x_n}{\omega_n |x - y|^n}.$$

□

Definition 3.12 (Poisson kernel for the half-space).

$$K(x, y) = \frac{\partial G}{\partial n}(x, y) = \frac{2x_n}{\omega_n |x - y|^n}$$

is the Poisson kernel for $\mathbb{R}_+^n$. Define

$$u(x) = \begin{cases} \int_{\partial\mathbb{R}_+^n} K(x, y) \varphi(y) d\sigma_y, & x \in \mathbb{R}_+^n, \\ \varphi(x), & x \in \partial\mathbb{R}_+^n. \end{cases}$$

Theorem 3.13. Assume φ is bounded and continuous on $\partial\mathbb{R}_+^n$. Then:

1. u solves

$$\begin{cases} \Delta u = 0 & \text{in } \mathbb{R}_+^n, \\ u = \varphi & \text{on } \partial\mathbb{R}_+^n; \end{cases}$$

2. $u \in C^\infty(\mathbb{R}_+^n) \cap L^\infty(\mathbb{R}_+^n)$;

3. $\lim_{\substack{x \rightarrow x^0 \\ x \in \mathbb{R}_+^n}} u(x) = \varphi(x^0)$ for every $x^0 \in \partial\mathbb{R}_+^n$.

Proof. (1) $\Delta_x u = \int_{\partial\mathbb{R}_+^n} \Delta_x K(x, y) \varphi(y) d\sigma_y = 0$, since $\Delta_x K(x, y) = 0$ for $x \in \mathbb{R}_+^n$ (the singularity would require $y = x$, which cannot occur for $y \in \partial\mathbb{R}_+^n$).

(2) Clear from the smoothness of K in $x \in \mathbb{R}_+^n$ and the boundedness of φ .

(3) Fix $x^0 \in \partial\mathbb{R}_+^n$. Given $\varepsilon > 0$, choose $\delta > 0$ so that $|\varphi(y) - \varphi(x^0)| < \varepsilon$ whenever $|y - x^0| < \delta$ and $y \in \partial\mathbb{R}_+^n$. Then

$$u(x) - \varphi(x^0) = \int_{\partial\mathbb{R}_+^n} K(x, y) (\varphi(y) - \varphi(x^0)) d\sigma_y,$$

and

$$\begin{aligned} |u(x) - \varphi(x^0)| &\leq \int_{\partial\mathbb{R}_+^n \cap \{|y - x^0| < \delta\}} K(x, y) |\varphi(y) - \varphi(x^0)| d\sigma_y \\ &\quad + \int_{\partial\mathbb{R}_+^n \cap \{|y - x^0| \geq \delta\}} K(x, y) |\varphi(y) - \varphi(x^0)| d\sigma_y \\ &\leq C\varepsilon + (\max |\varphi|) \int_{\partial\mathbb{R}_+^n \cap \{|y - x^0| \geq \delta\}} \frac{x_n}{\omega_n |x - y|^n} d\sigma_y \\ &\leq C\varepsilon + C x_n \int_{\substack{\partial\mathbb{R}_+^n \\ |y - x^0| \geq \delta}} |x - y|^{-n} d\sigma_y. \end{aligned}$$

As $x_n \rightarrow 0$ (with $x \rightarrow x^0$), the second term tends to 0, so $|u(x) - \varphi(x^0)| \leq C\varepsilon$ for x sufficiently close to x^0 . $\square$

3.3 Perron's method

3.3.1 Perron's method

- We have shown existence of

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega \end{cases}$$

for the special domains $\Omega = B_R(0)$ and $\Omega = \mathbb{R}_+^n$, by writing out an explicit formula.

- This is *not* possible for a general domain Ω .
- We will introduce two approaches to show existence:
 - Perron's method;
 - energy method (calculus of variations).

3.3.2 Subharmonic functions

Definition 3.14 (Subharmonic / superharmonic). A function $u \in C^2(\Omega)$ is *subharmonic* (respectively *superharmonic*) if $-\Delta u \leq 0$ (respectively $-\Delta u \geq 0$) in Ω .

Theorem 3.15 (Equivalent characterizations). *The following are equivalent:*

- (i) u is subharmonic: $-\Delta u \leq 0$ whenever Δu exists;
- (ii) for every ball $B \subset\subset \Omega$ and every $h \in C^2(B) \cap C(\overline{B})$ with $\Delta h = 0$ in B , if $u \leq h$ on ∂B then $u \leq h$ in B ;
- (iii)

$$u(x) \leq \frac{1}{|B_r|} \int_{B_r(x)} u(y) dy \quad \forall r > 0 \text{ with } B_r(x) \subset \Omega;$$

(iv)

$$u(x) \leq \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} u(y) d\sigma_y \quad \forall r > 0 \text{ with } B_r(x) \subset \Omega.$$

Sketch. The equivalences (i) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv) follow by the same arguments used for the mean value property of harmonic functions.

(ii) $\Leftarrow$ (iii)/(iv). For every $x \in \Omega$ take $B_r(x) \subset \Omega$ with $u \leq h$ on $\partial B_r(x)$. Then

$$u(x) \leq \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} u(y) d\sigma_y \leq \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} h(y) d\sigma_y = h(x).$$

(ii) $\Rightarrow$ (iv). Take the special harmonic function h solving

$$\begin{cases} \Delta h = 0 & \text{in } B, \\ h \equiv u & \text{on } \partial B. \end{cases}$$

Then

$$h(x) = \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} h(y) d\sigma_y = \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} u(y) d\sigma_y,$$

and $u \leq h$ on ∂B implies $u(x) \leq h(x)$ by (ii). Hence

$$u(x) \leq \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} u(y) d\sigma_y.$$

□

Lemma 3.16 (Maximum principle). *If u is subharmonic in Ω , then*

$$\sup_{\Omega} u = \max_{\partial\Omega} u,$$

or else $u \equiv C$.

Proof. Define

$$\Omega_1 := \left\{ x : u(x) = \sup_{\Omega} u =: M \right\} \subset \Omega,$$

which is closed. The aim is to show that Ω_1 is open. For every $x \in \Omega_1$ there exists $B_\rho(x) \subset \Omega$ such that, by subharmonicity,

$$M = u(x) \leq \frac{1}{|B_\rho|} \int_{B_\rho(x)} u(y) dy \leq M.$$

Hence $u = M$ a.e. in B_ρ , so $B_\rho(x) \subset \Omega_1$. Therefore Ω_1 is either empty or equal to Ω :

- if $\Omega_1 = \emptyset$, then $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$;
- if $\Omega_1 = \Omega$, then $u \equiv M$ in Ω .

□

Lemma 3.17. *Let u be subharmonic and v superharmonic, with $u - v \leq 0$ on $\partial\Omega$. Then either $u < v$ in Ω , or $u \equiv v$ in Ω .*

Proof. The difference $u - v$ is subharmonic. □

Remark 3.18. A harmonic function is both subharmonic and superharmonic.

3.3.3 Harmonic lifting

Let u be subharmonic in Ω , let $B \subset\subset \Omega$ be a fixed ball, and let $\bar{u}$ be harmonic in B with $\bar{u} = u$ on ∂B . Then (using the Green's function for the ball) either $u < \bar{u}$ in B , or $u \equiv \bar{u}$ in B .

Definition 3.19 (Harmonic lifting). The function

$$U(x) = \begin{cases} \bar{u}(x), & x \in B, \\ u(x), & x \in \Omega \setminus B, \end{cases}$$

is called a *harmonic lifting* of u in B .

Lemma 3.20. *The lifting U is subharmonic in Ω .*

Proof. Let $B' \subset\subset \Omega$ and let h be harmonic with $h \geq U$ on $\partial B'$. We need to show $U \leq h$ in B' . Write

$$\begin{aligned} C_1 &:= \partial B' \setminus B, \\ C_3 &:= \partial B' \cap B, \\ C_2 &:= \partial B \cap B'. \end{aligned}$$

Note that $\Delta h = 0$ and $\Delta \bar{u} = 0$.

On $C_1 = \partial B' \setminus B$ one has $h \geq U = u$ by assumption (on the whole of $\partial B'$). On $C_3 = \partial B' \cap B$ one has $h \geq U = \bar{u} \geq u$. Hence $h \geq u$ on $\partial B'$, and since h is harmonic and u is subharmonic,

$$h \geq u \quad \text{in } B'.$$

In particular, in $B' \setminus B$ we have $h \geq u = U$.

It remains to show $h \geq U$ in $B' \cap B$. On $C_2 = \partial B \cap B'$ one has $u = \bar{u} = U$, so $h \geq \bar{u}$. On $C_3 = \partial B' \cap B$ one has $h \geq U = \bar{u}$. Since both h and $\bar{u}$ are harmonic,

$$h \geq \bar{u} \quad \text{in } B' \cap B.$$

Thus $h \geq U$ on B' , and therefore U is subharmonic. □

Lemma 3.21. *If $u_1, \dots, u_N$ are subharmonic in Ω , then so is $\max\{u_1, \dots, u_N\}$.*

4 Perron's method, barriers, and the energy method

4.1 Equivalent characterizations of subharmonic functions

Recall the following are equivalent:

- (i) $-\Delta u \leq 0$ (if the Laplacian exists);
- (ii) for every ball $B \subset\subset \Omega$ and every harmonic h with $u \leq h$ on ∂B , one has $u \leq h$ in B ;
- (iii) $u(x) \leq \frac{1}{|B_r|} \int_{B_r(x)} u(y) dy$;
- (iv) $u(x) \leq \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} u(y) d\sigma_y$.

Proof of (i) $\Leftrightarrow$ (iii)/(iv). Define the spherical mean

$$\phi(r) = \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} u(y) d\sigma_y.$$

With the change of variables $y - x = rz$, $|z| = 1$, so that $d\sigma_y = r^{n-1} d\sigma_z$,

$$\phi(r) = \frac{1}{\omega_n r^{n-1}} r^{n-1} \int_{|z|=1} u(x + rz) d\sigma_z = \frac{1}{\omega_n} \int_{|z|=1} u(x + rz) d\sigma_z.$$

Differentiating in r ,

$$\phi'(r) = \frac{1}{\omega_n} \int_{|z|=1} z \cdot \nabla u(x + rz) d\sigma_z = \frac{1}{\omega_n r^{n-1}} \int_{\partial B_r} \frac{y - x}{r} \cdot \nabla u d\sigma_y = \frac{1}{\omega_n r^{n-1}} \int_{\partial B_r} \frac{\partial u}{\partial n} d\sigma_y = \frac{1}{\omega_n r^{n-1}} \int_{B_r} \Delta u dy \geq 0$$

where the last step uses the divergence theorem and $-\Delta u \leq 0$ (i.e. $\Delta u \geq 0$). Hence $\phi(r) \geq \phi(0)$, that is

$$\frac{1}{|\partial B_r|} \int_{\partial B_r} u(y) d\sigma_y \geq u(x).$$

Integrating in r yields the volume mean inequality

$$u(x) \leq \frac{1}{|B_r|} \int_{B_r(x)} u(y) dy.$$

□

Proof of (iii)/(iv) $\Rightarrow$ (ii). For every $x \in \Omega$ and every ball $B_r(x) \subset \subset \Omega$, if $u \leq h$ on $\partial B_r(x)$ with h harmonic, then

$$u(x) \leq \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} u(y) d\sigma_y \leq \frac{1}{|\partial B_r(x)|} \int_{\partial B_r(x)} h(y) d\sigma_y = h(x).$$

□

Proof of (ii) $\Rightarrow$ (iii)/(iv). For every $x \in \Omega$ and every $B_r(x) \subset \subset \Omega$, define h by

$$\begin{cases} \Delta h = 0 & \text{in } B_r(x), \\ h = u & \text{on } \partial B_r(x), \end{cases}$$

via the Green's function. Then by (ii),

$$u(x) \leq h(x) = \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} h d\sigma = \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} u d\sigma,$$

and likewise for the volume mean of the harmonic function h . □

4.2 Harmonic lifting and the idea of Perron's method

Harmonic lifting on a ball B . Define

$$U = \begin{cases} \bar{u} & x \in B, \\ u & x \in B^c, \end{cases}$$

where $\bar{u}$ is harmonic in B and $\bar{u} \equiv u$ on ∂B .

In one dimension, subharmonicity means $-u'' \leq 0$ (i.e. u is convex). On an interval $[a, b]$ the harmonic lifting is the linear interpolant

$$\bar{u}(x) = \frac{(x-a)u(b) + (b-x)u(a)}{b-a}.$$

Idea of Perron's method (in 1D). Consider

$$\begin{cases} -u'' = 0, \\ u(0) = u_0, \quad u(1) = u_1. \end{cases}$$

The exact solution is

$$u = u_0(1-x) + u_1x.$$

Starting from $v \in \mathcal{S}_\varphi$, one repeatedly replaces v on subintervals by its harmonic lifting; continually lifting subharmonic functions pushes them toward the solution.

4.3 Existence via Perron's method

We turn to existence of a solution to

$$\begin{cases} -\Delta u = 0 & \text{in } \Omega, \\ u = \varphi & \text{on } \partial\Omega. \end{cases}$$

Let

$$\mathcal{S}_\varphi = \{v \text{ subharmonic in } \Omega : v \leq \varphi \text{ on } \partial\Omega\}.$$

Theorem 4.1. *The function*

$$u \equiv \sup_{v \in \mathcal{S}_\varphi} v$$

is harmonic in Ω .

Proof. **Step 1:** u is well-defined.

- $\mathcal{S}_\varphi \neq \emptyset$: the constant $V \equiv \min_{\partial\Omega} \varphi$ is subharmonic.
- $u \not\equiv +\infty$: for every $v \in \mathcal{S}_\varphi$,

$$v(x) \leq \sup_{\partial\Omega} \varphi \quad \forall x \in \Omega.$$

In particular,

$$\inf_{\partial\Omega} \varphi \leq u(x) \leq \sup_{\partial\Omega} \varphi.$$

Step 2: Fix any $x \in \Omega$. There exists a sequence $\{v_n\} \subset \mathcal{S}_\varphi$ with $\lim_{n \rightarrow \infty} v_n(x) = u(x)$. Replacing v_k by $\max\{v_1, \dots, v_k, \inf \varphi\}$, we may assume $\{v_n\}$ is increasing and bounded from below. Let $B = B_R(x) \subset \subset \Omega$, and let $\bar{V}_n$ be the harmonic lifting of v_n in $B_R(x)$. Then $\bar{V}_n \in \mathcal{S}_\varphi$ and

$$\lim_{n \rightarrow \infty} \bar{V}_n(x) = u(x). \quad (*)$$

Proof of ().* By definition of lifting, $v_n(x) \leq \bar{V}_n(x)$, so $u(x) \leq \lim_n \bar{V}_n(x)$. On the other hand, $\bar{V}_n \in \mathcal{S}_\varphi$ implies $\bar{V}_n(x) \leq \sup_{v \in \mathcal{S}_\varphi} v = u(x)$, hence

$$u(x) \leq \lim_n \bar{V}_n(x) \leq u(x).$$

□

Step 3: The limit $v := \lim_{n \rightarrow \infty} \bar{V}_n$ is harmonic in $B_{R/2}(x)$. (Here $\{\bar{V}_n\}$ is harmonic in $B_R(x)$.) By the gradient estimate, for any harmonic h ,

$$\sup_{B_{R/2}(x)} |D^\alpha h| \leq C \sup_{B_R(x)} |h|.$$

Hence

$$\sup_n \sup_{B_{R/2}(x)} |D \bar{V}_n| \leq C.$$

A subsequence $\bar{V}_{n_k}$ therefore converges uniformly on $B_{R/2}(x)$ to v . By the MVP (uniform convergence is needed), v is harmonic.

Step 4: We show $u \equiv v$ in $B_{R/2}(x)$ (note $v \leq u$). Assume there exists $x' \in B_{R/2}(x)$ with $u(x') > v(x')$. Then there exists $u_0 \in \mathcal{S}_\varphi$ with

$$v(x') < u_0(x') < u(x').$$

Set $w_k := \max(u_0, \bar{V}_{n_k}) \geq u_0$, and let $\bar{w}_k$ be the harmonic lifting of w_k in $B_R(x)$. Then $\bar{w}_k$ converges uniformly on $B_{R/2}(x)$ to a harmonic function w , and

$$v(y) \leq w(y) \leq u(y) \quad \forall y \in B_{R/2}(x).$$

Evaluating at the center and using (*) gives $v(x) = w(x) = u(x)$. By the MVP,

$$w(x) = \frac{1}{|B_{R/2}(x)|} \int_{B_{R/2}(x)} w(y) dy \geq \frac{1}{|B_{R/2}(x)|} \int_{B_{R/2}(x)} v(y) dy = v(x) = w(x),$$

so $v \equiv w$ in $B_{R/2}(x)$. This contradicts $w(x') \geq u_0(x') > v(x')$. Hence $u \equiv v$ in $B_{R/2}(x)$, and u is harmonic in $B_{R/2}(x)$.

Step 5: Since $x \in \Omega$ is arbitrary, u is harmonic in Ω . □

4.4 Boundary behavior and barriers

Boundary behavior. Does $u = \varphi$ on $\partial\Omega$?

Definition 4.2 (Barrier). Let $\xi \in \partial\Omega$. A function $\beta \in C(\overline{\Omega})$ is called a *barrier* at ξ with respect to Ω if

- i) $\beta > 0$ in $\overline{\Omega} \setminus \{\xi\}$, and $\beta(\xi) = 0$;
- ii) β is superharmonic in Ω .

Definition 4.3 (Regular point). A point $\xi \in \partial\Omega$ is *regular* if there exists a barrier β at ξ with respect to Ω .

Remark 4.4. A barrier is a local property at $\partial\Omega$. If β is a barrier at ξ with respect to Ω , then for every ball $B_\rho(\xi)$ one may form

$$\tilde{\beta} \equiv \begin{cases} m \equiv \inf_{\Omega \setminus B_\rho} \beta & x \in \overline{\Omega} \setminus B_\rho, \\ \min(m, \beta) & x \in \overline{\Omega} \cap B_\rho, \end{cases}$$

which is again a barrier.

Theorem 4.5. Let $u \equiv \sup_{v \in \mathcal{S}_\varphi} v$ be as in Theorem 4.1. If ξ is a regular point of $\partial\Omega$ and φ is continuous at ξ , then

$$\lim_{\Omega \ni x \rightarrow \xi} u(x) = \varphi(\xi).$$

Proof. Let $\varepsilon > 0$. Then there exists $\delta > 0$ such that

$$|\varphi(x) - \varphi(\xi)| < \varepsilon \quad \text{if } x \in \partial\Omega \text{ and } |x - \xi| < \delta.$$

Let $M = \sup_{\partial\Omega} |\varphi|$, and let β be a barrier at ξ . Define $g : \partial\Omega \rightarrow \mathbb{R}$ by

$$g(x) = \liminf_{\substack{y \rightarrow x \\ y \in \Omega}} \beta(y).$$

Then g attains its minimum m_1 on $\{|x - \xi| \geq \delta\} \cap \partial\Omega$. Define

$$\overline{V}(y) = \varphi(\xi) + \varepsilon + \frac{\beta(y)}{m_1} (2M - \varphi(\xi)).$$

(Note $\beta(\xi) = 0$ implies $\overline{V}(\xi) = \varphi(\xi) + \varepsilon$.) Then $\overline{V}$ is superharmonic since β is.

Boundary values of $\overline{V}$.

- If $x \in \partial\Omega$ and $|x - \xi| < \delta$, then

$$\liminf_{\substack{y \rightarrow x \\ y \in \Omega}} \overline{V}(y) \geq \varphi(\xi) + \varepsilon \geq \varphi(x).$$

- If $x \in \partial\Omega$ and $|x - \xi| \geq \delta$, then

$$\liminf_{\substack{y \rightarrow x \\ y \in \Omega}} \overline{V}(y) \geq \varphi(\xi) + \varepsilon + 2M - \varphi(\xi) = 2M + \varepsilon > \varphi(x).$$

Thus $\overline{V}$ is superharmonic and $\overline{V} \geq \varphi$ on $\partial\Omega$. Recalling that u is harmonic with $u \leq \varphi$ on $\partial\Omega$, the strong maximum principle yields $u \leq \overline{V}$ in Ω , and

$$\limsup_{\substack{y \rightarrow \xi \\ y \in \Omega}} u(y) \leq \overline{V}(\xi) = \varphi(\xi) + \varepsilon.$$

Since ε is arbitrary,

$$\limsup_{\substack{y \rightarrow \xi \\ y \in \Omega}} u(y) \leq \varphi(\xi).$$

Similarly, define

$$\underline{V}(y) = \varphi(\xi) - \varepsilon - \frac{\beta(y)}{m_2} (\varphi(\xi) + 2M),$$

where $\limsup_{y \rightarrow x, y \in \Omega} \beta \leq m_2$ for $|x - \xi| \geq \delta$. One can show

$$\begin{cases} \underline{V} \text{ is subharmonic,} \\ \underline{V} \leq \varphi \text{ on } \partial\Omega, \end{cases}$$

i.e. $\underline{V} \in \mathcal{S}_\varphi$, whence $\underline{V} \leq u$. Therefore

$$\varphi(\xi) - \varepsilon \leq \liminf_{\substack{y \rightarrow \xi \\ y \in \Omega}} \underline{V}(y) \leq \liminf_{\substack{y \rightarrow \xi \\ y \in \Omega}} u(y).$$

Combining the liminf and limsup estimates,

$$\varphi(\xi) \leq \liminf_{\substack{y \rightarrow \xi \\ y \in \Omega}} u(y) \leq \limsup_{\substack{y \rightarrow \xi \\ y \in \Omega}} u(y) \leq \varphi(\xi),$$

so

$$\lim_{\substack{y \rightarrow \xi \\ y \in \Omega}} u(y) = \varphi(\xi).$$

□

Examples of barriers: exterior sphere condition. Suppose that for every $\xi \in \partial\Omega$ there exist $R > 0$ and $y \in \mathbb{R}^n$ such that

$$B_R(y) \cap \overline{\Omega} = \{\xi\}.$$

Define

$$\beta(x) = \begin{cases} R^{2-n} - |x - y|^{2-n} & n \geq 3, \\ \log(|x - y|/R) & n = 2. \end{cases}$$

Then:

- $\beta(x) > 0$ for all $x \in \overline{\Omega} \setminus \{\xi\}$;
- $\Delta_x \beta \leq 0$ in Ω (superharmonic);
- $\beta(\xi) = 0$.

Hence β is a barrier at ξ .

4.5 Energy method

Perron's method is not easy to solve numerically. An alternative is the *energy method*.

Dirichlet problem

Consider

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega, \end{cases} \quad (\text{Dirichlet problem}). \quad (*)$$

Uniqueness. Previously we used the maximum principle. Now assume u_1, u_2 are two solutions, and set $w := u_1 - u_2$. Then

$$\begin{cases} \Delta w = 0 & \text{in } \Omega, \\ w = 0 & \text{on } \partial\Omega. \end{cases}$$

Green's identity gives

$$0 = \int_{\Omega} w \Delta w \, dx = \int_{\partial\Omega} w \frac{\partial w}{\partial n} \, d\sigma - \int_{\Omega} \nabla w \cdot \nabla w \, dx.$$

The boundary integral vanishes since $w = 0$ on $\partial\Omega$, so

$$0 = \int_{\Omega} |\nabla w|^2 \, dx \geq 0 \implies \nabla w \equiv 0 \text{ in } \Omega \implies w \equiv C \text{ in } \Omega.$$

But $w = 0$ on $\partial\Omega$, hence $w \equiv 0$ in $\bar{\Omega}$, i.e. $u_1 \equiv u_2$ (energy method).

Energy functional. Define

$$E[w] := \int_{\Omega} \left(\frac{1}{2} |\nabla w|^2 - fw \right) dx,$$

subject to the admissible class

$$\mathcal{A}_0 = \{w \in C^2(\Omega) \cap C(\bar{\Omega}) : w = g \text{ on } \partial\Omega\}.$$

Theorem 4.6 (Dirichlet's principle). 1. If $u \in \mathcal{A}_0$ solves $(*)$, then

$$E[u] \leq E[w] \quad \forall w \in \mathcal{A}_0.$$

2. If $u \in \mathcal{A}_0$ satisfies $E[u] \leq E[w]$ for all $w \in \mathcal{A}_0$, then u solves $(*)$.

In other words,

$$u \in \operatorname{argmin}_{w \in \mathcal{A}_0} E[w], \quad \text{or} \quad \min_{w \in \mathcal{A}_0} E[w] = E[u].$$

Proof. (1) For every $w \in \mathcal{A}_0$,

$$\begin{aligned} 0 &= \int_{\Omega} (-\Delta u - f)(u - w) \, dx && (u - w|_{\partial\Omega} = 0) \\ &= \int_{\Omega} \nabla u \cdot \nabla(u - w) - f(u - w) \, dx. \end{aligned}$$

Hence

$$\int_{\Omega} (|\nabla u|^2 - uf) \, dx = \int_{\Omega} (\nabla u \cdot \nabla w - fw) \, dx \leq \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{2} |\nabla w|^2 - fw \right) dx,$$

where we used $\nabla u \cdot \nabla w \leq \frac{1}{2} |\nabla u|^2 + \frac{1}{2} |\nabla w|^2$. Rearranging yields

$$E[u] = \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 - uf \right) dx \leq \int_{\Omega} \left(\frac{1}{2} |\nabla w|^2 - fw \right) dx = E[w] \quad \forall w \in \mathcal{A}_0.$$

(2) Conversely, fix any $\varphi \in C_c^\infty(\Omega)$. Denote $F(s) := E[u + s\varphi]$ for $s \in \mathbb{R}$. Since $\varphi \in C_c^\infty(\Omega)$ one has $\varphi|_{\partial\Omega} = 0$, so $u + s\varphi \in \mathcal{A}_0$. Minimality gives $E[u] \leq E[u + s\varphi]$, i.e.

$$\begin{aligned} &\int_{\Omega} \frac{1}{2} (|\nabla u|^2 - |\nabla(u + s\varphi)|^2) - f(u - (u + s\varphi)) \, dx \leq 0 \\ &\implies \int_{\Omega} \frac{1}{2} (|\nabla u|^2 - |\nabla u|^2 - s^2 |\nabla \varphi|^2 - 2s \nabla u \cdot \nabla \varphi) + sf\varphi \, dx \leq 0 \\ &\implies \int_{\Omega} -\frac{s^2}{2} |\nabla \varphi|^2 + s\varphi(\Delta u + f) \, dx \leq 0 \\ &\implies \int_{\Omega} \varphi(\Delta u + f) \, dx \leq \int_{\Omega} \frac{|\nabla \varphi|^2}{2} s \, dx \quad \forall s > 0. \end{aligned}$$

Letting $s \rightarrow 0^+$ (and repeating with $-\varphi$) yields $-\Delta u = f$ in Ω .

Alternatively, $F'(0) = 0$ gives

$$\begin{aligned} 0 &= \lim_{s \rightarrow 0} \frac{F(s) - F(0)}{s} = \lim_{s \rightarrow 0} \frac{1}{s} \int_{\Omega} \left(\frac{s^2}{2} |\nabla \varphi|^2 + s \nabla u \cdot \nabla \varphi - s f \varphi \right) dx \\ &= \int_{\Omega} (\nabla u \cdot \nabla \varphi - f \varphi) dx = \int_{\Omega} \varphi (-\Delta u - f) dx. \end{aligned}$$

Since this holds for all $\varphi \in C_c^\infty(\Omega)$, we conclude $-\Delta u = f$ in Ω . $\square$

5 Dirichlet principle

5.1 Convexity and the direct method

Consider

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases} \quad (*)$$

Set

$$\Lambda = \{v \in C^2(\Omega) \cap C(\overline{\Omega}) \mid v = g \text{ on } \partial\Omega\}.$$

Then $u \in \Lambda$ solves $(*)$ if and only if

$$E[u] = \min_{v \in \Lambda} E[v],$$

where

$$E[v] = \int_{\Omega} \left(\frac{1}{2} |\nabla v|^2 - f v \right) dx.$$

Remark 5.1 (Convexity of E). The energy E is convex, i.e. for $\lambda \in [0, 1]$,

$$E[\lambda u + (1 - \lambda)v] \leq \lambda E[u] + (1 - \lambda)E[v].$$

One needs to check that $\lambda u + (1 - \lambda)v \in \Lambda$ whenever $u, v \in \Lambda$; this holds because

$$(\lambda u + (1 - \lambda)v) \Big|_{\partial\Omega} = \lambda g + (1 - \lambda)g = g.$$

To see convexity, expand

$$\begin{aligned} E[\lambda u + (1 - \lambda)v] &= \int_{\Omega} \left(\frac{1}{2} |\lambda \nabla u + (1 - \lambda) \nabla v|^2 - f(\lambda u + (1 - \lambda)v) \right) dx \\ &= \int_{\Omega} \left(\frac{1}{2} \lambda^2 |\nabla u|^2 + \frac{1}{2} (1 - \lambda)^2 |\nabla v|^2 + \lambda(1 - \lambda) \nabla u \cdot \nabla v - \lambda f u - (1 - \lambda) f v \right) dx. \end{aligned}$$

Hence

$$\begin{aligned} &E[\lambda u + (1 - \lambda)v] - \lambda E[u] - (1 - \lambda)E[v] \\ &= \int_{\Omega} \left(\frac{1}{2} (\lambda^2 - \lambda) |\nabla u|^2 + \frac{1}{2} ((1 - \lambda)^2 - (1 - \lambda)) |\nabla v|^2 + \lambda(1 - \lambda) \nabla u \cdot \nabla v \right) dx \\ &= -\lambda(1 - \lambda) \int_{\Omega} \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{2} |\nabla v|^2 - \nabla u \cdot \nabla v \right) dx \\ &= -\lambda(1 - \lambda) \int_{\Omega} \frac{1}{2} |\nabla u - \nabla v|^2 dx \leq 0. \end{aligned}$$

Outlook. Does E have a minimizer?

1. Later one shows that E has a minimizer in a larger space — a *Sobolev space*.
2. One then shows that the minimizer lies in Λ again (*regularity theory*).

Part II

Heat and wave equations

6 The heat equation

Consider the initial-value problem

$$\begin{cases} \partial_t u = \Delta u + f, & x \in \mathbb{R}^n, t > 0, \\ u(0, x) = g(x), & x \in \mathbb{R}^n \quad (t = 0). \end{cases}$$

6.0.1 Fundamental solution

Assume $f = 0$. By the Fourier transform,

$$\begin{cases} \hat{u}(\xi) = (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} u(x) dx, \\ u(x) = (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \hat{u}(\xi) d\xi. \end{cases}$$

Why $\widehat{\Delta u}(\xi) = -|\xi|^2 \hat{u}(\xi)$? First,

$$\begin{aligned} \widehat{\partial_j u}(\xi) &= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \partial_j u e^{-ix \cdot \xi} dx \\ &= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \left(\partial_{x_j} (u e^{-i\xi \cdot x}) - u (-i\xi_j) e^{-i\xi \cdot x} \right) dx \\ &= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} i\xi_j u e^{-i\xi \cdot x} dx \\ &= i\xi_j \hat{u}(\xi). \end{aligned}$$

Therefore

$$\widehat{\Delta u}(\xi) = \sum_{j=1}^n \widehat{\partial_{jj} u}(\xi) = \sum_{j=1}^n (i\xi_j)^2 \hat{u}(\xi) = \sum_{j=1}^n -(\xi_j)^2 \hat{u}(\xi) = -|\xi|^2 \hat{u}(\xi).$$

Consequently the heat equation becomes an ODE in t :

$$\begin{cases} \frac{d}{dt} \hat{u}(t, \xi) = -|\xi|^2 \hat{u}(t, \xi), \\ \hat{u}(0, \xi) = \hat{g}(\xi), \end{cases}$$

so

$$\frac{d}{dt} \log \hat{u}(t, \xi) = -|\xi|^2 \quad \implies \quad \hat{u}(t, \xi) = \hat{g}(\xi) e^{-t|\xi|^2}.$$

Equivalently,

$$\hat{u}(t, \xi) = e^{-t|\xi|^2} \hat{g}(\xi) \quad (\text{or } u(t, x) = e^{t\Delta} g(x)).$$

Recall: convolution theorem.

$$\widehat{\varphi * \psi}(\xi) = \hat{\varphi}(\xi) \hat{\psi}(\xi) (2\pi)^{\frac{n}{2}},$$

where

$$(\varphi * \psi)(x) = \int_{\mathbb{R}^n} \varphi(x - y) \psi(y) dy.$$

Indeed,

$$\begin{aligned}\widehat{\varphi * \psi}(\xi) &= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} \varphi(x-y) \psi(y) dy \right) e^{-ix \cdot \xi} dx \\ &= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \varphi(x-y) \psi(y) e^{-ix \cdot \xi} dy dx.\end{aligned}$$

By Fubini and the change of variables $z = x - y$,

$$\begin{aligned}\widehat{\varphi * \psi}(\xi) &= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \varphi(z) \psi(y) e^{-iz \cdot \xi} e^{-iy \cdot \xi} dy dz \\ &= (2\pi)^{-\frac{n}{2}} \left(\int_{\mathbb{R}^n} \varphi(z) e^{-iz \cdot \xi} dz \right) \left(\int_{\mathbb{R}^n} \psi(y) e^{-iy \cdot \xi} dy \right) \\ &= \hat{\varphi}(\xi) \hat{\psi}(\xi) (2\pi)^{\frac{n}{2}}.\end{aligned}$$

Now

$$\hat{u}(t, \xi) = e^{-t|\xi|^2} \hat{g}(\xi) = (2\pi)^{\frac{n}{2}} \hat{F}(\xi) \hat{g}(\xi),$$

so

$$\hat{F}(\xi) = (2\pi)^{-\frac{n}{2}} e^{-t|\xi|^2}.$$

Inverting the Fourier transform,

$$\begin{aligned}\Gamma(x) &= (2\pi)^{-\frac{n}{2}} (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} e^{-t|\xi|^2} e^{ix \cdot \xi} d\xi \\ &= (2\pi)^{-n} \int_{\mathbb{R}^n} e^{-t|\xi|^2 + ix \cdot \xi} d\xi.\end{aligned}$$

Completing the square (completing the square),

$$-t|\xi|^2 + ix \cdot \xi = -\left(\sqrt{t}\xi - \frac{ix}{2\sqrt{t}}\right)^2 - \frac{|x|^2}{4t},$$

hence

$$\Gamma(x) = (2\pi)^{-n} \int_{\mathbb{R}^n} e^{-\left(\sqrt{t}\xi - \frac{ix}{2\sqrt{t}}\right)^2} e^{-\frac{|x|^2}{4t}} d\xi.$$

Let

$$\eta = \sqrt{t}\xi - \frac{ix}{2\sqrt{t}} \quad \implies \quad d\xi = (t^{-1/2})^n d\eta.$$

Then

$$\Gamma(x) = (2\pi)^{-n} e^{-\frac{|x|^2}{4t}} t^{-\frac{n}{2}} \int_{\mathbb{R}^n} e^{-|\eta|^2} d\eta.$$

The Gaussian factors as a product of one-dimensional integrals:

$$\int_{\mathbb{R}^n} e^{-|\eta|^2} d\eta = \left(\int_{-\infty}^{\infty} e^{-\eta_1^2} d\eta_1 \right) \cdots \left(\int_{-\infty}^{\infty} e^{-\eta_n^2} d\eta_n \right) = (\sqrt{\pi})^n = \pi^{\frac{n}{2}}.$$

Therefore

$$\Gamma(x) = (4\pi t)^{-\frac{n}{2}} e^{-\frac{|x|^2}{4t}}.$$

In summary,

$$\hat{u}(t, \xi) = e^{-t|\xi|^2} \hat{g}(\xi) \quad \implies \quad u(t, x) = (F_t * g)(x),$$

where

$$F_t(x) = (4\pi t)^{-\frac{n}{2}} e^{-\frac{|x|^2}{4t}},$$

so

$$u(t, x) = \int_{\mathbb{R}^n} (4\pi t)^{-\frac{n}{2}} e^{-\frac{|x-y|^2}{4t}} g(y) dy.$$

The integrand is the *heat kernel* (Gaussian)

$$K(x, y, t) := (4\pi t)^{-\frac{n}{2}} e^{-\frac{|x-y|^2}{4t}},$$

which is singular at $t = 0$.

Theorem 6.1. *Let $g \in C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$ and set*

$$u(t, x) = \int_{\mathbb{R}^n} K(x, y, t) g(y) dy$$

(the representation exhibits a smoothing effect). Then:

- (i) $u \in C^\infty$;
- (ii) $(\partial_t - \Delta_x)u = 0$ for all $(t, x) \in (0, \infty) \times \mathbb{R}^n$;
- (iii)

$$\lim_{(t,x) \rightarrow (0,x^0)} u(t, x) = g(x^0) \quad \forall x^0 \in \mathbb{R}^n.$$

Remark 6.2 (Properties of K). The proof follows from basic properties of K :

- (a) $K \in C^\infty$;
- (b) $(\partial_t - \Delta_x)K(x, y, t) = 0$ for $t > 0$;
- (c) $K(x, y, t) > 0$ for $t > 0$;
- (d) $\int_{\mathbb{R}^n} K(x, y, t) dy = 1$ for all $x \in \mathbb{R}^n, t > 0$;
- (e) for every $\delta > 0$,

$$\lim_{t \rightarrow 0} \int_{|y-x| > \delta} K(x, y, t) dy = 0$$

uniformly in $x \in \mathbb{R}^n$.

Proof of Theorem 6.1. (i) & (ii). Since $K \in C^\infty$ with uniformly bounded derivatives of all orders on $[\delta, \infty) \times \mathbb{R}^n$ for each $\delta > 0$, one has $u \in C^\infty$. Moreover

$$\partial_t u - \Delta u = \int_{\mathbb{R}^n} (\partial_t - \Delta_x)K(x, y, t) g(y) dy = 0,$$

because $(\partial_t - \Delta_x)K = 0$.

(iii). Fix $x^0 \in \mathbb{R}^n$ and $\varepsilon > 0$. Choose $\delta > 0$ small enough that

$$|g(y) - g(x^0)| < \varepsilon \quad \text{for all } |y - x^0| < \delta.$$

Also, for $|x - x^0| < \delta/2$,

$$u(t, x) - g(x^0) = \int_{\mathbb{R}^n} K(x, y, t) (g(y) - g(x^0)) dy,$$

because $\int_{\mathbb{R}^n} K(x, y, t) dy = 1$. (One way to see the latter: if

$$\begin{cases} \partial_t u = \Delta u, \\ u(0, x) = 1 \end{cases} \quad (g \equiv 1),$$

then $u \equiv 1$, since $\partial_t u = 0$.)

Split

$$\begin{aligned} |u(t, x) - g(x^0)| &\leq \int_{|y-x^0|<\delta} K(x, y, t) |g(y) - g(x^0)| dy \\ &\quad + \int_{|y-x^0|\geq\delta} K(x, y, t) |g(y) - g(x^0)| dy =: I_1 + I_2. \end{aligned}$$

For the near-field integral,

$$I_1 := \int_{|y-x^0|<\delta} K(x, y, t) |g(y) - g(x^0)| dy \leq \varepsilon \int_{\mathbb{R}^n} K(x, y, t) dy = \varepsilon.$$

For the far-field integral: if $|x - x^0| < \delta/2$ and $|y - x^0| \geq \delta$, then

$$|y - x^0| \leq |y - x| + |x - x^0| < |y - x| + \frac{\delta}{2},$$

and hence

$$\delta \leq |y - x^0| \leq 2|y - x| \quad (\text{in particular } |y - x| \geq \frac{\delta}{2}).$$

Therefore

$$\begin{aligned} I_2 &\leq \int_{|y-x|\geq\delta/2} 2 \max |g| (4\pi t)^{-\frac{n}{2}} e^{-\frac{|x-y|^2}{4t}} dy \\ &\leq C t^{-\frac{n}{2}} \int_{|y-x|\geq\delta/2} e^{-\frac{|x-y|^2}{4t}} dy. \end{aligned}$$

Let $\frac{y-x}{\sqrt{4t}} = z$, so $dy = (\sqrt{4t})^n dz$. Then

$$I_2 \lesssim C \int_{|z|\geq\frac{\delta}{4\sqrt{t}}} e^{-|z|^2} dz \xrightarrow{t\rightarrow 0^+} 0$$

(exponentially fast).

Combining the estimates for I_1 and I_2 yields (iii). $\square$

7 Inhomogeneous heat equation: Duhamel, maximum principle, and uniqueness

7.1 Inhomogeneous heat equation

Consider

$$\begin{cases} \partial_t u - \Delta u = f(t, x) & t > 0, x \in \mathbb{R}^n, \\ u(0, \cdot) = g(x) & x \in \mathbb{R}^n. \end{cases} \quad (*)$$

The candidate solution is

$$u = \int_{\mathbb{R}^n} K(x, y, t) g(y) dy + \int_0^t \int_{\mathbb{R}^n} K(x, y, t-s) f(s, y) dy ds.$$

Aim: u solves $(*)$ and

$$\lim_{(t,x)\rightarrow(0,x^0)} u(t, x) = g(x^0).$$

By Duhamel's principle, set

$$\begin{aligned} u_1(t, x) &= \int_{\mathbb{R}^n} K(x, y, t) g(y) dy, \\ u_2(t, x) &= \int_0^t \int_{\mathbb{R}^n} K(x, y, t-s) f(s, y) dy ds. \end{aligned}$$

By previous arguments,

$$(\partial_t - \Delta_x)u_1 = 0 \quad \text{and} \quad \lim_{(t,x) \rightarrow (0,x^0)} u_1(t,x) = g(x^0).$$

It suffices to show the following.

Theorem 7.1. Assume $f \in C_c^\infty([0, \infty) \times \mathbb{R}^n)$ (with compact support). Define

$$u_2(t, x) = \int_0^t \int_{\mathbb{R}^n} K(x, y, t-s) f(s, y) dy ds.$$

Then:

- (i) $u_2 \in C^0([0, \infty) \times \mathbb{R}^n)$;
- (ii) $(\partial_t - \Delta_x)u_2 = f(t, x)$ for all $(t, x) \in (0, \infty) \times \mathbb{R}^n$;
- (iii) $\lim_{(t,x) \rightarrow (0,x^0)} u_2(t, x) = 0$ for all $x^0 \in \mathbb{R}^n$.

Proof. (i). Note that $K(x-y, t)$ is singular at $t = 0$; we move derivatives onto f to avoid this. That is,

$$\begin{aligned} u_2(t, x) &= \int_0^t \int_{\mathbb{R}^n} (4\pi(t-s))^{-\frac{n}{2}} e^{-\frac{|x-y|^2}{4(t-s)}} f(s, y) dy ds \\ &= \tilde{K} * f \quad \text{in } \mathbb{R}^+ \times \mathbb{R}^n \quad (\text{space-time convolution}). \end{aligned}$$

Change of variables $(s, y) \mapsto (t-s, x-y)$ yields

$$u_2(t, x) = \int_0^t \int_{\mathbb{R}^n} (4\pi s)^{-\frac{n}{2}} e^{-\frac{|y|^2}{4s}} f(t-s, x-y) dy ds.$$

Differentiating in t ,

$$\begin{aligned} \partial_t u_2(t, x) &= \int_0^t \int_{\mathbb{R}^n} (4\pi s)^{-\frac{n}{2}} e^{-\frac{|y|^2}{4s}} \partial_t f(t-s, x-y) dy ds \\ &\quad + \int_{\mathbb{R}^n} (4\pi t)^{-\frac{n}{2}} e^{-\frac{|y|^2}{4t}} f(0, x-y) dy. \end{aligned}$$

Claim.

$$\frac{d}{dt} \int_0^t F(t, s) ds = \int_0^t \partial_t F(t, s) ds + F(t, t).$$

Indeed, let $s = t\tau$, so $s \in [0, t]$ iff $\tau \in [0, 1]$ and $ds = t d\tau$. Then

$$\int_0^t F(t, s) ds = \int_0^1 F(t, t\tau) t d\tau,$$

and

$$\begin{aligned} \frac{d}{dt} \int_0^t F(t, s) ds &= \int_0^1 [t \partial_t F(t, t\tau) + t\tau \partial_s F(t, t\tau) + F(t, t\tau)] d\tau \\ &= \int_0^t \partial_t F(t, s) ds + \int_0^t \frac{s}{t} \partial_s F(t, s) ds + \int_0^t F(t, s) \frac{1}{t} ds. \end{aligned}$$

Integrating by parts,

$$\int_0^t \frac{s}{t} \partial_s F(t, s) ds = \left[F(t, s) \frac{s}{t} \right]_{s=0}^t - \int_0^t \frac{1}{t} F(t, s) ds = F(t, t) - \int_0^t \frac{1}{t} F(t, s) ds.$$

Hence

$$\frac{d}{dt} \int_0^t F(t, s) ds = \int_0^t \partial_t F(t, s) ds + F(t, t).$$

In our case

$$F(t, s) = \int_{\mathbb{R}^n} K(y, s) f(t - s, x - y) dy,$$

so

$$\partial_t u_2(t, x) = \int_0^t \int_{\mathbb{R}^n} K(y, s) \partial_t f(t - s, x - y) dy ds + \int_{\mathbb{R}^n} K(y, t) f(0, x - y) dy.$$

Similarly,

$$\partial_{x_i} u_2(t, x) = \int_0^t \int_{\mathbb{R}^n} K(y, s) \partial_{x_i} f(t - s, x - y) dy ds.$$

Repeating the process yields

$$u_2 \in C^\infty((0, \infty); C^\infty(\mathbb{R}^n)).$$

(ii). We have

$$\begin{aligned} (\partial_t - \Delta_x) u_2 &= \int_0^t \int_{\mathbb{R}^n} K(y, s) (\partial_t f - \Delta_x f)(t - s, x - y) dy ds \\ &\quad + \int_{\mathbb{R}^n} K(y, t) f(0, x - y) dy. \end{aligned}$$

Idea: split $(0, t) = (0, \varepsilon) \cup (\varepsilon, t)$, using smallness on $(0, \varepsilon)$ and differentiability on (ε, t) . Write

$$(\partial_t - \Delta_x) u_2 = I_1 + I_2 + I_3,$$

where

$$\begin{aligned} I_1 &= \int_0^\varepsilon \int_{\mathbb{R}^n} K(y, s) (\partial_t f - \Delta_x f)(t - s, x - y) dy ds, \\ I_2 &= \int_\varepsilon^t \int_{\mathbb{R}^n} K(y, s) (\partial_t f - \Delta_x f)(t - s, x - y) dy ds, \\ I_3 &= \int_{\mathbb{R}^n} K(y, t) f(0, x - y) dy. \end{aligned}$$

For I_1 ,

$$|I_1| \leq \max(|\partial_t f| + |\Delta_x f|) \int_0^\varepsilon \int_{\mathbb{R}^n} K(y, s) dy ds.$$

Since $\int_{\mathbb{R}^n} K(y, s) dy = 1$,

$$|I_1| < C \cdot \varepsilon \xrightarrow{\varepsilon \rightarrow 0} 0.$$

For I_2 ,

$$\begin{aligned}
I_2 &= \int_{\varepsilon}^t \int_{\mathbb{R}^n} K(y, s) (\partial_t f - \Delta_x f)(t - s, x - y) dy ds \\
&= \int_{\varepsilon}^t \int_{\mathbb{R}^n} K(y, s) (-\partial_s f - \Delta_y f)(t - s, x - y) dy ds \\
&= \int_{\varepsilon}^t \int_{\mathbb{R}^n} (-\Delta_y K(y, s)) f(t - s, x - y) dy ds \\
&\quad + \int_{\varepsilon}^t \int_{\mathbb{R}^n} \partial_s K(y, s) f(t - s, x - y) dy ds \\
&\quad - \left[\int_{\mathbb{R}^n} K(y, s) f(t - s, x - y) dy \right]_{s=\varepsilon}^t \\
&= \int_{\varepsilon}^t \int_{\mathbb{R}^n} (\partial_s - \Delta_y) K(y, s) f(t - s, x - y) dy ds \\
&\quad + \int_{\mathbb{R}^n} K(y, \varepsilon) f(t - \varepsilon, x - y) dy - \int_{\mathbb{R}^n} K(y, t) f(0, x - y) dy.
\end{aligned}$$

For $s > 0$ one has $(\partial_s - \Delta_y)K(y, s) = 0$, and the last term equals I_3 . Thus

$$I_2 + I_3 = \int_{\mathbb{R}^n} K(y, \varepsilon) f(t - \varepsilon, x - y) dy = \int_{\mathbb{R}^n} K(x, y, \varepsilon) f(t - \varepsilon, y) dy.$$

Therefore

$$\begin{aligned}
\lim_{\varepsilon \rightarrow 0} (I_1 + I_2 + I_3) &= \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n} K(x, y, \varepsilon) f(t - \varepsilon, y) dy \\
&= \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n} K(x, y, \varepsilon) f(t, y) dy = f(t, x),
\end{aligned}$$

by Theorem 6.1 (recall: $\lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n} K(x, y, \varepsilon) f(y) dy = f(x)$).

(iii).

$$|u_2(t, x)| \leq \|f\|_{L_t^\infty L_x^\infty} \int_0^t \int_{\mathbb{R}^n} K(y, s) dy ds \leq \|f\|_{L_t^\infty L_x^\infty} \int_0^t ds = \|f\|_{L_t^\infty L_x^\infty} \cdot t \xrightarrow{t \rightarrow 0} 0.$$

□

Thus

$$u(t, x) = \int_{\mathbb{R}^n} K(x, y, t) g(y) dy + \int_0^t \int_{\mathbb{R}^n} K(x, y, t - s) f(s, y) dy ds$$

solves the inhomogeneous problem (*Duhamel's principle*).

Remark 7.2 (Differentiating under a moving domain). Let $\Omega(t) \subseteq \mathbb{R}^n$ and let $v(t, x)$ be a velocity field. Then

$$\frac{d}{dt} \int_{\Omega(t)} f dx = \int_{\partial\Omega(t)} f v \cdot n d\sigma_x + \int_{\Omega(t)} \partial_t f dx.$$

By Green's formula this equals

$$\int_{\Omega(t)} (\operatorname{div}(fv) + \partial_t f) dx.$$

In fluid mechanics, $\partial_t f + \operatorname{div}(fv)$ is the material derivative Df (material derivative).

Recall the *transport equation*

$$\partial_t f + v \cdot \nabla_x f = 0,$$

which is equivalent to

$$\partial_t f + \operatorname{div}(fv) = 0$$

if $\operatorname{div} v = 0$ (incompressibility), since $\operatorname{div}(fv) = v \cdot \nabla f + f \operatorname{div} v$.

7.2 Maximum principle

Let

$$\begin{aligned}\Omega_T &= \{(t, x) \mid x \in \Omega, t \in (0, T)\}, \\ \partial' \Omega_T &= \{(t, x) \mid x \in \partial\Omega, t \in [0, T] \text{ or } x \in \Omega, t = 0\} \quad (\text{parabolic boundary}), \\ \partial'' \Omega_T &= \{(t, x) \mid x \in \Omega, t = T\}.\end{aligned}$$

Theorem 7.3. *If $\partial_t u - \Delta u = f(t, x) \leq 0$, then*

$$\max_{\bar{\Omega}_T} u = \max_{\partial' \Omega_T} u.$$

Proof. Step 1. Assume $f < 0$. Let the maximum of u on $\bar{\Omega}_T$ be attained at (t_0, x_0) .

Case 1. If $t_0 \in (0, T)$ and $x_0 \in \Omega$, then

$$\begin{cases} \partial_t u(t_0, x_0) = 0, \\ \Delta u(t_0, x_0) \leq 0, \end{cases}$$

so

$$(\partial_t u - \Delta u)(t_0, x_0) \geq 0,$$

contradicting $f < 0$.

Case 2. If $t_0 = T$ and $x_0 \in \Omega$, then

$$\partial_t u(T, x_0) \geq 0, \quad \Delta u(T, x_0) \leq 0,$$

so again

$$(\partial_t u - \Delta u)(T, x_0) \geq 0,$$

contradicting $f < 0$.

Hence $t_0 = 0$ with $x_0 \in \Omega$, or $x_0 \in \partial\Omega$, i.e. $(t_0, x_0) \in \partial' \Omega_T$.

Step 2. Now assume $f \leq 0$. Set

$$u_\varepsilon(t, x) = u(t, x) + \varepsilon|x|^2.$$

Then

$$\partial_t u_\varepsilon - \Delta u_\varepsilon = \partial_t u - \Delta u - 2n\varepsilon = f - 2n\varepsilon < 0.$$

By Step 1,

$$\max_{\bar{\Omega}_T} u_\varepsilon = \max_{\partial' \Omega_T} u_\varepsilon.$$

Let $M := \max_{\partial' \Omega_T} u$ and assume $\Omega \subset B_R(0)$. Then

$$\max_{\bar{\Omega}_T} u_\varepsilon = \max_{\partial' \Omega_T} u_\varepsilon \leq M + \varepsilon R^2,$$

so

$$\max_{\bar{\Omega}_T} u \leq \max_{\bar{\Omega}_T} u_\varepsilon \leq M + \varepsilon R^2 \quad \forall \varepsilon > 0.$$

Letting $\varepsilon \rightarrow 0$ yields

$$\max_{\bar{\Omega}_T} u \leq M,$$

i.e.

$$\max_{\bar{\Omega}_T} u = \max_{\partial' \Omega_T} u.$$

□

7.3 Uniqueness

There is at most one solution to

$$\begin{cases} \partial_t u - \Delta u = f(t, x) & x \in \Omega, t \in (0, \infty), \\ u(0, x) = g(x), \\ u|_{\partial\Omega} = \varphi(t, x). \end{cases}$$

Proof via the maximum principle. Assume there exist two solutions u_1, u_2 , and set $w = u_1 - u_2$. Then w solves

$$\begin{cases} \partial_t w - \Delta w = 0, \\ w(0, x) = 0, \\ w|_{\partial\Omega} = 0. \end{cases}$$

By the maximum principle,

$$\max_{\overline{\Omega}_T} w = 0.$$

Considering $-w$ likewise yields $\max_{\overline{\Omega}_T} (-w) = 0$, hence $w \equiv 0$ in Ω_T . $\square$

Uniqueness from energy. Again let $w = u_1 - u_2$, so $w|_{\partial\Omega} = 0$. From $\partial_t w - \Delta w = 0$, integrate $(\partial_t w - \Delta w)w$ over Ω :

$$\begin{aligned} 0 &= \int_{\Omega} (\partial_t w - \Delta w) w \, dx = \int_{\Omega} \partial_t w \cdot w - (\Delta w) w \, dx \\ &= \int_{\Omega} \frac{1}{2} \partial_t |w|^2 + |\nabla w|^2 \, dx - \int_{\partial\Omega} w \frac{\partial w}{\partial n} \, d\sigma_x \\ &= \frac{1}{2} \frac{d}{dt} \int_{\Omega} |w|^2 \, dx + \int_{\Omega} |\nabla w|^2 \, dx, \end{aligned}$$

where the boundary integral vanishes since $w = 0$ on $\partial\Omega$. Hence

$$\frac{d}{dt} \int_{\Omega} |w|^2(t, x) \, dx = -2 \int_{\Omega} |\nabla w|^2 \, dx \leq 0.$$

Since $w(0, x) = 0$, one has $\int_{\Omega} |w|^2(0, x) \, dx = 0$, and therefore

$$0 \leq \int_{\Omega} |w|^2(t, x) \, dx \leq \int_{\Omega} |w|^2(0, x) \, dx = 0.$$

Thus $w \equiv 0$ for all $t > 0$.

Energy and backwards uniqueness. Define

$$E(w(t)) = \int_{\Omega} |w|^2(t, x) \, dx.$$

Then

$$\frac{d}{dt} E(w(t)) = -2 \int_{\Omega} |\nabla w|^2(t, x) \, dx \leq 0.$$

Theorem 7.4 (Backwards uniqueness). *Suppose $u_1, u_2 \in C^2(\overline{\Omega}_T)$ solve the heat equation in Ω_T , agree on the lateral boundary $\partial\Omega \times [0, T]$, and $u_1(\cdot, T) = u_2(\cdot, T)$. Then $u_1 \equiv u_2$ in Ω_T .*

Proof. Set $w = u_1 - u_2$. Then

$$\frac{d}{dt} E(w(t)) = -2 \int_{\Omega} |\nabla w|^2 \, dx,$$

and

$$\begin{aligned}\frac{d^2}{dt^2}E(w(t)) &= -2\frac{d}{dt}\int_{\Omega}|\nabla w|^2 dx = -4\int_{\Omega}\nabla w \cdot \nabla w_t dx \\ &= -4\int_{\Omega}\nabla w \cdot \nabla(\Delta w) dx = 4\int_{\Omega}(\Delta w)^2 dx,\end{aligned}$$

where we used $w_t = \Delta w$.

On the other hand,

$$-2\int_{\Omega}|\nabla w|^2 dx = 2\int_{\Omega}w\Delta w dx \leq 2\left(\int_{\Omega}w^2 dx\right)^{1/2}\left(\int_{\Omega}(\Delta w)^2 dx\right)^{1/2}$$

by Hölder's inequality. This shows

$$(E'(t))^2 \leq E(t)E''(t).$$

Let $f(t) = \log E(t)$. Then

$$f'(t) = \frac{E'(t)}{E(t)}, \quad f''(t) = \frac{E''(t)E(t) - (E'(t))^2}{(E(t))^2} \geq 0,$$

so f is convex. For $t \in [t_1, t_2]$ with $t = (1 - \lambda)t_1 + \lambda t_2$,

$$f((1 - \lambda)t_1 + \lambda t_2) \leq (1 - \lambda)f(t_1) + \lambda f(t_2),$$

hence

$$E(t) = E((1 - \lambda)t_1 + \lambda t_2) \leq E(t_1)^{1-\lambda} E(t_2)^\lambda.$$

If $t_2 = T$ and $E(T) = 0$, then $E(t) \equiv 0$ for all $0 \leq t \leq T$. □

8 The wave equation

8.1 One-dimensional homogeneous case

$$\begin{cases} \partial_{tt}u - \partial_{xx}u = 0, \\ (u, \partial_t u)(0, x) = (u_0, u_1). \end{cases} \quad (14)$$

Recall by d'Alembert's method:

$$u(t, x) = \frac{1}{2}(u_0(x+t) + u_0(x-t)) + \frac{1}{2}\int_{x-t}^{x+t} u_1(y) dy. \quad (+)$$

Theorem 8.1. Assume $(u_0, u_1) \in C^2(\mathbb{R}) \times C^1(\mathbb{R})$ and u is defined by (+). Then

- (i) $u \in C^2(\mathbb{R}^+ \times \mathbb{R})$;
- (ii) $\partial_{tt}u - \partial_{xx}u = 0$ in $\mathbb{R}^+ \times \mathbb{R}$;
- (iii) $\lim_{(t,x) \rightarrow (0,x_0)} u(t, x) = u_0(x_0)$;
- (iv) $\lim_{(t,x) \rightarrow (0,x_0)} \partial_t u(t, x) = u_1(x_0)$.

Proof. (i) Differentiating (+),

$$\begin{aligned}\partial_t u &= \frac{1}{2}(u'_0(x+t) - u'_0(x-t)) + \frac{1}{2}(u_1(x+t) + u_1(x-t)), \\ \partial_{tt}u &= \frac{1}{2}(u''_0(x+t) + u''_0(x-t)) + \frac{1}{2}(u'_1(x+t) - u'_1(x-t)).\end{aligned}$$

Similarly,

$$\begin{aligned}\partial_x u &= \frac{1}{2}(u'_0(x+t) + u'_0(x-t)) + \frac{1}{2}(u_1(x+t) - u_1(x-t)), \\ \partial_{xx} u &= \frac{1}{2}(u''_0(x+t) + u''_0(x-t)) + \frac{1}{2}(u'_1(x+t) - u'_1(x-t)).\end{aligned}$$

Since $u_0 \in C^2(\mathbb{R})$ and $u_1 \in C^1(\mathbb{R})$, one has $\partial_{tt}u, \partial_{xx}u \in C(\mathbb{R}^+ \times \mathbb{R})$.

(ii) Comparing the expressions above yields $\partial_{tt}u - \partial_{xx}u = 0$.

(iii)

$$\lim_{(t,x) \rightarrow (0,x_0)} u(t,x) = \frac{1}{2}(u_0(x_0) + u_0(x_0)) + \frac{1}{2} \int_{x_0}^{x_0} u_1(y) dy = u_0(x_0).$$

(iv)

$$\lim_{(t,x) \rightarrow (0,x_0)} \partial_t u = \lim_{(t,x) \rightarrow (0,x_0)} \left(\frac{1}{2}u'_0(x+t) - \frac{1}{2}u'_0(x-t) + \frac{1}{2}(u_1(x+t) + u_1(x-t)) \right) = u_1(x_0).$$

□

8.2 One-dimensional inhomogeneous case

$$\begin{cases} \partial_{tt}u - \partial_{xx}u = f, \\ (u, \partial_t u)|_{t=0} = (u_0, u_1). \end{cases} \quad (15)$$

Remark 8.2 (Duhamel's principle for ODEs). For $\frac{d^2}{dt^2}u + u = f$,

$$u(t) = S'(t)u_0 + S(t)u_1 + \int_0^t S(t-s)f(s) ds,$$

where $S(t) = \sin t$.

By Duhamel's principle we claim

$$\begin{aligned} u(t,x) &= \underbrace{\frac{1}{2}(u_0(x+t) + u_0(x-t)) + \frac{1}{2} \int_{x-t}^{x+t} u_1(y) dy}_{u_{\text{hom}}} \\ &\quad + \underbrace{\frac{1}{2} \int_0^t \int_{x-(t-s)}^{x+(t-s)} f(s,y) dy ds}_{u_{\text{inh}}}. \end{aligned} \quad (*)$$

Theorem 8.3. Assume $(u_0, u_1, f) \in C^2(\mathbb{R}) \times C^1(\mathbb{R}) \times C^1(\mathbb{R}^+ \times \mathbb{R})$. Then u from $(*)$ satisfies

(i) $u \in C^2(\mathbb{R}^+ \times \mathbb{R})$;

(ii) $\partial_{tt}u - \partial_{xx}u = f$ in $\mathbb{R}^+ \times \mathbb{R}$;

(iii) $\lim_{(t,x) \rightarrow (0,x_0)} u(t,x) = u_0(x_0)$ and $\lim_{(t,x) \rightarrow (0,x_0)} \partial_t u(t,x) = u_1(x_0)$.

Proof. It suffices to consider

$$V(t,x) = \frac{1}{2} \int_0^t \int_{x-(t-s)}^{x+(t-s)} f(s,y) dy ds$$

(the inhomogeneous part u_{inh}). We will show

(i) $V \in C^2(\mathbb{R}^+ \times \mathbb{R})$;

$$(ii) \quad \partial_{tt}V - \partial_{xx}V = f \text{ in } \mathbb{R}^+ \times \mathbb{R};$$

$$(iii) \quad \lim_{(t,x) \rightarrow (0,x_0)} V(t,x) = \lim_{(t,x) \rightarrow (0,x_0)} \partial_t V(t,x) = 0.$$

Remark 8.4. $u = u_{\text{hom}} + \underbrace{u_{\text{inh}}}_{:=V}$; the homogeneous part u_{hom} is treated above.

Set

$$U(t, s, x) = \int_{x-(t-s)}^{x+(t-s)} f(s, y) dy,$$

so that $V(t, x) = \frac{1}{2} \int_0^t U(t, s, x) ds$. Then

$$\partial_t V = \frac{1}{2} \frac{d}{dt} \int_0^t U(t, s, x) ds = \frac{1}{2} \int_0^t \partial_t U(t, s, x) ds + \frac{1}{2} U(t, t, x).$$

Since $U(t, t, x) = 0$,

$$\partial_t V = \frac{1}{2} \int_0^t \partial_t U(t, s, x) ds.$$

Differentiating again,

$$\partial_{tt}V = \frac{1}{2} \int_0^t \partial_{tt}U(t, s, x) ds + \frac{1}{2} \partial_t U(t, t, x).$$

Note

$$\partial_t U(t, s, x) = \frac{d}{dt} \int_{x-(t-s)}^{x+(t-s)} f(s, y) dy = f(s, x + (t - s)) + f(s, x - (t - s)).$$

Hence

$$\partial_t U(t, t, x) = 2f(t, x),$$

and

$$\partial_{tt}U(t, s, x) = \partial_x f(s, x + (t - s)) - \partial_x f(s, x - (t - s)).$$

Therefore

$$\partial_{tt}V = f(t, x) + \frac{1}{2} \int_0^t \left(\partial_x f(s, x + (t - s)) - \partial_x f(s, x - (t - s)) \right) ds.$$

Also

$$\begin{aligned} \partial_x V &= \frac{1}{2} \int_0^t \left(f(s, x + (t - s)) - f(s, x - (t - s)) \right) ds, \\ \partial_{xx}V &= \frac{1}{2} \int_0^t \left(\partial_x f(s, x + (t - s)) - \partial_x f(s, x - (t - s)) \right) ds. \end{aligned}$$

Consequently

$$\partial_{tt}V - \partial_{xx}V = f(t, x).$$

Moreover $V \in C^2(\mathbb{R}^+ \times \mathbb{R})$ as long as $f \in C^1(\mathbb{R}^+ \times \mathbb{R})$.

For the initial conditions,

$$\begin{aligned} \lim_{(t,x) \rightarrow (0,x_0)} V(t,x) &= \lim_{t \rightarrow 0} \frac{1}{2} \int_0^t \int_{x_0-(t-s)}^{x_0+(t-s)} f(s, y) dy ds = 0, \\ \lim_{(t,x) \rightarrow (0,x_0)} \partial_t V(t,x) &= \lim_{t \rightarrow 0} \frac{1}{2} \int_0^t \left(f(s, x_0 + (t - s)) + f(s, x_0 - (t - s)) \right) ds = 0. \end{aligned}$$

□

8.3 Three-dimensional wave equation

$$\begin{cases} \partial_{tt}u - \Delta u = 0 & (t, x) \in (0, \infty) \times \mathbb{R}^3, \\ (u, \partial_t u)|_{t=0} = (u_0, u_1) & x \in \mathbb{R}^3. \end{cases} \quad (16)$$

8.3.1 Spherical mean method

Define the spherical mean

$$\bar{U}(t, x, r) := \frac{1}{4\pi r^2} \int_{\partial B_r(x)} u(t, y) d\sigma_y,$$

i.e. the average of u on the sphere $\partial B_r(x)$. In particular,

$$u(x) = \lim_{r \rightarrow 0^+} \bar{U}.$$

Similarly,

$$\begin{cases} \bar{U}_0(x, r) = \frac{1}{4\pi r^2} \int_{\partial B_r(x)} u_0(y) d\sigma_y, \\ \bar{U}_1(x, r) = \frac{1}{4\pi r^2} \int_{\partial B_r(x)} u_1(y) d\sigma_y. \end{cases} \quad (17)$$

We will show that $\bar{U}(t, x, r)$ solves the radial wave equation (Euler–Poisson–Darboux)

$$\begin{cases} \partial_{tt}\bar{U} - \partial_{rr}\bar{U} - \frac{2}{r}\partial_r\bar{U} = 0, & t, r \in (0, \infty), \\ (\bar{U}, \partial_t\bar{U})|_{t=0} = (\bar{U}_0, \bar{U}_1). \end{cases} \quad (\star)$$

Proof. Write

$$\bar{U}(t, x, r) = \frac{1}{4\pi r^2} \int_{\partial B_r(x)} u(t, y) d\sigma_y = \frac{1}{4\pi} \int_{|z|=1} u(t, x + rz) d\sigma_z.$$

Differentiating in r ,

$$\begin{aligned} \partial_r \bar{U}(t, x, r) &= \frac{1}{4\pi} \int_{|z|=1} z \cdot \nabla u(t, x + rz) d\sigma_z = \frac{1}{4\pi} \int_{|z|=1} \frac{\partial u}{\partial n}(t, x + rz) d\sigma_z \\ &= \frac{1}{4\pi r^2} \int_{\partial B_r(x)} \frac{\partial u}{\partial n}(t, y) d\sigma_y = \frac{1}{4\pi r^2} \int_{B_r(x)} \Delta u(t, y) dy \quad (\text{by Green's formula}) \\ &= \frac{1}{4\pi r^2} \int_{B_r(x)} \partial_{tt}u(t, y) dy \quad (\text{since } \partial_{tt}u - \Delta u = 0). \end{aligned}$$

Hence

$$\begin{aligned} 2r \partial_r \bar{U} + r^2 \partial_{rr} \bar{U} &= \partial_r (r^2 \partial_r \bar{U}) = \partial_r \left(\frac{1}{4\pi} \int_{B_r(x)} \partial_{tt}u(t, y) dy \right) \\ &= \frac{1}{4\pi} \int_{\partial B_r(x)} \partial_{tt}u(t, y) d\sigma_y \quad (\text{moving-domain differentiation}) \\ &= r^2 \partial_{tt} \bar{U}. \end{aligned}$$

Dividing by r^2 yields $(\star)$. □

Then define $\tilde{U} = r\bar{U}$ (radiation field). We find

$$\begin{aligned} \partial_{tt}\tilde{U} - \partial_{rr}\tilde{U} &= r\partial_{tt}\bar{U} - \partial_r(\bar{U} + r\partial_r\bar{U}) \\ &= r\partial_{tt}\bar{U} - r\partial_{rr}\bar{U} - 2\partial_r\bar{U} = 0. \end{aligned}$$

Therefore

$$\begin{cases} \partial_{tt}\tilde{U} - \partial_{rr}\tilde{U} = 0 & (1D \text{ wave}), \\ (\tilde{U}, \partial_t\tilde{U})|_{t=0} = (\tilde{U}_0, \tilde{U}_1), \\ \tilde{U}|_{r=0} = 0. \end{cases} \quad (18)$$

By the 1D d'Alembert formula,

$$\tilde{U} = \frac{1}{2}(\tilde{U}_0(r+t) + \tilde{U}_0(r-t)) + \frac{1}{2} \int_{r-t}^{r+t} \tilde{U}_1(\rho) d\rho.$$

Then

$$\begin{aligned} u(t, x) &= \lim_{r \rightarrow 0^+} \bar{U} = \lim_{r \rightarrow 0^+} (r^{-1} \tilde{U}) \\ &= \frac{1}{2} \lim_{r \rightarrow 0^+} \left(r^{-1} (\tilde{U}_0(r+t) + \tilde{U}_0(r-t)) + r^{-1} \int_{t-r}^{t+r} \tilde{U}_1(\rho) d\rho \right). \end{aligned}$$

Since $\tilde{U}_0(0) = 0$, take the odd extension $\tilde{U}_0(t-r) = -\tilde{U}_0(r-t)$. The limit becomes

$$u(t, x) = \tilde{U}'_0(t) + \tilde{U}_1(t).$$

Now

$$\tilde{U}_0(t) = \frac{1}{4\pi t} \int_{\partial B_t(x)} u_0(y) d\sigma_y = \frac{t}{4\pi} \int_{|z|=1} u_0(x+tz) d\sigma_z,$$

so

$$\begin{aligned} \tilde{U}'_0(t) &= \frac{1}{4\pi} \int_{|z|=1} u_0(x+tz) d\sigma_z + \frac{t}{4\pi} \int_{|z|=1} z \cdot \nabla u_0(x+tz) d\sigma_z \\ &= \frac{1}{4\pi t^2} \int_{\partial B_t(x)} [u_0(y) + (y-x) \cdot \nabla u_0(y)] d\sigma_y. \end{aligned}$$

Therefore Kirchhoff's formula reads

$$u(t, x) = \frac{1}{4\pi t^2} \int_{\partial B_t(x)} [u_0(y) + (y-x) \cdot \nabla u_0(y) + t u_1(y)] d\sigma_y. \quad (19)$$

Remark 8.5.

$$u(t_0, x_0) = \frac{1}{4\pi t_0^2} \int_{\partial B_{t_0}(x_0)} (u_0 + (y-x_0) \cdot \nabla u_0 + t_0 u_1) d\sigma_y.$$

One needs information on $|y-x_0| = t_0$ (not the solid ball $|y-x_0| \leq t_0$): the light cone. The value at (t_0, x_0) depends on data on the sphere of radius t_0 (domain of dependence); conversely, data at a point x_1 influence the forward light cone (region of influence).

9 Method of descent and energy methods for the wave equation

9.1 2D homogeneous wave equation: method of descent

For $x \in \mathbb{R}^2$, set $\bar{x} = (x, x_3) \in \mathbb{R}^3$ and $\bar{u}(t, \bar{x}) = u(t, x)$, so that $\partial_{x_3}\bar{u} = 0$. If

$$\partial_{tt}u - \Delta_{\mathbb{R}^2}u = 0 \quad \text{for } x \in \mathbb{R}^2,$$

then

$$\partial_{tt}\bar{u} - \Delta_{\mathbb{R}^3}\bar{u} = 0 \quad \text{for } \bar{x} \in \mathbb{R}^3.$$

Thus $u(t, x) = \bar{u}(t, x, 0)$ is given by the fundamental solution in $\mathbb{R}^3$.

Now take $x_3 = 0$. For the ball $B_t(\bar{x}) \subset \mathbb{R}^3$, points on the sphere $|w - \bar{x}| = t$ may be written

$$w = (y, \pm \sqrt{t^2 - |y - x|^2}).$$

Hence

$$u(t, x) = \frac{1}{4\pi t^2} \int_{|w-\bar{x}|=t} (\bar{u}_0(w) + (w - \bar{x}) \cdot D_{\bar{x}} \bar{u}_0(w) + t \bar{u}_1(w)) d\sigma_w.$$

Let $\gamma(y) = \sqrt{t^2 - |y - x|^2}$. By symmetry in w_3 (using $\partial_{x_3} \bar{u}_0 = 0$),

$$u(t, x) = \frac{1}{2\pi t^2} \int_{\substack{|w-\bar{x}|=t \\ w_3 \geq 0}} (\bar{u}_0(w) + (w - \bar{x}) \cdot D_{\bar{x}} \bar{u}_0(w) + t \bar{u}_1(w)) d\sigma_w.$$

Recall: surface integral. With $w(y) = (y, \gamma(y))$,

$$\iint_S f(w) d\sigma_w = \iint_A f(w(y)) \sqrt{1 + |\nabla \gamma(y)|^2} dy.$$

Then

$$\nabla \gamma(y) = \frac{-(y - x)}{\sqrt{t^2 - |y - x|^2}},$$

so

$$1 + |\nabla \gamma(y)|^2 = 1 + \frac{|y - x|^2}{t^2 - |y - x|^2} = \frac{t^2}{t^2 - |y - x|^2}.$$

Moreover

$$S = \{|w - \bar{x}| = t, w_3 \geq 0\} \iff A = \{|y - x| \leq t\}.$$

Thus

$$\int_S (\bar{u}_0(w) + t \bar{u}_1(w)) d\sigma_w = \int_{|y-x| \leq t} (u_0(y) + t u_1(y)) \frac{t}{\sqrt{t^2 - |x - y|^2}} dy,$$

and, noting $\partial_{x_3} \bar{u}_0 = 0$,

$$\int_S (w - \bar{x}) \cdot \nabla_{\bar{x}} \bar{u}_0(w) d\sigma_w = \int_{|y-x| \leq t} (y - x) \cdot \nabla u_0(y) \frac{t}{\sqrt{t^2 - |x - y|^2}} dy.$$

Therefore

$$u(t, x) = \frac{1}{2\pi t^2} \int_{|y-x| \leq t} (u_0 + (y - x) \cdot Du_0 + t u_1) \frac{t}{\sqrt{t^2 - |x - y|^2}} dy.$$

Remark 9.1 (Duhamel). Duhamel's principle yields the inhomogeneous case

$$\begin{cases} \partial_{tt} u - \Delta u = f, \\ (u, \partial_t u)|_{t=0} = (u_0, u_1). \end{cases}$$

In 2D one may write

$$u(t, x) = \frac{1}{2\pi t} \int_{|y-x| \leq t} \frac{u_0 + (y - x) \cdot Du_0 + t u_1}{\sqrt{t^2 - |x - y|^2}} dy$$

and, equivalently,

$$\begin{aligned} u(t, x) &= \frac{1}{2\pi} \partial_t \int_{|y-x| \leq t} \frac{u_0(y)}{\sqrt{t^2 - |x - y|^2}} dy + \frac{1}{2\pi} \int_{|y-x| \leq t} \frac{u_1(y)}{\sqrt{t^2 - |x - y|^2}} dy \\ &\quad + \frac{1}{2\pi} \int_0^t \int_{|y-x| \leq t-s} \frac{f(s, y)}{\sqrt{(t-s)^2 - |x - y|^2}} dy ds. \end{aligned}$$

9.2 Use of the Fourier transform

Consider

$$\begin{cases} \partial_{tt}u - \Delta u = 0, \\ (u, \partial_t u)|_{t=0} = (u_0, u_1). \end{cases} \quad (20)$$

In Fourier space,

$$\begin{cases} \partial_{tt}\hat{u}(\xi) + |\xi|^2 \hat{u}(\xi) = 0, \\ (\hat{u}(\xi), \partial_t \hat{u}(\xi))|_{t=0} = (\hat{u}_0(\xi), \hat{u}_1(\xi)). \end{cases} \quad (21)$$

Hence

$$\hat{u}(\xi) = A \sin(|\xi|t) + B \cos(|\xi|t).$$

The initial conditions give $\hat{u}_0(\xi) = B$ and $\hat{u}_1(\xi) = A|\xi|$, so $A = \hat{u}_1(\xi)/|\xi|$. Thus

$$\hat{u}(\xi) = \frac{\hat{u}_1(\xi)}{|\xi|} \sin(|\xi|t) + \hat{u}_0(\xi) \cos(|\xi|t).$$

Equivalently, in operator form,

$$u(t, x) = \frac{\sin(|\nabla|t)}{|\nabla|} u_1 + \cos(|\nabla|t) u_0.$$

9.3 Energy; uniqueness

Theorem 9.2 (Uniqueness). *Let v_1, v_2 solve*

$$\begin{cases} \square u := \partial_{tt}u - \Delta u = f & \text{in } \mathbb{R}^n, \\ u \text{ compactly supported,} \\ (u, \partial_t u)|_{t=0} = (u_0, u_1). \end{cases} \quad (22)$$

Then $v_1 \equiv v_2$.

Proof. The difference $w = v_1 - v_2$ satisfies

$$\begin{cases} \square w = \partial_{tt}w - \Delta w = 0 & \text{in } \mathbb{R}^n, \\ (w, \partial_t w)|_{t=0} = (0, 0). \end{cases} \quad (23)$$

Define

$$E(t) := \frac{1}{2} \int_{\mathbb{R}^n} (|\partial_t w|^2 + |\nabla w|^2) dx.$$

Then

$$\begin{aligned} E'(t) &= \frac{1}{2} \int_{\mathbb{R}^n} (2 \partial_t w \cdot \partial_{tt}w + 2 \nabla w \cdot \nabla \partial_t w) dx \\ &= \frac{1}{2} \int_{\mathbb{R}^n} (2 \partial_t w (\Delta w) - 2 \Delta w \partial_t w) dx = 0, \end{aligned}$$

using $\partial_{tt}w = \Delta w$ and integration by parts (compact support). Hence $E(t) = E(0) = 0$, so $w \equiv 0$. $\square$

Theorem 9.3 (Backward uniqueness in the domain of dependence). *If $u = \partial_t u = 0$ on $B_{t_0}(x_0) \times \{t = 0\}$, then $u \equiv 0$ inside the backward cone with tip (t_0, x_0) (domain of dependence).*

Proof. For $t \in [0, t_0]$ define

$$E(t) := \frac{1}{2} \int_{B_{t_0-t}(x_0)} (|\partial_t u|^2 + |\nabla u|^2)_{(t,x)} dx.$$

Then $E(0) = 0$. Differentiating the integral over the shrinking ball,

$$\begin{aligned}
 E'(t) &= \int_{B_{t_0-t}(x_0)} (\partial_{tt}u \cdot \partial_t u + Du \cdot \partial_t Du) dx \\
 &\quad - \frac{1}{2} \int_{\partial B_{t_0-t}(x_0)} (|\partial_t u|^2 + |Du|^2) d\sigma_x \\
 &= \int_{B_{t_0-t}(x_0)} \partial_t u (\partial_{tt}u - \Delta u) dx + \int_{\partial B_{t_0-t}(x_0)} \frac{\partial u}{\partial n} \partial_t u d\sigma_x \\
 &\quad - \frac{1}{2} \int_{\partial B_{t_0-t}(x_0)} (|\partial_t u|^2 + |Du|^2) d\sigma_x \\
 &= -\frac{1}{2} \int_{\partial B_{t_0-t}(x_0)} \left(|\partial_t u|^2 + |Du|^2 - 2 \frac{\partial u}{\partial n} \partial_t u \right) d\sigma_x.
 \end{aligned}$$

Since $|Du|^2 \geq |\nabla u \cdot n|^2 = |\partial u / \partial n|^2$,

$$E'(t) \leq -\frac{1}{2} \int_{\partial B_{t_0-t}(x_0)} \left| \partial_t u - \frac{\partial u}{\partial n} \right|^2 d\sigma_x \leq 0.$$

Hence $E(t) \leq E(0) = 0$, so $E(t) = 0$ and $u \equiv 0$ in the cone. □

Part III

Sobolev spaces

10 Weak derivatives

10.0.1 Distributions and weak derivatives

Definition 10.1 (L^1_{loc}).

$$L^1_{\text{loc}}(\mathbb{R}^n) := \{\text{locally integrable functions } f : \mathbb{R}^n \rightarrow \mathbb{R}\},$$

i.e. $\int_K |f| < +\infty$ for every compact $K \subset \mathbb{R}^n$. The support of a function is

$$\text{supp}(\phi) = \overline{\{x \mid \phi(x) \neq 0\}}.$$

If $f \in C^1$ and $\phi \in C_c^\infty$, then

$$\int_{\mathbb{R}^n} D_i f \phi \, dx = - \int_{\mathbb{R}^n} f \partial_i \phi \, dx$$

(integration by parts); the right-hand side is well-defined for $\phi \in C_c^\infty$.

Definition 10.2 (Weak derivative). A weak derivative of f is a locally integrable function g such that

$$\int_{\mathbb{R}^n} g(x) \phi \, dx = - \int_{\mathbb{R}^n} f(x) \partial_i \phi \, dx$$

for all $\phi \in C_c^\infty(\mathbb{R}^n)$. We write $g = D_i f$. (Arbitrary changes on a set of measure zero do not affect weak derivatives.)

Exercise 10.3 (Example 1). Let

$$f(x) = \begin{cases} 0 & x \leq 0, \\ x & x > 0. \end{cases}$$

What is the weak derivative of f ?

Solution. By definition we need g with

$$\int_{-\infty}^{\infty} f(x) \phi'(x) \, dx = - \int_{-\infty}^{\infty} g(x) \phi(x) \, dx.$$

Compute

$$\int_{-\infty}^{\infty} f(x) \phi'(x) \, dx = \int_0^{\infty} x \phi'(x) \, dx = [x\phi(x)]_0^{\infty} - \int_0^{\infty} \phi(x) \, dx.$$

The boundary term vanishes since $\phi \in C_c^\infty(\mathbb{R})$, so

$$\int_{-\infty}^{\infty} f \phi' \, dx = - \int_0^{\infty} \phi \, dx = - \left(\int_{-\infty}^0 0 \cdot \phi(x) \, dx + \int_0^{\infty} 1 \cdot \phi(x) \, dx \right).$$

Therefore

$$g(x) = \begin{cases} 0 & x \leq 0, \\ 1 & x > 0 \end{cases}$$

(the Heaviside function). □

Exercise 10.4 (Example 2). What is the weak derivative of the Heaviside function?

It does not exist as an L^1_{loc} function. Indeed, suppose $h \in L^1_{\text{loc}}$ satisfies, for all $\phi \in C_c^\infty(\mathbb{R})$,

$$\int_{-\infty}^{\infty} h \phi \, dx = - \int_{-\infty}^{\infty} g \phi' \, dx = - \int_0^{\infty} \phi'(x) \, dx = \phi(0).$$

Thus $\int_{-\infty}^{\infty} h \phi = \phi(0)$ for all $\phi \in C_c^\infty$. Choose (smooth approximations of)

$$\Phi_n = \begin{cases} 1 & x \in [-\frac{1}{n}, \frac{1}{n}], \\ 0 & \text{otherwise,} \end{cases}$$

so that $\int h \Phi_n = \Phi_n(0) = 1$, but

$$\left| \int_{-\infty}^{\infty} h \Phi_n \, dx \right| \leq \int_{-1/n}^{1/n} |h| \, dx.$$

Since $h \in L^1_{\text{loc}}$, the right-hand side tends to 0 as $n \rightarrow \infty$, a contradiction.

Remark 10.5. The weak derivative of the Heaviside function is the Dirac delta, which is not an L^1_{loc} function:

- $\delta_0(x) = \begin{cases} \infty & x = 0, \\ 0 & x \neq 0 \end{cases}$ (not a function);
- $\int_{-\infty}^{\infty} \delta_0(x) \, dx = 1$;
- $\int_{-\infty}^{\infty} f(x) \delta_0(x) \, dx = f(0)$.

10.0.2 Properties of weak derivatives in $\mathbb{R}^n$ and $W^{k,p}$ spaces

Theorem 10.6 (Uniqueness of weak derivatives). *If $g_1, g_2 = D^i f$ are two weak derivatives of f , then*

$$\int_{\mathbb{R}^n} g_1 \phi \, dx = - \int_{\mathbb{R}^n} f \partial^i \phi \, dx = \int_{\mathbb{R}^n} g_2 \phi \, dx,$$

so

$$\int_{\mathbb{R}^n} (g_1 - g_2) \phi \, dx = 0 \quad \forall \phi \in C_c^\infty(\mathbb{R}^n),$$

hence $g_1 \equiv g_2$.

Theorem 10.7 (Commutativity). $D^\alpha(D^\beta f) = D^\beta(D^\alpha f) = D^{\alpha+\beta} f$.

Proof.

$$\int D^\alpha(D^\beta f) \phi \, dx = (-1)^{|\alpha|} \int D^\beta f \partial^\alpha \phi \, dx = \dots = (-1)^{|\alpha|+|\beta|} \int f \partial^{\alpha+\beta} \phi \, dx.$$

□

Theorem 10.8 (Stability under L^1_{loc} limits). *If $f_n \rightarrow f$ in L^1_{loc} and $D^\alpha f_n \rightarrow g$ in L^1_{loc} , then $D^\alpha f = g$.*

Proof. For every $\phi \in C_c^\infty(\mathbb{R}^n)$ we aim to show

$$(-1)^{|\alpha|} \int f \partial^\alpha \phi \, dx = \int g \phi \, dx.$$

We know

$$(-1)^{|\alpha|} \int f_n \partial^\alpha \phi \, dx = \int D^\alpha f_n \phi \, dx.$$

Since $f_n \rightarrow f$ in L^1_{loc} ,

$$\lim_n (-1)^{|\alpha|} \int f_n \partial^\alpha \phi = (-1)^{|\alpha|} \int f \partial^\alpha \phi.$$

Since $D^\alpha f_n \rightarrow g$ in L^1_{loc} ,

$$\lim_n \int D^\alpha f_n \phi = \int g \phi.$$

□

11 Sobolev spaces $W^{k,p}$: approximation, extension, and embeddings

11.1 Sobolev spaces $W^{k,p}(\mathbb{R}^n)$

11.1.1 L^p spaces ($1 \leq p \leq \infty$)

$$\|f\|_{L^p(\mathbb{R}^n)} = \left(\int_{\mathbb{R}^n} |f|^p \, dx \right)^{1/p}, \quad \|f\|_{L^\infty(\mathbb{R}^n)} = \operatorname{ess\,sup}_{x \in \mathbb{R}^n} |f|$$

(in the measure-theoretic sense).

Important inequalities.

- Hölder's inequality: if $\frac{1}{p} + \frac{1}{q} = 1$, then

$$\int_{\mathbb{R}^n} |uv| \, dx \leq \left(\int_{\mathbb{R}^n} |u|^p \, dx \right)^{1/p} \left(\int_{\mathbb{R}^n} |v|^q \, dx \right)^{1/q},$$

i.e. $\|uv\|_{L^1(\mathbb{R}^n)} \leq \|u\|_{L^p(\mathbb{R}^n)} \|v\|_{L^q(\mathbb{R}^n)}$. On a finite-measure domain one also has $\|u\|_{L^{p_1}(\Omega)} \leq C \|u\|_{L^{p_2}(\Omega)}$ (for $p_1 \leq p_2$).

- Minkowski inequality:

$$\|u + v\|_{L^p(\Omega)} \leq \|u\|_{L^p(\Omega)} + \|v\|_{L^p(\Omega)}.$$

Remark 11.1 (L^1_{loc} vs. L^1).

$$\begin{aligned} L^1_{\text{loc}}(\mathbb{R}^n) : \quad & \forall \text{ compact } K \subset \mathbb{R}^n, \quad \int_K |f| \, dx < +\infty, \\ L^1(\mathbb{R}^n) : \quad & \int_{\mathbb{R}^n} |f| \, dx < +\infty. \end{aligned}$$

Example: $f(x) = x$ on $\mathbb{R}$ satisfies $f \notin L^1(\mathbb{R})$ since $\int_{-\infty}^{\infty} |x| \, dx = +\infty$, but for every $[a, b] \subset \mathbb{R}$ one has $\int_a^b |x| \, dx < +\infty$, so $f \in L^1_{\text{loc}}(\mathbb{R})$. Indeed $L^1(\mathbb{R}^n) \subset L^1_{\text{loc}}(\mathbb{R}^n)$: for compact $K \subset \mathbb{R}^n$,

$$\int_K |f| \, dx \leq \int_{\mathbb{R}^n} |f| \, dx = \|f\|_{L^1(\mathbb{R}^n)} < +\infty.$$

Similarly one can define $L^p(\Omega)$ for $\Omega \subseteq \mathbb{R}^n$.

11.1.2 Definition of $W^{k,p}(\Omega)$

$$W^{k,p}(\Omega) = \{u \mid D^\alpha u \in L^p(\Omega) \forall |\alpha| \leq k\},$$

with

$$\|u\|_{W^{k,p}(\Omega)} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)},$$

or equivalently

$$\left(\int_{\Omega} \sum_{|\alpha| \leq k} |D^\alpha u|^p dx \right)^{1/p}.$$

Define

$$W_0^{k,p}(\Omega) := \overline{C_c^\infty(\Omega)}^{\|\cdot\|_{W^{k,p}}},$$

i.e. the closure of C_c^∞ functions under the $W^{k,p}$ norm. Thus for every $f \in W_0^{k,p}(\Omega)$ there exist $f_n \in C_c^\infty(\Omega)$ with $\|f_n - f\|_{W^{k,p}(\Omega)} \rightarrow 0$.

Example 11.2. Take $\Omega = (-1, 1)$ and $u = |x|$. Then

$$Du = \begin{cases} 1 & x > 0, \\ 0 & x = 0, \\ -1 & x < 0, \end{cases} \implies Du \in L^p(\Omega) \quad \forall p \geq 1,$$

but D^2u does not exist in any L^p . Indeed, for every $\phi \in C_c^\infty((-1, 1))$,

$$\begin{aligned} \int_{-1}^1 D^2u \phi dx &= - \int_{-1}^1 Du \phi'(x) dx \\ &= - \int_{-1}^0 (-1) \phi'(x) dx - \int_0^1 \phi'(x) dx \\ &= \int_{-1}^0 \phi'(x) dx - \int_0^1 \phi'(x) dx \\ &= (\phi(0) - \phi(-1)) - (\phi(1) - \phi(0)) = 2\phi(0), \end{aligned}$$

since $\phi(1) = \phi(-1) = 0$. Hence $\int_{-1}^1 D^2u \phi dx = 2\phi(0)$, i.e. $D^2u = 2\delta_0$.

Example 11.3. Let $\Omega = B_1(0) \subseteq \mathbb{R}^n$ and $u(x) = |x|^{-\alpha}$. Then

$$u \in W^{1,p}(\Omega) \iff \alpha < \frac{n-p}{p}.$$

Proof.

(1) $u \in L^p(\Omega)$ iff $\int_{B_1(0)} (|x|^{-\alpha})^p dx < +\infty$. In polar coordinates,

$$\int_{B_1(0)} |x|^{-\alpha p} dx = \int_0^1 \int_{\partial B_\rho(0)} \rho^{-\alpha p} dS d\rho = C \int_0^1 \rho^{-\alpha p} \rho^{n-1} d\rho = C [\rho^{-\alpha p + n}]_{\rho=0}^1.$$

As $\rho \rightarrow 0$ this is meaningful iff $-\alpha p + n \geq 0$, i.e. $\alpha \leq n/p$.

(2) $Du \in L^p(\Omega)$: $Du = (-\alpha)|x|^{-\alpha-1}D|x|$, and $D_i|x| = x_i/|x|$ for $x \neq 0$ (cf. the 1D case $D|x| = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$). Thus $Du = (-\alpha)|x|^{-\alpha-2}x$, and

$$\begin{aligned} Du \in L^p(\Omega) &\iff \int_{B_1(0)} ||x|^{-\alpha-2}x|^p dx < +\infty \iff \int_{B_1(0)} |x|^{-\alpha p - p} dx < +\infty \\ &\iff \int_0^1 \rho^{-\alpha p - p + n - 1} d\rho < +\infty \iff [\rho^{-\alpha p - p + n}]_{\rho=0}^1 < +\infty, \end{aligned}$$

i.e. $-\alpha p - p + n \geq 0$, or $\alpha \leq (n-p)/p$.

Remark 11.4. $W^{k,p}(\Omega)$ is a Banach space (a complete normed space):

$$\|cu\| = |c| \|u\| \quad \forall c \in \mathbb{C}, \quad \|u\| = 0 \iff u = 0, \quad \|u+v\| \leq \|u\| + \|v\|.$$

Special case $p = 2$: $H^k := W^{k,2}$. Completeness means every Cauchy sequence converges; in particular L^p is complete.

11.2 Approximation by smooth functions

Define the standard mollifier

$$\eta(x) = \begin{cases} C \exp\left(\frac{1}{|x|^2 - 1}\right) & \text{if } |x| \leq 1, \\ 0 & \text{if } |x| > 1, \end{cases}$$

so that $\exp(1/(|x|^2 - 1)) \rightarrow e^{-\infty} = 0$ as $|x| \rightarrow 1$. Choose $C > 0$ so that $\int_{\mathbb{R}^n} \eta dx = 1$. Then set

$$\eta^\varepsilon(x) = \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right) \quad (\text{height } \uparrow, \text{ width } \downarrow).$$

A change of variables $y = x/\varepsilon$ gives

$$\int_{\mathbb{R}^n} \eta^\varepsilon(x) dx = \int_{\mathbb{R}^n} \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right) dx = \int_{|x| \leq \varepsilon} \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right) dx = \int_{|y| \leq 1} \eta(y) dy = 1.$$

We now define $u^\varepsilon = \eta^\varepsilon * u$ (mollifier).

Theorem 11.5. $u^\varepsilon \in C^\infty(\Omega_\varepsilon)$, where $\Omega_\varepsilon = \{d(x, \partial\Omega) > \varepsilon\}$, and $u^\varepsilon \rightarrow u$ in $W_{\text{loc}}^{k,p}(\Omega)$.

Proof.

$$u^\varepsilon(x) = \eta^\varepsilon * u(x) = \int_{|y-x| \leq \varepsilon} \frac{1}{\varepsilon^n} \eta\left(\frac{y-x}{\varepsilon}\right) u(y) dy,$$

and

$$u^\varepsilon(x) - u(x) = \int_{|y-x| \leq \varepsilon} \frac{1}{\varepsilon^n} \eta\left(\frac{y-x}{\varepsilon}\right) (u(y) - u(x)) dy.$$

If u is continuous, then for every $\delta > 0$ one has $|u(y) - u(x)| < \delta$ whenever $|y - x| < \varepsilon$ (for ε small enough), whence

$$|u^\varepsilon(x) - u(x)| \leq \delta \int_{|y-x| \leq \varepsilon} \frac{1}{\varepsilon^n} \eta\left(\frac{y-x}{\varepsilon}\right) dy = \delta \rightarrow 0.$$

Thus $u^\varepsilon(x) \rightarrow u(x)$ pointwise as $\varepsilon \rightarrow 0$. Also

$$\partial_{x_i} u^\varepsilon(x) = \int_{|y-x| \leq \varepsilon} \frac{1}{\varepsilon^n} \cdot \frac{1}{\varepsilon} \partial_{y_i} \eta\left(\frac{y-x}{\varepsilon}\right) u(y) dy$$

(bounded derivatives), so $u^\varepsilon \in C^\infty(\Omega_\varepsilon)$. Similarly $D^\alpha u^\varepsilon \rightarrow D^\alpha u$ pointwise; the dominated convergence theorem finishes the argument. $\square$

Theorem 11.6 (Global approximation of $W^{k,p}(\Omega)$ by C^∞). Assume Ω is bounded, $u \in W^{k,p}(\Omega)$, and $p \in [1, \infty)$. Then there exist $u_n \in C^\infty(\bar{\Omega}) \cap W^{k,p}(\Omega)$ such that $\|u_n - u\|_{W^{k,p}(\Omega)} \rightarrow 0$.

(Proof omitted; density theorem.)

Remark 11.7. $C_c^\infty(\mathbb{R}^n)$ is dense in $W^{k,p}(\mathbb{R}^n)$.

11.3 Extension and trace

Extension.

Theorem 11.8 (Extension). *Assume Ω is bounded and $\partial\Omega$ is C^1 . Select an open bounded set V with $\Omega \subset\subset V$. Then for any $u \in W^{1,p}(\Omega)$ there exists an extension $Eu \in W^{1,p}(\mathbb{R}^n)$ such that*

- (i) $Eu = u$ a.e. in Ω ;
- (ii) $\text{supp}(Eu) \subset V$;
- (iii) $\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(\Omega)}$.

In particular E is a bounded linear operator.

Idea of the proof. *Step 1.* Fix $x^0 \in \partial\Omega$ and suppose $\partial\Omega$ is flat near x^0 (locally flatten). Let $B = B_r(x^0)$ and

$$B^+ := B \cap \{x_n \geq 0\} \subseteq \bar{B}, \quad B^- := B \cap \{x_n \leq 0\}.$$

The function u is defined in B^+ ; the aim is to define Eu on B^- . Set

$$\bar{u}(x) := \begin{cases} u(x) & x = (x_1, \dots, x_n) \in B^+, \\ -3u(x', -x_n) + 4u(x', -\frac{x_n}{2}) & x = (x_1, \dots, x_n) \in B^-, \end{cases}$$

where $x' = (x_1, \dots, x_{n-1})$. Continuity across $\{x_n = 0\}$:

$$\lim_{x_n \rightarrow 0^+} \bar{u}(x) = \lim_{x_n \rightarrow 0^-} \bar{u}(x),$$

since at $x_n = 0$ one has $-3u(x', 0) + 4u(x', 0) = u(x', 0)$. Moreover $\partial_{x_n} \bar{u}$ is continuous across $\{x_n = 0\}$: on B^- ,

$$\partial_{x_n} \left(-3u(x', -x_n) + 4u(x', -\frac{x_n}{2}) \right) = 3\partial_{x_n} u(x', -x_n) - 2\partial_{x_n} u(x', -\frac{x_n}{2}),$$

which equals $\partial_{x_n} u$ at $x_n = 0$. It is also not hard to check $\|\bar{u}\|_{W^{1,p}(B)} \leq C\|u\|_{W^{1,p}(B^+)}$.

Step 2. $\partial\Omega$ is C^1 : locally straighten (differential geometry).

Step 3. $\partial\Omega$ is compact: covers by finitely many balls B .

Trace. Question: given $u \in W^{1,p}(\Omega)$, how do we define $u|_{\partial\Omega}$? Here Ω is n -dimensional while $\partial\Omega$ is $(n-1)$ -dimensional (and $\partial\Omega$ has measure zero in $\mathbb{R}^n$).

Theorem 11.9 (Trace). *Assume Ω is bounded and $\partial\Omega$ is C^1 . Then there exists a bounded linear operator $T : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega)$ such that*

(1) $Tu = u|_{\partial\Omega}$ if $u \in W^{1,p}(\Omega) \cap C(\bar{\Omega})$;

(2) $\|Tu\|_{L^p(\partial\Omega)} \leq C\|u\|_{W^{1,p}(\Omega)}$.

Idea of the proof. Assume $x^0 \in \partial\Omega$ and $\partial\Omega$ is flat near x^0 . Let $B = B_r(x^0)$, $\hat{B} = B_{r/2}(x^0)$, and pick $\zeta \in C_c^\infty(B)$ with

$$\zeta = \begin{cases} 1 & \text{in } \hat{B}, \\ 0 & \text{outside } B. \end{cases}$$

Write $\Gamma := \partial\Omega \cap \hat{B}$ and $x' = (x_1, \dots, x_{n-1})$. Then

$$\begin{aligned} \int_{\Gamma} |u|^p dx' &\leq \int_{\{x_n=0\}} \zeta |u|^p dx' = \int_{\{x_n=0\}} 1 \cdot \zeta |u|^p dx' \quad (\text{functionality}) \\ &= - \int_{B^+} \partial_{x_n} (\zeta |u|^p) dx \quad (\text{Green's formula}) \\ &= - \int_{B^+} \left[|u|^p \partial_{x_n} \zeta + p |u|^{p-1} \operatorname{sgn}(u) \cdot \partial_{x_n} u \cdot \zeta \right] dx \\ &\leq C \int_{B^+} |u|^p + |Du|^p dx. \end{aligned}$$

(Young: $ab \leq a^\alpha/\alpha + b^\beta/\beta$ with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, taking $\alpha = p$ and $\beta = p/(p-1)$.)

Then for non-flat $\partial\Omega$:

- straighten $\Rightarrow \int_{\Gamma} |u|^p d\sigma \leq C \int_{\Omega} |u|^p + |Du|^p dx$;
- compactness, finite cover $\Rightarrow \|u\|_{L^p(\partial\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}$.

Theorem 11.10 (Trace-zero functions in $W^{1,p}(\Omega)$). *Assume Ω is bounded, $\partial\Omega$ is C^1 , and $u \in W^{1,p}(\Omega)$. Then $u \in W_0^{1,p}(\Omega)$ if and only if $Tu = 0$ on $\partial\Omega$.*

Recall $W_0^{1,p}(\Omega) := \overline{C_c^\infty(\Omega)}^{\|\cdot\|_{W^{1,p}}}$.

Proof. ($\Rightarrow$) If $u \in W_0^{1,p}(\Omega)$, there exist $u_m \in C_c^\infty(\Omega)$ with $u_m \rightarrow u$ in $W^{1,p}$, i.e. $\|u_m - u\|_{W^{1,p}(\Omega)} \rightarrow 0$. Then

$$\|Tu_m - Tu\|_{L^p(\partial\Omega)} \leq C \|u_m - u\|_{W^{1,p}(\Omega)} \rightarrow 0.$$

Since $Tu_m = 0$ (as $u_m \in C_c^\infty(\Omega)$), one gets $Tu = 0$ on $\partial\Omega$.

($\Leftarrow$) If $Tu = 0$ on $\partial\Omega$, aim to construct $u_m \in C_c^\infty(\Omega)$ with $u_m \rightarrow u$ in $W^{1,p}$. WLOG (locally flat + extension) we assume

- (1) $u \in W^{1,p}(\mathbb{R}_+^n)$ has compact support;
- (2) $Tu = 0$ on $\{x_n = 0\}$.

Then apply a mollifier to construct $u_m \dots$ (omitted). □

11.4 Sobolev embeddings

$X \subset Y$ embedding means $\|f\|_Y \leq C \|f\|_X$, so $f \in X$ with $\|f\|_X < +\infty$ implies $f \in Y$ with $\|f\|_Y < +\infty$.

Question: $W^{1,p}(\Omega) \subset ?$ depends on p :

$$\begin{cases} 1 \leq p < n, \\ p = n, \\ n < p \leq \infty. \end{cases}$$

11.4.1 Gagliardo–Nirenberg–Sobolev inequality

Case $1 \leq p < n$: we want

$$\|u\|_{L^q(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)} \quad (*)$$

where C and q do not depend on u .

Motivation. If $(*)$ holds for every u , it must hold for $u_\lambda(x) = u(\lambda x)$, $x \in \mathbb{R}^n$, i.e.

$$\|u_\lambda\|_{L^q(\mathbb{R}^n)} \leq C \|Du_\lambda\|_{L^p(\mathbb{R}^n)}.$$

Now

$$\int_{\mathbb{R}^n} |u_\lambda|^q dx = \int_{\mathbb{R}^n} |u(\lambda x)|^q dx = \int_{\mathbb{R}^n} |u(y)|^q \frac{dy}{\lambda^n},$$

and

$$\int_{\mathbb{R}^n} |Du_\lambda|^p dx = \int_{\mathbb{R}^n} \lambda^p |Du(\lambda x)|^p dx = \lambda^{p-n} \int_{\mathbb{R}^n} |Du(y)|^p dy.$$

Thus $(*)$ becomes

$$\lambda^{-n/q} \|u\|_{L^q(\mathbb{R}^n)} \leq C \lambda^{(p-n)/p} \|Du\|_{L^p(\mathbb{R}^n)},$$

or

$$\|u\|_{L^q(\mathbb{R}^n)} \leq C \lambda^{1 - \frac{n}{p} + \frac{n}{q}} \|Du\|_{L^p(\mathbb{R}^n)}.$$

This should be independent of λ ! So

$$1 - \frac{n}{p} + \frac{n}{q} = 0$$

(otherwise let $\lambda \rightarrow \infty$ or $\lambda \rightarrow 0$). Hence

$$\frac{1}{q} = \frac{1}{p} - \frac{1}{n} \quad \text{or} \quad q = \frac{np}{n-p} =: p^*,$$

which requires $1 \leq p < n$.

Theorem 11.11 (Gagliardo–Nirenberg–Sobolev). *Assume $1 \leq p < n$. Then*

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)} \quad \forall u \in C_c^1(\mathbb{R}^n).$$

Remark 11.12. This version needs u to have compact support! (Counterexample: $u \equiv 1$ gives LHS = $+\infty$, RHS = 0.) But C does not depend on the support.

Proof. Step 1. Assume $p = 1$ (so $1^* = n/(n-1)$). By the fundamental theorem of calculus,

$$u(x) = \int_{-\infty}^{x_i} \partial_{x_i} u(x_1, \dots, x_{i-1}, y_i, x_{i+1}, \dots, x_n) dy_i,$$

hence

$$|u(x)| \leq \int_{-\infty}^{\infty} |Du(x_1, \dots, y_i, \dots, x_n)| dy_i \quad (i = 1, \dots, n).$$

So

$$|u(x)|^{n/(n-1)} \leq \prod_{i=1}^n \left(\int_{-\infty}^{\infty} |Du(x_1, \dots, y_i, \dots, x_n)| dy_i \right)^{1/(n-1)}. \quad (*)$$

Integrate $(*)$ with respect to x_1 :

$$\begin{aligned} \int_{-\infty}^{\infty} |u|^{n/(n-1)} dx_1 &\leq \int_{-\infty}^{\infty} \prod_{i=1}^n \left(\int_{-\infty}^{\infty} |Du| dy_i \right)^{1/(n-1)} dx_1 \\ &= \left(\int_{-\infty}^{\infty} |Du| dy_1 \right)^{1/(n-1)} \int_{-\infty}^{\infty} \prod_{i=2}^n \left(\int_{-\infty}^{\infty} |Du| dy_i \right)^{1/(n-1)} dx_1 \end{aligned}$$

(independent of x_1 ; apply Hölder in x_1 with $p_i = n-1$). Thus

$$\leq \left(\int_{-\infty}^{\infty} |Du| dy_1 \right)^{1/(n-1)} \prod_{i=2}^n \left(\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du| dy_i dx_1 \right)^{1/(n-1)}.$$

Again integrate with respect to x_2 (factor out; $i = 2$ term independent of x_2 ; apply Hölder in x_2):

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |u|^{n/(n-1)} dx_1 dx_2 \\ & \leq \left(\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du| dx_1 dy_2 \right)^{1/(n-1)} \left(\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du| dy_1 dx_2 \right)^{1/(n-1)} \\ & \quad \cdot \prod_{i=3}^n \left(\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |Du| dx_1 dx_2 dy_i \right)^{1/(n-1)}. \end{aligned}$$

Repeating the process,

$$\int_{\mathbb{R}^n} |u|^{n/(n-1)} dx \leq \left(\prod_{i=1}^n \int_{\mathbb{R}^n} |Du| dx \right)^{1/(n-1)} = \left(\int_{\mathbb{R}^n} |Du| dx \right)^{n/(n-1)},$$

i.e. $\|u\|_{L^{1^*}(\mathbb{R}^n)} \leq \|Du\|_{L^1(\mathbb{R}^n)}$.

Step 2. Now for $1 < p < n$, apply the $p = 1$ case to $v = |u|^\gamma$:

$$\begin{aligned} \|v\|_{L^{n/(n-1)}(\mathbb{R}^n)} &= \left(\int_{\mathbb{R}^n} |u|^{\gamma n/(n-1)} dx \right)^{(n-1)/n}, \\ \|Dv\|_{L^1(\mathbb{R}^n)} &= \int_{\mathbb{R}^n} \gamma |u|^{\gamma-1} |Du| dx \\ &\leq \gamma \left(\int_{\mathbb{R}^n} |u|^{(\gamma-1)p/(p-1)} dx \right)^{(p-1)/p} \|Du\|_{L^p(\mathbb{R}^n)}. \end{aligned}$$

Choose $\gamma n/(n-1) = (\gamma-1)p/(p-1)$, i.e. $\gamma = p(n-1)/(n-p) > 1$. Then

$$\frac{\gamma n}{n-1} = \frac{np}{n-p} = p^*,$$

and

$$\left(\int_{\mathbb{R}^n} |u|^{p^*} dx \right)^{(n-1)/n - (p-1)/p} \leq \|Du\|_{L^p(\mathbb{R}^n)}.$$

Since $(n-1)/n - (p-1)/p = (n-p)/(np) = 1/p^*$, we obtain

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)}.$$

□

Corollary 11.13. *Let $\Omega \subseteq \mathbb{R}^n$ be bounded with $\partial\Omega$ of class C^1 , and $1 \leq p < n$. Then for every $q \in [1, p^*]$ with $p^* = np/(n-p)$,*

$$\|f\|_{L^q(\Omega)} \leq C \|f\|_{W^{1,p}(\Omega)}.$$

Proof. Let $\tilde{\Omega} = \{x \in \mathbb{R}^n : d(x, \Omega) < 1\}$ be an open neighborhood of Ω . Apply the extension theorem:

$$\|f\|_{L^q(\Omega)} \leq C_1 \|f\|_{L^{p^*}(\Omega)} \leq C_2 \|Ef\|_{L^{p^*}(\mathbb{R}^n)} \leq C_3 \|f\|_{W^{1,p}(\Omega)}$$

(by Hölder, then GNS).

□

11.4.2 Morrey's estimate

Question: what about $p = n$ and $p > n$? Borderline $p = n$: $p^* = np/(n-p) \rightarrow +\infty$ as $p \rightarrow n$. Does $W^{1,n}(\Omega) \subseteq L^\infty(\Omega)$? **No for $n > 1$.**

Example 11.14. Take $n = 2$ and $u = \log \log(1 + 1/|x|)$. Then $u \notin L^\infty(B_1(0))$ (as $x \rightarrow 0$, $u \rightarrow \infty$), but $u \in W^{1,2}(B_1(0))$.

1°. $u \in L^2(B_1(0))$:

$$\begin{aligned} \int_{B_1(0)} [\log \log(1 + 1/|x|)]^2 dx &= \int_0^{2\pi} \int_0^1 (\log \log(1 + 1/\rho))^2 \rho d\rho d\theta \\ &\leq C \int_0^1 (\log \log(1 + 1/\rho))^2 \rho d\rho \quad (\text{bounded}). \end{aligned}$$

Indeed

$$\lim_{\rho \rightarrow 0} (\log \log(1 + 1/\rho))^2 \rho = \lim_{\rho \rightarrow 0} \frac{(\log \log(1 + 1/\rho))^2}{1/\rho} = 0$$

(by L'Hôpital twice). Or: $\log$ grows slower than any positive power of ρ , so $u \in L^2(B_1(0))$ (integrand bounded).

2°. $\nabla u \in L^2(B_1(0))$:

$$\nabla u = \frac{1}{\log(1 + 1/|x|)} \cdot \frac{1}{1 + 1/|x|} \cdot \left(-\frac{1}{|x|^2}\right) \cdot \frac{x}{|x|}.$$

Thus

$$\begin{aligned} \int_{B_1(0)} |\nabla u|^2 dx &\leq C \int_0^1 \frac{1}{\log^2(1 + 1/\rho) (1 + 1/\rho)^2 \rho^3} d\rho \\ &\leq C \int_0^1 \frac{1}{\log^2(1 + 1/\rho)} \frac{1}{\rho} d\rho. \end{aligned}$$

Set $s = \log(1 + 1/\rho)$, so $1/\rho = e^s - 1$ and $\rho : 0 \rightarrow 1$ gives $s : \infty \rightarrow \log 2$. Then

$$I \leq C \int_{\log 2}^\infty \frac{1}{s^2} \left(\frac{1}{e^s - 1} + 1 \right) ds \leq C \int_{\log 2}^\infty \frac{1}{s^2} ds < +\infty.$$

(The $p = n$ case is complicated!)

11.4.3 The case $p > n$

Definition 11.15 (Hölder spaces $C^{k,\gamma}(\Omega)$). • $\|u\|_{C(\Omega)} := \sup_{x \in \Omega} |u(x)| = \|u\|_{L^\infty(\Omega)}$.

- γ -th Hölder seminorm ($\gamma \in (0, 1]$):

$$[u]_{C^{0,\gamma}(\Omega)} := \sup_{x \neq y \in \Omega} \frac{|u(x) - u(y)|}{|x - y|^\gamma}.$$

- γ -th Hölder norm:

$$\|u\|_{C^{0,\gamma}(\Omega)} := \|u\|_{C(\Omega)} + [u]_{C^{0,\gamma}(\Omega)}.$$

Remark 11.16. What does $[u]_{C^{0,\gamma}(\Omega)}$ describe?

Example 11.17 ($\gamma = 1$: Lipschitz). If $|u(x) - u(y)| \leq L|x - y|$ for all $x, y \in \Omega$, then $[u]_{C^{0,1}} \leq L$. Lipschitz functions need not be differentiable everywhere (e.g. $f(x) = |x|$ is not differentiable at $x = 0$).

Example 11.18 ($C^1(\Omega) \subseteq C^{0,1}(\Omega)$ if Ω is bounded).

$$\sup_{x \neq y \in \Omega} \frac{|f(x) - f(y)|}{|x - y|} \leq C \sup_{x \in \Omega} |f'(x)|.$$

Locally: in a neighborhood of x_0 , $f(x)$ is contained between $\pm|x|$.

Example 11.19 ($\gamma \in (0, 1)$). $|u(x) - u(y)| \leq C|x - y|^\gamma$. Locally $f(x)$ is contained between $\pm|x|^\gamma$.

Inclusions:

$$C^1(\Omega) \subseteq C^{0,1}(\Omega) \subseteq C^{0,\gamma}(\Omega),$$

since

$$\frac{|f(x) - f(y)|}{|x - y|^\gamma} \leq \frac{|f(x) - f(y)|}{|x - y|} \cdot |x - y|^{1-\gamma} \leq \frac{|f(x) - f(y)|}{|x - y|} \cdot \text{diam}(\Omega)^{1-\gamma}.$$

Definition 11.20.

$$\|u\|_{C^{k,\gamma}(\Omega)} = \|u\|_{C^k(\Omega)} + \sum_{|\alpha|=k} [\partial^\alpha u]_{C^{0,\gamma}(\Omega)}.$$

Informally $C^{k,\gamma}(\Omega) \approx W^{k+\gamma,\infty}(\Omega)$ (fractional derivatives).

Theorem 11.21 (Morrey's estimate). Assume $n < p \leq \infty$. Then there exists $C > 0$ such that

$$\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(\mathbb{R}^n)}, \quad \gamma = 1 - \frac{n}{p}.$$

In particular ($p = \infty$): $W^{1,\infty}(\mathbb{R}^n) \subset C^{0,1}(\mathbb{R}^n)$.

Proof. Step 1. We show

$$\frac{1}{|B_r(x)|} \int_{B_r(x)} |u(y) - u(x)| dy \leq C \int_{B_r(x)} \frac{|Du(y)|}{|y - x|^{n-1}} dy$$

for every $B_r(x) \subseteq \mathbb{R}^n$. Fix $w \in \partial B_1(0)$. By FTC,

$$u(x + sw) - u(x) = \int_0^s Du(x + tw) \cdot w dt,$$

so $|u(x + sw) - u(x)| \leq \int_0^s |Du(x + tw)| dt$. Integrating over $\partial B_1(0)$,

$$\begin{aligned} \int_{\partial B_1(0)} |u(x + sw) - u(x)| d\sigma_w &= \frac{1}{s^{n-1}} \int_{\partial B_s(x)} |u(y) - u(x)| d\sigma_y \\ &\leq \int_{\partial B_1(0)} \int_0^s |Du(x + tw)| dt d\sigma_w \\ &= \int_0^s \int_{\partial B_t(x)} \frac{|Du(y)|}{t^{n-1}} d\sigma_y dt = \int_{B_s(x)} \frac{|Du(y)|}{|x - y|^{n-1}} dy. \end{aligned}$$

Hence

$$\int_{\partial B_s(x)} |u(z) - u(x)| d\sigma_z \leq s^{n-1} \int_{B_r(x)} \frac{|Du(y)|}{|y - x|^{n-1}} dy.$$

Integrate s from 0 to r :

$$\int_{B_r(x)} |u(y) - u(x)| dy \leq \frac{r^n}{n} \int_{B_r(x)} \frac{|Du(y)|}{|y - x|^{n-1}} dy.$$

Step 2.

$$u(x) = \frac{1}{|B_1(x)|} \int_{B_1(x)} (u(x) - u(y)) dy + \frac{1}{|B_1(x)|} \int_{B_1(x)} u(y) dy,$$

so

$$\begin{aligned} |u(x)| &\leq C \int_{B_1(x)} \frac{|Du(y)|}{|x-y|^{n-1}} dy + C \|u\|_{L^p(B_1(x))} \quad (\text{bounded domain}) \\ &\leq C \|Du\|_{L^p(\mathbb{R}^n)} \left(\int_{B_1(x)} |x-y|^{-(n-1)p/(p-1)} dy \right)^{(p-1)/p} + \|u\|_{L^p(\mathbb{R}^n)}. \end{aligned}$$

The radial integral

$$\int_0^1 \rho^{-(n-1)p/(p-1)} \rho^{n-1} d\rho = \int_0^1 \rho^{-(n-1)/(p-1)} d\rho = C [\rho^{(p-n)/(p-1)}]_0^1$$

converges when $p > n$. Thus $|u(x)| \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}$ for all $x \in \mathbb{R}^n$, i.e. $\|u\|_{C(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}$.

Steps for Hölder continuity. Let $r = |x-y|$ and $W = B_r(x) \cap B_r(y)$. Then

$$|u(x) - u(y)| \leq \frac{1}{|W|} \int_W |u(x) - u(z)| dz + \frac{1}{|W|} \int_W |u(z) - u(y)| dz =: I_1 + I_2.$$

Now

$$\begin{aligned} I_1 &\leq C \frac{1}{|B_r(x)|} \int_{B_r(x)} |u(x) - u(z)| dz \leq C \int_{B_r(x)} \frac{|Du(z)|}{|x-z|^{n-1}} dz \\ &\leq C \|Du\|_{L^p(\mathbb{R}^n)} \left(\int_{B_r(x)} |x-z|^{-(n-1)p/(p-1)} dz \right)^{(p-1)/p} \\ &\leq C \|Du\|_{L^p(\mathbb{R}^n)} r^{(p-n)/p} = C \|Du\|_{L^p(\mathbb{R}^n)} r^{1-n/p}. \end{aligned}$$

Similarly $I_2 \leq C \|Du\|_{L^p(\mathbb{R}^n)} r^{1-n/p}$. Hence

$$|u(x) - u(y)| \leq C |x-y|^{1-n/p} \|Du\|_{L^p(\mathbb{R}^n)},$$

so $[u]_{C^{0,1-n/p}} \leq C \|Du\|_{L^p(\mathbb{R}^n)}$, and therefore

$$\|u\|_{C^{0,1-n/p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}.$$

□

Corollary 11.22. *If Ω is bounded and $n < p \leq \infty$, then*

$$\|u\|_{C^{0,1-n/p}(\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}$$

(by extension).

11.4.4 General Sobolev inequalities

Summary of embeddings for $W^{k,p}$:

- $kp < n$: $W^{k,p} \hookrightarrow L^q$ with $q = np/(n-kp)$.
- $kp > n$: $W^{k,p} \hookrightarrow C^{k-[n/p]-1,\gamma}$, where

$$\gamma = \begin{cases} [n/p] + 1 - n/p & \text{if } n/p \text{ is not an integer,} \\ \text{any } \gamma < 1 & \text{if } n/p \text{ is an integer.} \end{cases}$$

- $kp = n$: $W^{k,p} \hookrightarrow \{u : \int e^{C|u|^q} < +\infty\}$ (Trudinger / exponential integrability; related to BMO).

12 Compact embeddings, Poincaré inequalities, and difference quotients

12.1 Compact embedding

By GNS, $W^{1,p}(U) \hookrightarrow L^q(U)$ for all $1 \leq p < n$ and $q \in [1, p^*]$. Goal: $W^{1,p}(U) \hookrightarrow L^q(U)$ for all $1 \leq q < p^*$.

Definition 12.1 (Compact embedding). Let X, Y be Banach spaces with $X \subset Y$. We say X is compactly embedded in Y ($X \hookrightarrow Y$) if $X \hookrightarrow Y$ and every bounded subset of X is (pre)compact in Y .

Theorem 12.2 (Rellich–Kondrachov). Let $U \subset \mathbb{R}^n$ be open and bounded with ∂U of class C^1 , and $1 \leq p < n$. Then

$$W^{1,p}(U) \hookrightarrow L^q(U) \quad \forall 1 \leq q < p^*.$$

Remark 12.3. Since $p^* > p$ and $p^* \rightarrow \infty$ as $p \rightarrow n$, one has in particular $W^{1,p}(U) \hookrightarrow L^p(U)$.

12.2 Poincaré inequalities and difference quotients

12.2.1 Poincaré's inequalities

Theorem 12.4 (Poincaré's inequality). Let $U \subset \mathbb{R}^n$ be open, bounded, and connected, with ∂U of class C^1 , and $1 \leq p \leq \infty$. Then

$$\left\| u - \frac{1}{|U|} \int_U u \, dy \right\|_{L^p(U)} \leq C \|Du\|_{L^p(U)} \quad \forall u \in W^{1,p}(U),$$

where C depends on n, p, U .

Proof (by contradiction). If the claim fails, there exists $\{u_k\} \subset W^{1,p}(U)$ with

$$\left\| u_k - \frac{1}{|U|} \int_U u_k \, dx \right\|_{L^p(U)} > k \|Du_k\|_{L^p(U)}.$$

Define

$$v_k = \frac{u_k - \frac{1}{|U|} \int_U u_k \, dy}{\left\| u_k - \frac{1}{|U|} \int_U u_k \, dy \right\|_{L^p}}.$$

Then $\frac{1}{|U|} \int_U v_k \, dy = 0$, $\|v_k\|_{L^p} = 1$, and $\|Dv_k\|_{L^p} < 1/k$ (by the assumption). In particular $\{v_k\}$ is bounded in $W^{1,p}$. By Rellich–Kondrachov ($W^{1,p}(U) \subset\subset L^p(U)$), there exist a subsequence v_{k_j} and $v \in L^p(U)$ with $v_{k_j} \rightarrow v$ in $L^p(U)$.

Moreover

$$\left| \frac{1}{|U|} \int_U v \, dy \right| \leq C \|v - v_{k_j}\|_{L^p(U)} \rightarrow 0$$

(by Hölder), so $\frac{1}{|U|} \int_U v \, dy = 0$. Also $\|v\|_{L^p(U)} = 1$.

For every $\phi \in C_c^\infty(U)$,

$$\begin{aligned} \left| \int_U v \partial_{x_i} \phi \, dx \right| &= \lim_{j \rightarrow \infty} \left| \int_U v_{k_j} \partial_{x_i} \phi \, dx \right| = \lim_{j \rightarrow \infty} \left| - \int_U D_{x_i} v_{k_j} \phi \, dx \right| \\ &\leq \lim_{j \rightarrow \infty} \frac{1}{k_j} \|\phi\|_{L^\infty(U)} = 0. \end{aligned}$$

Hence $Dv = 0$ a.e., so $v \in W^{1,p}(U)$ with $Dv = 0$. Since U is connected, $v \equiv C$ a.e.; the mean-zero condition forces $v \equiv 0$, contradicting $\|v\|_{L^p} = 1$. $\square$

Remark 12.5 (Case $n = 1$). Suppose $u \in W^{1,p}((0,1))$ and $Du = 0$ a.e. in $(0,1)$. Prove u is constant a.e. Assume toward a contradiction

$$u(x) = \begin{cases} C_1 & x \in (0, 1/2), \\ C_2 & x \in (1/2, 1) \end{cases}$$

with $C_1 \neq C_2$. Choose $\phi \in C_c^\infty((0,1))$ with $\phi(1/2) \neq 0$. Then

$$\begin{aligned} 0 &= \int_0^1 Du \phi \, dx = - \int_0^1 u D\phi \, dx \\ &= -(C_1\phi(1/2) - C_1\phi(0)) - (C_2\phi(1) - C_2\phi(1/2)) = (C_2 - C_1)\phi(1/2), \end{aligned}$$

a contradiction.

Corollary 12.6 (Poincaré inequality for a ball). Assume $1 \leq p \leq \infty$. There exists $C = C(n, p)$ such that

$$\left\| u - \frac{1}{|B(x, r)|} \int_{B(x, r)} u(z) \, dz \right\|_{L^p(B(x, r))} \leq Cr \|Du\|_{L^p(B(x, r))}$$

for every ball $B(x, r) \subset \mathbb{R}^n$ and every $u \in W^{1,p}(B(x, r))$.

Remark 12.7. Relative to the usual Poincaré inequality, the constant here depends on the domain $V = B(x, r)$.

Proof. Set $v(y) := u(x + ry)$ for $y \in B(0, 1)$. Then $u \in W^{1,p}(B(x, r))$ implies $v \in W^{1,p}(B(0, 1))$. By Poincaré on the unit ball (with C independent of x, r),

$$\left\| v - \frac{1}{|B(0, 1)|} \int_{B(0, 1)} v \, dy \right\|_{L^p(B(0, 1))} \leq C \|Dv\|_{L^p(B(0, 1))}.$$

The change of variables $z = x + ry$ and $D_y v(y) = r D_z u(z)$ produces the factor Cr . $\square$

12.2.2 Difference quotients

Assume $u : U \rightarrow \mathbb{R}$ is locally summable and $V \subset\subset U$.

Definition 12.8. (i) The i -th difference quotient of size h is

$$D_i^h u(x) = \frac{u(x + he_i) - u(x)}{h}, \quad i = 1, \dots, n,$$

for all $x \in V$ and $0 < |h| < \text{dist}(V, \partial U)$ (ensures $x + he_i$ stays in U).

(ii) $D^h u := (D_1^h u, \dots, D_n^h u)$.

Theorem 12.9 (Difference quotients and weak derivatives). (i) Suppose $1 \leq p < \infty$ and $u \in W^{1,p}(U)$. Then for every $V \subset\subset U$,

$$\|D_i^h u\|_{L^p(V)} \leq C \|Du\|_{L^p(U)}$$

for some $C > 0$ and all $0 < |h| < \frac{1}{2} \text{dist}(V, \partial U)$.

(ii) Assume $1 < p < \infty$, $u \in L^p(U)$, and there exists $C > 0$ such that

$$\|D_i^h u\|_{L^p(V)} \leq C$$

for all $0 < |h| < \frac{1}{2} \text{dist}(V, \partial U)$. Then $u \in W^{1,p}(V)$.

Proof. First assume u is smooth. Then

$$u(x + he_i) - u(x) = \int_0^1 \partial_{x_i} u(x + the_i) \cdot h \, dt,$$

so

$$|D_i^h u(x)| \leq \int_0^1 |Du(x + the_i)| \, dt.$$

Hence (elementary inequality)

$$\begin{aligned} |D^h u| &\leq \sum_{i=1}^n \int_0^1 |Du(x + the_i)| \, dt, \\ |D^h u|^p &\leq C \sum_{i=1}^n \left(\int_0^1 |Du(x + the_i)| \, dt \right)^p \leq C \sum_{i=1}^n \int_0^1 |Du(x + the_i)|^p \, dt \end{aligned}$$

(by Jensen: if f is convex and γ is a probability measure, then $f(\int u \, d\gamma) \leq \int f(u) \, d\gamma$). Consequently

$$\begin{aligned} \int_V |D^h u|^p \, dx &\leq C \sum_{i=1}^n \int_0^1 \int_V |Du(x + the_i)|^p \, dx \, dt \\ &\leq C \int_U |Du|^p \, dx \end{aligned}$$

(let $y = x + the_i$). For general u , use the local approximation theorem (approximate using this identity).

(ii). For $\phi \in C_c^\infty(V)$ and h small,

$$\int_V u D_i^h \phi \, dx = - \int_V (D_i^{-h} u) \phi \, dx$$

(discrete integration by parts). Since $\|D_i^h u\|_{L^p(V)} \leq C$, one has $\sup_h \|D_i^{-h} u\|_{L^p(V)} < +\infty$. For $1 < p < \infty$ (L^p reflexive), Kakutani's theorem yields $v_i \in L^p(V)$ and $h_k \rightarrow 0$ with $D_i^{-h_k} u \rightharpoonup v_i$ weakly in $L^p(V)$. (Kakutani: a Banach space E is reflexive iff the closed unit ball is weakly compact.)

Then for $\phi \in C_c^\infty(V)$,

$$\begin{aligned} \int_V u \partial_{x_i} \phi \, dx &= \lim_{h_k \rightarrow 0} \int_V u D_i^{h_k} \phi \, dx \\ &= - \lim_{h_k \rightarrow 0} \int_V (D_i^{-h_k} u) \phi \, dx = - \int_V v_i \phi \, dx, \end{aligned}$$

where the first limit uses

$$\left| \int_U u (D_i^{h_k} \phi - \partial_{x_i} \phi) \, dx \right| \leq \|u\|_{L^p} \|D_i^{h_k} \phi - \partial_{x_i} \phi\|_{L^{p'}} \rightarrow 0$$

(Lebesgue dominated convergence theorem). Thus $v_i = D_{x_i} u$. Since $u, v_i \in L^p(V)$, one has $u \in W^{1,p}(V)$. $\square$

Part IV

Weak solutions and regularity

13 Weak solutions of second-order elliptic equations

13.1 Compact embedding: counterexamples

Let U be bounded in $\mathbb{R}^n$. Then

$$W^{1,p}(U) \hookrightarrow L^q(U), \quad p < n,$$

if $1 \leq q < p^* = \frac{np}{n-p}$.

Counterexamples.

°1. $U = \mathbb{R}^n$. Let

$$f(x) = \begin{cases} 0 & x \leq 0, \\ x & 0 < x < 1, \\ 2-x & 1 \leq x \leq 2, \\ 0 & x > 2, \end{cases} \quad Df(x) = \begin{cases} 0 & x < 0, \\ 1 & 0 < x < 1, \\ -1 & 1 < x < 2, \\ 0 & x > 2. \end{cases}$$

Define $f_n(x) = f(x-n)$ (translation). Then $\|Df_n\|_{L^p(\mathbb{R}^n)} \leq C$, so $\|f_n\|_{W^{1,p}(\mathbb{R}^n)} \leq C$ is bounded, but f_n does not converge in $L^q(\mathbb{R}^n)$ because $\|f_n\|_{L^q(\mathbb{R}^n)} = \|f\|_{L^q(\mathbb{R}^n)} \not\rightarrow 0$, while $f_n \rightarrow 0$.

°2. $q = p^* = \frac{np}{n-p}$, assuming $\text{supp } u \subset B(0, r) \subset U$. Set $u_\varepsilon(x) := \varepsilon^{-n/p^*} u(x/\varepsilon)$. Then

$$\begin{aligned} \|u_\varepsilon\|_{L^{p^*}(U)} &= \|u\|_{L^{p^*}(U)}, \\ \|\nabla u_\varepsilon\|_{L^p(U)} &= \varepsilon^{n/p-1+n/p^*} \|\nabla u\|_{L^p(U)} = \|\nabla u\|_{L^p(U)}. \end{aligned}$$

Therefore $\|u_\varepsilon\|_{W^{1,p}(U)} \leq C$ is bounded as $\varepsilon \rightarrow 0$, $u_\varepsilon \rightarrow 0$, but $\|u_\varepsilon\|_{L^{p^*}(U)} \not\rightarrow 0$. Note that

$$\|u_\varepsilon\|_{L^q(U)} = \varepsilon^{n/q-n/p^*} \|u\|_{L^q(U)} \xrightarrow{\varepsilon \rightarrow 0} 0 \quad \text{if } q < p^*.$$

13.2 L^2 theory for weak solutions

In this section we consider

$$\begin{cases} Lu = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (24)$$

with Ω a bounded domain.

Remark 13.1 (Recall). Long time ago,

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (25)$$

energy methods gave uniqueness, but existence needs a weaker space and improved regularity.

Definition 13.2.

$$Lu = -\partial_i(a^{ij}(x)\partial_j u) + b^i(x)\partial_i u + c(x)u \quad (1)$$

or

$$Lu = -a^{ij}(x)\partial_{ij}u + b^i(x)\partial_i u + c(x)u. \quad (2)$$

Form (1) is called *divergence form*; form (2) is *non-divergence form*. (Einstein summation: repeated indices are summed.) Here $a^{ij}(x)$ is a matrix of measurable functions.

Uniform ellipticity.

$$\lambda|\xi|^2 \leq a^{ij}(x)\xi_i\xi_j \leq \Lambda|\xi|^2, \quad \lambda \leq 1 \leq \Lambda,$$

and $a^{ij}(x), b^i(x), c(x) \in L^\infty(\overline{\Omega})$.

Remark 13.3. ① If $a^{ij} = \delta_{ij}$ and $b^i \equiv c \equiv 0$, then $Lu = -\Delta u$. The operator $-\Delta$ is uniformly elliptic because $\delta_{ij}\xi_i\xi_j = |\xi|^2$, so $\lambda = \Lambda = 1$.

② Assume symmetry $a^{ij}(x) = a^{ji}(x)$. For example

$$\partial_{11}u + 2\partial_{12}u + \partial_{22}u = \partial_{11}u + \partial_{12}u + \partial_{21}u + \partial_{22}u.$$

- ③ If $a^{ij} \in C^1$ then L may be written in either divergence or non-divergence form. But if a^{ij} is not differentiable, these two forms are not equivalent.
- ④ In matrix form, $a^{ij}(x)\xi_i\xi_j = \langle \xi, A\xi \rangle$. So L is uniformly elliptic if and only if A is positive definite with lowest eigenvalue $\lambda_0(x) \geq \theta > 0$ for all x . (A is real symmetric, so it has n real eigenvalues.)

13.3 Existence of weak solutions

Definition 13.4 (Bilinear form). For all $u, v \in H_0^1(\Omega)$,

$$B[u, v] := \int_{\Omega} \sum_{i,j} a^{ij} \partial_i u \partial_j v + \sum_{i=1}^n b^i \partial_{x_i} u v + c u v \, dx.$$

Here

$$H_0^1(\Omega) = \overline{C_c^\infty(\Omega)} \quad (\text{closed under } \|\cdot\|_{H^1})$$

or

$$\{u \in H^1(\Omega) : u|_{\partial\Omega} = 0\}$$

(the boundary condition via Trace).

14 Lax–Milgram theory, Fredholm alternative, and H^2 regularity

14.1 Problem set

Exercise 14.1. Sobolev embedding ($n \geq 2$):

- $W^{1,p}(\mathbb{R}^n) \hookrightarrow L^{p^*}(\mathbb{R}^n)$ for $p < n$, $p^* = ?$
- $W^{1,p}(\mathbb{R}^n) \hookrightarrow C^{0,\gamma}(\mathbb{R}^n)$ for $p > n$, $\gamma = ?$
- $n = 1$ case: $\|u\|_{L^\infty([a,b])} \leq C\|u\|_{W^{1,1}((a,b))}$?

Exercise 14.2. Let u be a solution to $\partial_t u = \Delta u + u^3$ in $\mathbb{R}^n$ ($n = 1, 2, 3$).

- $u_\lambda = \lambda^s u(\lambda^2 t, \lambda x)$ is again a solution. Determine $s = ?$ (scaling invariance of the equation)
 - Determine what $\dot{H}^k$ space satisfies $\|u_\lambda\|_{\dot{H}^k} = \|u\|_{\dot{H}^k}$? (scaling invariance of the norm)
- ① $\partial_t u_\lambda \sim \lambda^{s+2} \partial_t u$, $\Delta u_\lambda \sim \lambda^{s+2} \Delta u$, and $u_\lambda^3 \sim \lambda^{3s} u^3$. Hence $s + 2 = 3s \Rightarrow s = 1$, so $u_\lambda = \lambda u(\lambda^2 t, \lambda x)$.

②

$$\|u_\lambda\|_{\dot{H}^k} = \lambda \cdot \lambda^k \cdot \lambda^{-n/2} \|u\|_{\dot{H}^k} = \lambda^{1+k-n/2} \|u\|_{\dot{H}^k} \implies k = \frac{n}{2} - 1.$$

14.2 Application of Lax–Milgram

We now apply Lax–Milgram. Take

$$Lu = -\partial_j(a^{ij}\partial_i u) + b^i \partial_i u + cu.$$

In our case we want $B(u, v) = \langle f, v \rangle$ for all $v \in H_0^1(\Omega)$.

- ① $|B(u, v)| \leq C\|u\|_{H^1}\|v\|_{H^1}$ (boundedness).
- ② Not positive-definite! ($c < 0$ case). Instead we have, for some $\gamma > 0$,

$$\frac{\lambda}{2} \int_{\Omega} (|\nabla u|^2 + |u|^2) \leq B(u, u) + \gamma \|u\|_{L^2}^2 \quad (*)$$

hence

$$B(u, u) + \gamma \|u\|_{L^2}^2 \geq \frac{\lambda}{2} \|u\|_{H^1}^2.$$

Theorem 14.3 (First existence theorem). *There exists $\gamma \geq 0$ such that for all $\mu \geq \gamma$ and all $f \in H^{-1}(\Omega)$ there exists a unique weak solution $u \in H_0^1(\Omega)$ to*

$$\begin{cases} Lu + \mu u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases} \quad (26)$$

Proof. Define $B_\mu(u, v) := B(u, v) + \mu\langle u, v \rangle$. Check:

- ① $B_\mu(u, v)$ is bilinear;
- ② $|B_\mu(u, v)| \leq |B(u, v)| + \mu|\langle u, v \rangle| \leq C\|u\|_{H^1}\|v\|_{H^1}$;
- ③ by (*),

$$B_\mu(u, u) = B(u, u) + \mu(u, u) \geq B(u, u) + \gamma\|u\|_{L^2}^2 \geq \frac{\lambda}{2} \int_{\Omega} (|\nabla u|^2 + |u|^2) = \frac{\lambda}{2} \|u\|_{H^1}^2,$$

so $B_\mu(u, u)$ is positive definite.

Then Lax–Milgram applies. □

Remark 14.4 (Example). Let $\Omega = (0, \pi)$ and $Lu = -\frac{d^2}{dx^2}u + cu$. Then $u_k = \sin kx \Rightarrow Lu_k = (k^2 + c)u_k$. If $c = -k_0^2$ for some k_0 , then $Lu_{k_0} = 0$.

- $B(u, v)$ is not positive definite: $B(u_{k_0}, u_{k_0}) = 0$.
- Therefore Lax–Milgram cannot apply. No uniqueness: if $Lu = 0$ then $L(u + u_{k_0}) = 0$.
- No existence for some f ! For $f = u_{k_0}$: if there exists u with $Lu = u_{k_0}$, then

$$\|u_{k_0}\|_{L^2}^2 = (Lu, u_{k_0}) = (u, Lu_{k_0}) = 0,$$

a contradiction.

14.3 Fredholm alternative

Question: what happens without adding μu ?

Formal adjoint. E.g. $Lu = -\frac{d^2}{dx^2}u - u$ on $(0, \pi)$. Then $Lu = f$ has no solution if $\|f\|_{L^2}^2 = (Lu, f) = (u, L^*f)$ forces a contradiction whenever f is not orthogonal to $\ker L^*$ (here $L^* = L$ and $\ker L = \text{span}\{\sin x\}$). Here L^* is the formal adjoint of L .

For divergence form

$$Lu = -\partial_i(a^{ij}\partial_j u) + b^i\partial_i u + cu,$$

$$\begin{aligned} (Lu, v) &= \int a^{ij}\partial_j u \partial_i v + b^i\partial_i u v + cuv \, dx \\ &= \int a^{ij}\partial_i v \partial_j u \boxed{-u \partial_i(b^i v)} + cuv \, dx =: (u, L^*v). \end{aligned}$$

(The first term is symmetric.) Therefore

$$L^*v = -\partial_i(a^{ij}\partial_j v + b^i v) + cv.$$

Definition 14.5 (Compact operators). Let X, Y be Banach spaces. A bounded operator $K : X \rightarrow Y$ is compact if any bounded sequence $\{x_k\} \subset X$ has a subsequence $\{x_{k_j}\}$ such that Kx_{k_j} converges in Y .

Remark 14.6. E.g. $\text{id} : W^{1,p}(\Omega) \rightarrow L^q(\Omega)$ for $q < \frac{np}{n-p}$ is compact. For an adjoint operator $K : H \rightarrow H$, $\langle Kx, y \rangle = \langle x, K^*y \rangle$.

Theorem 14.7 (Fredholm alternative). *Let H be a Hilbert space and $K : H \rightarrow H$ a compact linear operator. Then*

- (i) $\dim N(I - K) = \dim N(I - K^*) < +\infty$;
- (ii) $R(I - K)$ is closed and $R(I - K) = N(I - K^*)^\perp$.

In particular, $R(I - K) = H$ if and only if $N(I - K^) = \{0\}$. Either $(I - K)u = v$ has a unique solution for all $v \in H$ and $(I - K)^{-1}$ is bounded, or $(I - K)u = 0$ has a nontrivial solution and $(I - K)u = v$ has a solution if and only if $(v, w) = 0$ for all w with $w - K^*w = 0$.*

Theorem 14.8 (Second existence theorem). *Assume Ω is bounded, $a^{ij}, b^i, c \in L^\infty(\Omega)$, and L is uniformly elliptic. Exactly one of the following holds:*

- *Either*

$$\begin{cases} Lu = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega \end{cases} \quad (27)$$

has a unique weak solution $u \in H_0^1(\Omega)$ for all $f \in L^2$, and L^{-1} is bounded;

- *or*

$$\begin{cases} Lu = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega \end{cases} \quad (28)$$

has a nonzero weak solution, and

$$\begin{cases} Lu = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega \end{cases} \quad (29)$$

*has a solution if and only if $(f, v) = 0$ for all $v \in H_0^1(\Omega)$ with $L^*v = 0$.*

Proof. Let γ be from the first existence theorem. We know $(L + \gamma)^{-1} : H^{-1}(\Omega) \rightarrow H_0^1(\Omega)$ is a bounded linear operator, because $Lu + \gamma u = g$ in Ω can be solved uniquely. Now since $Lu = f$ if and only if $Lu + \gamma u = f + \gamma u$,

$$u = (L + \gamma)^{-1}(\gamma u + f), \quad \text{i.e.} \quad u - \gamma(L + \gamma)^{-1}u = (L + \gamma)^{-1}f. \quad (*)$$

Let $K = \gamma(L + \gamma)^{-1}$ and $\tilde{f} = (L + \gamma)^{-1}f$. Taking $H = L^2$, we have $K : L^2(\Omega) \rightarrow H_0^1(\Omega)$ bounded linear. We check K is compact as a map $L^2(\Omega) \rightarrow L^2(\Omega)$:

$$\tilde{K} = \text{id}_{H_0^1(\Omega) \hookrightarrow L^2(\Omega)} \circ K_{L^2(\Omega) \rightarrow H_0^1(\Omega)}$$

is compact (composition of a compact embedding and a bounded map). Now K is compact, so by the Fredholm alternative theorem: either $(*)$ has a solution $u \in H$ for all $\tilde{f} \in H$, or $(*)$ has a nonzero solution for $f = 0$, and $(*)$ has a solution if and only if $(\tilde{f}, v) = 0$ for all $v \in N(I - K^*)$, i.e. $L^*v = 0$. $\square$

14.4 Regularity of weak solutions

14.4.1 H^2 -regularity

Let u be a weak solution to $-\Delta u = f$ in Ω . Formally

$$\int_{\Omega} f^2 = \int_{\Omega} (-\Delta u)^2 = \int_{\Omega} \sum \partial_{ii} u \partial_{jj} u = - \sum \int_{\Omega} \partial_i u \partial_{ijj} u = \sum \int_{\Omega} \partial_{ij} u \partial_{ij} u.$$

So if $u \in H^2(\Omega)$ we have $\sum \|\partial_{ij} u\|_{L^2} \leq \|f\|_{L^2}$. The problem is to prove $u \in H^2(\Omega)$ (for general L), where

$$Lu = -\partial_j(a^{ij}\partial_i u) + b^i\partial_i u + cu.$$

Theorem 14.9 (Interior regularity). Assume $a^{ij} \in C^1(\Omega)$, $b^i, c \in L^\infty(\Omega)$, $f \in L^2(\Omega)$. Let u be a weak solution to $Lu = f$ in Ω . Then $u \in H_{\text{loc}}^2$, i.e. $u \in H^2(V)$ for all $V \subset\subset \Omega$, and

$$\|u\|_{H^2(V)} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}).$$

Remark 14.10. We can show $Lu = f$ a.e. in Ω (almost everywhere). Indeed $B(u, v) = (f, v)$ for all $v \in C_c^\infty(\Omega)$, where

$$B(u, v) := \int_{\Omega} a^{ij} \partial_i u \partial_j v + b^i \partial_i u \cdot v + cuv \, dx.$$

Since $u \in H_{\text{loc}}^2$, we can integrate by parts ($v \in C_c^\infty(\Omega)$):

$$B(u, v) = \int_{\Omega} (-\partial_j(a^{ij} \partial_i u) + b^i \partial_i u + cu)v \, dx = (Lu, v).$$

Hence $(Lu, v) = (f, v)$ for all $v \in C_c^\infty(\Omega)$, so $Lu = f$ a.e.

Remark 14.11. No boundary conditions are needed here (interior).

Remark 14.12. The RHS term $\|u\|_{L^2(\Omega)}$ is necessary: e.g. $u \equiv \text{a constant}$ and $L = -\Delta$, $f = 0$.

Proof. Take a cut-off η with $\eta = 1$ on V and $\eta = 0$ on W^c , where $V \subset\subset W \subset\subset \Omega$. The weak solution satisfies $B(u, v) = \langle f, v \rangle$ for all $v \in C_c^\infty(\Omega)$, so

$$\int_{\Omega} a^{ij} \partial_i u \partial_j v = \int \tilde{f} v = \int [f - \sum b^i \partial_i u - cu] v.$$

Note that since $u \in H^1(\Omega)$, we have $\tilde{f} \in L^2(\Omega)$. Let $|h| > 0$ be small and $k \in \{1, \dots, n\}$. Approximate $\partial_k u$ by defining

$$v := -D_k^{-h}(\eta^2 D_k^h u) \in H_0^1(\Omega)$$

(why H_0^1 ?), where

$$D_k^h u = \frac{u(x + he_k) - u(x)}{h}$$

is the difference quotient. Then

$$\begin{aligned} \int a^{ij} \partial_i u \partial_j v &= - \int a^{ij} \partial_i u \partial_j (D_k^{-h}(\eta^2 D_k^h u)) \\ &= \int D_k^h(a^{ij} \partial_i u) \partial_j (\eta^2 D_k^h u) \end{aligned}$$

(by the summation-by-parts property of difference quotients: $\int_{\Omega} v D_k^{-h} u = - \int_{\Omega} D_k^h v \cdot u$). Expanding,

$$\begin{aligned} &= \int_{\Omega} a^{ij} D_k^h(\partial_i u) \partial_j (\eta^2 D_k^h u) + \int_{\Omega} D_k^h(a^{ij}) \partial_i u \partial_j (\eta^2 D_k^h u) \\ &= \int_{\Omega} [a^{ij} D_k^h(\partial_i u) + D_k^h(a^{ij}) \partial_i u] [\eta^2 D_k^h \partial_j u + 2\eta \partial_j \eta D_k^h u] \\ &= \underbrace{\int_{\Omega} a^{ij} (D_k^h \partial_i u) (D_k^h \partial_j u) \eta^2}_{I_1} + I_2 + I_3 + I_4, \end{aligned}$$

where (writing $\xi_i = D_k^h \partial_i u$, $\xi_j = D_k^h \partial_j u$)

$$\begin{aligned} I_2 &= \int_{\Omega} a^{ij} D_k^h(\partial_i u) \cdot 2\eta \partial_j \eta D_k^h u, \\ I_3 + I_4 &= \int_{\Omega} D_k^h(a^{ij}) \partial_i u (\eta^2 D_k^h \partial_j u + 2\eta \partial_j \eta D_k^h u). \end{aligned}$$

By uniform ellipticity, $I_1 \geq \lambda \int \eta^2 |D_k^h Du|^2$. The rest:

$$\begin{aligned} |I_2| &\leq C \|D_k^h Du \cdot \eta\|_2 \|\nabla \eta D_k^h u\|_2 \leq \varepsilon \int_{\Omega} |D_k^h Du \cdot \eta|^2 + \frac{C}{\varepsilon} \int_{\Omega} |\nabla \eta|^2 |D_k^h u|^2, \\ |I_3| &\leq \varepsilon \int_{\Omega} (|D_k^h Du|^2 \eta^2) + \frac{C}{\varepsilon} \int_{\Omega} |Du|^2 \eta^2, \\ |I_4| &\leq C \int_{\Omega} |Du|^2 \eta^2 + C \int_{\Omega} |D_k^h u|^2 \eta^2. \end{aligned}$$

Therefore

$$\left| \int_{\Omega} a^{ij} \partial_i u \partial_j v \right| \geq \lambda \int_{\Omega} \eta^2 |D_k^h Du|^2 - 2\varepsilon \int_{\Omega} \eta^2 |D_k^h Du|^2 - \frac{C}{\varepsilon} \int_{\Omega} (|Du|^2 + |D_k^h u|^2) \eta^2.$$

Take $\varepsilon = \lambda/4$; also note $\|D_k^h u\|_2 \leq C \|Du\|_2$. Then

$$\int_{\Omega} a^{ij} \partial_i u \partial_j v \, dx \geq \frac{\lambda}{2} \int_{\Omega} \eta^2 |D_k^h Du|^2 - C \int_{\Omega} |Du|^2.$$

Using $\int_{\Omega} a^{ij} \partial_i u \partial_j v = \langle f, v \rangle$,

$$\frac{\lambda}{2} \int_{\Omega} \eta^2 |D_k^h Du|^2 \leq C \int_{\Omega} (f^2 + u^2 + |Du|^2) \, dx.$$

Hence $\int_V |D_k^h Du|^2$ is controlled, so $\sup_h \|D_k^h Du\|_{L^2(V)} < +\infty$ (this bounds $D^2 u$). Therefore there exists $\tilde{v} \in L^2(V)$ such that $D_k^h Du \rightharpoonup \tilde{v}$ weakly in $L^2(V)$. Thus $u \in H^2(V)$ and

$$\|u\|_{H^2(V)} \leq C(\|u\|_{H^1(\Omega)} + \|f\|_{L^2(\Omega)}).$$

(The H^1 term can be simplified to L^2 : testing $v = \eta u$ gives $\|Du\|_{L^2(V)} \leq \|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}$.) $\square$

Corollary 14.13 (Higher interior regularity). *If $a^{ij}, b^i, c \in C^{m+1}(\Omega)$ and $f \in H^m(\Omega)$, then $u \in H_{\text{loc}}^{m+2}(\Omega)$ and*

$$\|u\|_{H^{m+2}(V)} \leq C(\|f\|_{H^m(\Omega)} + \|u\|_{L^2(\Omega)}).$$

15 Boundary regularity and De Giorgi local boundedness

15.1 Interior regularity (sketch)

Assume $a^{ij} \in C^1(\Omega)$, $b^i, c \in L^\infty(\Omega)$, $f \in L^2(\Omega)$. Let u be a weak solution to $Lu = f$ in Ω . Then $u \in H_{\text{loc}}^2$, i.e. $u \in H^2(V)$ for all $V \subset\subset \Omega$, and

$$\|u\|_{H^2(V)} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}).$$

Idea. We show $\|u\|_{H^1(\Omega)} \leq C(\|f\|_{L^2} + \|u\|_{L^2})$. To control $\|u\|_{H^1}$, test with $u \in H_0^1(\Omega)$:

$$B(u, u) = \langle f, u \rangle \implies \int_{\Omega} a^{ij} \partial_i u \partial_j u + b^i \partial_i u \cdot u + cu^2 = \langle f, u \rangle,$$

hence

$$\begin{aligned} \int_{\Omega} a^{ij} \partial_i u \partial_j u &= \int_{\Omega} fu - b^i \partial_i u \cdot u - cu^2 \, dx \\ &\implies \lambda \|\nabla u\|_{L^2}^2 \leq \|f\|_{L^2}^2 + \|u\|_{L^2}^2 + \|b\|_{\infty} (\varepsilon \|\nabla u\|_{L^2}^2 + C(\varepsilon) \|u\|_{L^2}^2) + \|c\|_{\infty} \|u\|_{L^2}^2. \end{aligned}$$

Take $\varepsilon \|b\|_{\infty} = \lambda/2$ to obtain

$$\frac{\lambda}{2} \|\nabla u\|_2^2 \leq \|f\|_2^2 + C \|u\|_2^2 \implies \|u\|_{H^1(\Omega)}^2 \leq C(\|f\|_{L^2(\Omega)}^2 + \|u\|_{L^2(\Omega)}^2).$$

For $\|u\|_{H^2}^2$, try the test function $v = -\partial_k(\eta^2 \partial_k u)$ (formally; otherwise use $D_k^h Du$). Why $v = \partial_k(\eta^2 \partial_k u)$? Because $\partial_k u|_{\partial\Omega} \neq 0$ in general, so a cut-off is needed.

$$\begin{aligned} - \int a^{ij} \partial_i u \partial_j (\partial_k (\eta^2 \partial_k u)) &= \langle f - b^i \partial_i u - cu, v \rangle, \\ \text{LHS} &= \int a^{ij} \partial_{ik} u \partial_j (\eta^2 \partial_k u) + \int \partial_k (a^{ij}) \partial_i u \partial_j (\eta^2 \partial_k u) \\ &= \int a^{ij} \partial_{ik} u (\partial_{jk} u \cdot \eta^2 + 2\eta \partial_j \eta \partial_k u) \\ &\quad + \int \partial_k (a^{ij}) \partial_i u (\partial_{jk} u \cdot \eta^2 + 2\eta \partial_j \eta \partial_k u). \end{aligned}$$

Rearranging,

$$\begin{aligned} \int a^{ij} \partial_{ik} u \partial_{jk} u \cdot \eta^2 &= \langle f - b^i \partial_i u - cu, v \rangle - \int 2a^{ij} \partial_{ik} u \eta \partial_j \eta \partial_k u \\ &\quad - \int \partial_k (a^{ij}) \partial_i u (\partial_{jk} u \cdot \eta^2 + 2\eta \partial_j \eta \partial_k u), \end{aligned}$$

hence

$$\lambda \int |D^2 u|^2 \eta^2 \leq \varepsilon \int |D^2 u|^2 \eta^2 + C \|u\|_{H^1}^2 + C \|f\|_{L^2}^2.$$

Therefore

$$\|D^2 u\|_{L^2(V)} \leq C(\|f\|_{L^2}^2 + \|u\|_{H_0^1}^2) \leq C(\|f\|_{L^2}^2 + \|u\|_{L^2}^2).$$

15.2 Boundary regularity

Theorem 15.1. Let $a^{ij} \in C^1(\bar{\Omega})$, $b^i, c \in L^\infty(\Omega)$, $f \in L^2(\Omega)$. Assume $\partial\Omega \in C^2$ and $u \in H_0^1(\Omega)$ is a weak solution. Then

$$\|u\|_{H^2(\Omega)} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)})$$

(H^2 of the whole Ω).

Sketch.

1) Use boundary straightening to the standard case (straightening): $x = \Psi(y)$, $y = \Phi(x)$. Then $a^{ij} \mapsto \tilde{a}^{kl} \dots$ give new coefficients, still uniformly elliptic.

2) Model case: $\Omega = B(0, 1) \cap \mathbb{R}_+^n$.

“Assume” $u \in H^2$ (otherwise use $D_k^h Du$).

- Tangential derivatives only ($l \neq n$): by a similarity argument,

$$\sum_{(l,m) \neq (n,n)} \|\partial_{lm} u\|_{L^2(\Omega)}^2 \lesssim \|f\|_{L^2(\Omega)}^2 + \|u\|_{L^2(\Omega)}^2.$$

- For $\partial_{nn} u$:

$$a^{nn} \partial_{nn} u = - \sum_{(l,m) \neq (n,n)} a^{lm} \partial_{lm} u + (b^j - \partial_k a^{jk}) \partial_j u + cu - f,$$

with $a^{nn}(x) \geq \lambda > 0$. Hence

$$|\partial_{nn} u| \leq C \sum_{(l,m) \neq (n,n)} |\partial_{lm} u| + |Du| + |u| + |f|,$$

and

$$\|\partial_{nn} u\|_{L^2(\Omega)} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}).$$

Summary. If $Lu = f$ with weak solution $u \in H_0^1(\Omega)$, i.e.

$$\int_{\Omega} a^{ij} \partial_i u \partial_j v + b^i \partial_i u v + cuv = \int_{\Omega} f v \quad \forall v \in H_0^1(\Omega),$$

we have: two existence theorems, and H^2 -regularity. Recall: classical solutions need $u \in C^2(\Omega)$; there remains an $H^2 \rightarrow C^2$ gap.

15.3 Local boundedness

Theorem 15.2. Suppose $a^{ij} \in L^\infty(B_1)$, $c \in L^q(B_1)$ for some $q > n/2$, with $a^{ij} \xi_i \xi_j \geq \lambda |\xi|^2$ and $\|a^{ij}\|_{L^\infty} + \|c\|_q \leq \Lambda$. Suppose $u \in H^1(B_1)$ is a subsolution ($b = 0$):

$$\int_{B_1} a^{ij} \partial_i u \partial_j \varphi + cu\varphi \leq \int_{B_1} f \varphi \quad \forall \varphi \in H_0^1(B_1), \varphi \geq 0.$$

If $f \in L^q(B_1)$ then $u^+ \in L_{\text{loc}}^\infty(B_1)$ and for all $p > 0$,

$$\|u^+\|_{L^\infty(B_\theta)} \leq C \left\{ \frac{1}{(1-\theta)^\beta} \|u^+\|_{L^p(B_1)} + \|f\|_{L^q(B_1)} \right\}.$$

Method 1: De Giorgi. It suffices to consider the $\theta = \frac{1}{2}$ case:

$$\sup_{B_{1/2}} u^+ \leq C(2^{-n/2} \|u^+\|_{L^2(B_1)} + \|f\|_{L^q(B_1)}).$$

Let $v = (u - k)^+$ for $k \geq 0$ (in the end let $k \rightarrow +\infty$), and let $\eta \in C_c^\infty(B_1)$ to be chosen later. Take $\varphi = v\eta^2$ as the test function. Note

$$Dv = \begin{cases} Du & \text{a.e. in } \{u > k\}, \\ 0 & \text{a.e. in } \{u \leq k\}. \end{cases}$$

Then

$$\begin{aligned} \int a^{ij} D_i u D_j \varphi &= \int a^{ij} D_i u D_j (v\eta^2) = \int a^{ij} D_i u (D_j v \eta^2 + 2\eta D_j \eta v) \\ &\geq \lambda \int_{\{u > k\}} |Dv|^2 \eta^2 - 2\Lambda \int_{\{u > k\}} |Du| |D\eta| v \eta \\ &\geq \frac{\lambda}{2} \int_{\{u > k\}} |Dv|^2 \eta^2 - \frac{2\Lambda^2}{\lambda} \int |D\eta|^2 v^2. \end{aligned}$$

By the subsolution inequality,

$$\int a^{ij} D_i u D_j (v\eta^2) + cu v \eta^2 \leq \int f v \eta^2,$$

and (using $u \leq v + k$)

$$\int |Du|^2 \eta^2 \leq C \left\{ \int v^2 |D\eta|^2 + |c| v^2 \eta^2 + |c| k^2 \eta^2 + |f| v \eta^2 dx \right\}.$$

Hence

$$\int |D(v\eta)|^2 \leq C \left\{ \int v^2 |D\eta|^2 + \int |c| v^2 \eta^2 + k^2 \int |c| \eta^2 + \int |f| v \eta^2 \right\}. \quad (\star)$$

Recall by Sobolev embedding $W^{1,2} \hookrightarrow L^{2^*}$ with $2^* = \frac{2n}{n-2}$. Estimate the RHS of $(\star)$ by Hölder: with $\frac{1}{q} + \frac{1}{2^*} + \frac{1}{q'} = 1$,

$$\begin{aligned} \int |f|v\eta^2 &= \int_{\{v\eta \neq 0\}} |f|v\eta \cdot \eta \\ &\leq \|f\|_{L^q} \|v\eta\|_{L^{2^*}} \left(\int_{\{v\eta \neq 0\}} \eta^{q'} \right)^{1/q'} \\ &\leq \|f\|_{L^q} \|v\eta\|_{L^{2^*}} |\{v\eta \neq 0\}|^{1/2+1/n-1/q} \\ &\leq \varepsilon \|v\eta\|_{L^{2^*}}^2 + C(\varepsilon) \|f\|_{L^q}^2 |\{v\eta \neq 0\}|^{1+2/n-2/q} \\ &\leq \delta \|D(v\eta)\|_{L^2}^2 + C(\delta) \|f\|_{L^q}^2 |\{v\eta \neq 0\}|^{1-1/q+2/n}. \end{aligned}$$

Note $1 + 2/n - 2/q > 1 - 1/q$ as long as $q > n/2$. Similarly

$$\begin{aligned} \int |c|v^2\eta^2 &\leq \|c\|_{L^q} \|v\eta\|_{L^{2^*}}^2 |\{v\eta \neq 0\}|^{2/n-1/q} \\ &\leq \|c\|_{L^q} \|D(v\eta)\|_{L^2}^2 |\{v\eta \neq 0\}|^{2/n-1/q}, \end{aligned}$$

and

$$\int_{\{v\eta \neq 0\}} k^2 |c|\eta^2 \leq k^2 \|c\|_{L^q} |\{v\eta \neq 0\}|^{1-1/q}.$$

Combining (and using $2/n > 1/q$),

$$\begin{aligned} \|D(v\eta)\|_{L^2}^2 &\leq C \left\{ \int v^2 |D\eta|^2 + (k^2 + \|f\|_{L^q}^2) |\{v\eta \neq 0\}|^{1-1/q} \right. \\ &\quad \left. + \|D(v\eta)\|_{L^2}^2 |\{v\eta \neq 0\}|^{2/n-1/q} \right\}. \end{aligned}$$

Now if $|\{v\eta \neq 0\}| < \varepsilon_0 \ll 1$, then $|\{v\eta \neq 0\}|^{2/n-1/q} < \varepsilon_0 \ll 1$, so

$$\|D(v\eta)\|_{L^2}^2 \leq C \left\{ \int v^2 |D\eta|^2 + (k^2 + \|f\|_{L^q}^2) |\{v\eta \neq 0\}|^{1-1/q} \right\}.$$

Again by Sobolev embedding and Hölder,

$$\begin{aligned} \int (v\eta)^2 &\leq \left(\int (v\eta)^{2^*} \right)^{2/2^*} |\{v\eta \neq 0\}|^{1-2/2^*} = \|v\eta\|_{L^{2^*}}^2 |\{v\eta \neq 0\}|^{2/n} \\ &\leq C \|D(v\eta)\|_{L^2}^2 |\{v\eta \neq 0\}|^{2/n} \\ &\leq C \left(\int v^2 |D\eta|^2 dx |\{v\eta \neq 0\}|^{2/n} + (k^2 + \|f\|_{L^q}^2) |\{v\eta \neq 0\}|^{1+2/n-1/q} \right). \end{aligned}$$

Thus

$$\int (v\eta)^2 \leq C \left[\int v^2 |D\eta|^2 |\{v\eta \neq 0\}|^{2/n} + (k^2 + \|f\|_{L^q}^2) |\{v\eta \neq 0\}|^{1+\varepsilon} \right]$$

for $\varepsilon = 2/n - 1/q$, if $|\{v\eta \neq 0\}|$ is small.

Now choose $\eta \in C_0^\infty(B_R)$ with $\eta = 1$ on B_r , $\eta = 0$ on B_R^c , and $|D\eta| \leq \frac{2}{R-r}$. Set $A(k, r) :=$

$\{x \in B_r : u \geq k\}$ (so $\{v\eta \neq 0\} \subseteq A(k, R)$). Then for $0 < r < R < 1$ and $k > 0$,

$$\begin{aligned} \int_{A(k,r)} (u-k)^2 &\leq \int v^2 \eta^2 \\ &\leq C \left[\frac{1}{(R-r)^2} \int v^2 dx |\{v\eta \neq 0\}|^\varepsilon + (k^2 + \|f\|_q^2) |\{v\eta \neq 0\}|^{1+\varepsilon} \right] \\ &\leq C \left[\frac{1}{(R-r)^2} \int_{A(k,R)} (u-k)^2 dx |A(k, R)|^\varepsilon + (k^2 + \|f\|_q^2) |A(k, R)|^{1+\varepsilon} \right], \end{aligned}$$

provided $|A(k, R)|$ is small. Also

$$|A(k, R)| \leq \frac{1}{k} \int_{A(k,R)} u^+ \leq \frac{1}{k} \|u^+\|_{L^2},$$

so $|A(k, R)|$ is small if $k \geq k_0$ for some k_0 .

Take any $h > k \geq k_0$ and $0 < r < R$ such that $A(k, r) \supset A(h, r)$ and $\int_{A(h,r)} (u-h)^2 \leq \int_{A(k,r)} (u-k)^2$, and

$$|A(h, r)| \leq |\{u-k > h-k\}| \leq \frac{1}{(h-k)^2} \int_{A(k,r)} (u-k)^2.$$

Therefore

$$\begin{aligned} \int_{A(h,r)} (u-h)^2 &\leq C |A(h, R)|^\varepsilon \left[\frac{1}{(R-r)^2} \int_{A(h,R)} (u-h)^2 + (h^2 + \|f\|_{L^q}^2) |A(h, R)| \right] \\ &\leq \frac{C}{(h-k)^{2\varepsilon}} \left(\int_{A(k,R)} (u-k)^2 \right)^\varepsilon \left[\frac{1}{(R-r)^2} \int_{A(k,R)} (u-k)^2 \right. \\ &\quad \left. + \frac{h^2 + \|f\|_{L^q}^2}{(h-k)^2} \int_{A(k,R)} (u-k)^2 \right] \\ &\leq \frac{C}{(h-k)^{2\varepsilon}} \left(\int_{A(k,R)} (u-k)^2 \right)^{1+\varepsilon} \left[\frac{1}{(R-r)^2} + \frac{h^2 + \|f\|_{L^q}^2}{(h-k)^2} \right]. \end{aligned}$$

Hence

$$\|(u-h)^+\|_{L^2(B_r)} \leq C \left(\frac{1}{R-r} + \frac{h + \|f\|_{L^q}}{h-k} \right) \frac{\|(u-k)^+\|_{L^2(B_R)}^{1+\varepsilon}}{(h-k)^\varepsilon}.$$

Set $y(k, r) = \|(u-k)^+\|_{L^2(B_r)}$, $k_\ell = k_0 + k(1 - \frac{1}{2^\ell})$, $r_\ell = \frac{1}{2} + \frac{1}{2^{\ell+1}}$, so $k_\ell - k_{\ell-1} = \frac{k}{2^\ell}$ and $r_{\ell-1} - r_\ell = \frac{1}{2^{\ell+1}}$. Setting $R = r_{\ell-1}$, $r = r_\ell$,

$$y(k_\ell, r_\ell) \leq C \left[2^{\ell+1} + \frac{2^\ell}{k} \left(k_0 + k(1 - \frac{1}{2^\ell}) + \|f\|_{L^q} \right) \right] \cdot \frac{2^{\ell\varepsilon}}{k^\varepsilon} (y(k_{\ell-1}, r_{\ell-1}))^{1+\varepsilon}. \quad (**)$$

Inductively one shows

$$y(k_\ell, r_\ell) \leq \frac{y(k_0, r_0)}{\gamma^\ell} \quad \text{for some } \gamma > 1.$$

Then

$$(y(k_{\ell-1}, r_{\ell-1}))^{1+\varepsilon} \leq \left(\frac{y(k_0, r_0)}{\gamma^{\ell-1}} \right)^{1+\varepsilon} = \frac{y(k_0, r_0)^\varepsilon}{\gamma^{(\ell-1)(1+\varepsilon)}} \cdot \frac{y(k_0, r_0)}{\gamma^{\ell-1}}.$$

By (**),

$$y(k_\ell, r_\ell) \leq \frac{C\gamma^{1+\varepsilon}(k_0 + k + \|f\|_q)}{k^{1+\varepsilon}} 2^{\ell\varepsilon} \cdot \frac{y(k_{\ell-1}, r_{\ell-1})^\varepsilon}{\gamma^{\varepsilon\ell}} \cdot \frac{y(k_0, r_0)}{\gamma^\ell}.$$

We want

$$C\gamma^{1+\varepsilon} \left(\frac{k_0 + k + \|f\|_q}{k^{1+\varepsilon}} \right) \left(2^{\ell(1+\varepsilon)} \frac{y(k_0, r_0)^\varepsilon}{\gamma^{\varepsilon\ell}} \right) < 1.$$

Take $\gamma^\varepsilon = 2^{1+\varepsilon}$ ($\gamma > 1$). Then it suffices that

$$C\gamma^{1+\varepsilon} \left(\frac{y(k_0, r_0)}{k} \right)^\varepsilon \frac{k_0 + k + \|f\|_q}{k} < 1,$$

which holds by taking k large. Now

$$y(k_\ell, r_\ell) \leq \frac{y(k_0, r_0)}{\gamma^\ell} \longrightarrow 0 \quad \text{as } \ell \rightarrow \infty,$$

and $k_\ell \rightarrow k_0 + k$, $r_\ell \rightarrow \frac{1}{2}$, i.e. $y(k_0 + k, \frac{1}{2}) = 0$. Thus

$$\|(u - (k_0 + k))^+\|_{L^2(B_{1/2})} = 0 \implies u^+ \leq k_0 + k \quad \text{in } B_{1/2},$$

so

$$\sup_{B_{1/2}} u^+ \leq C\{k_0 + \|f\|_{L^q} + y(k_0, r_0)\},$$

where $k_0 + \|f\|_{L^q}$ and $y(k_0, r_0)$ are both controlled by $C\|u^+\|_{L^2(B_1)}$. $\square$

16 Nash–Moser iteration, comparison principles, and moving planes

16.1 Method 2 (Nash–Moser)

Assume $b^i = c = f = 0$. The idea is to show the reverse Hölder inequality

$$\|u\|_{L^p(B_{r_1})} \leq C\|u\|_{L^{p_0}(B_{r_2})} \quad \text{for } p > p_0, \quad r_1 < r_2$$

(issue: $C \sim \frac{1}{r_2 - r_1}$).

Step 1.

For $m > 0$ define

$$\bar{u}_m = \begin{cases} u^+ & \text{if } u < m, \\ m & \text{if } u \geq m \end{cases}$$

(will let $m \rightarrow +\infty$ later). Then $D\bar{u}_m = 0$ in $\{u < 0\}$ (where $u^+ = 0$) and in $\{u > m\}$ (where $\bar{u}_m \equiv m$ is constant). Also $\bar{u}_m \leq u^+$. Set the test function

$$\varphi = \eta^2 (\bar{u}_m^\beta u^+) \in H_0^1(B_1) \quad \text{for some } \beta \geq 0,$$

with $\eta \geq 0 \in C_c^\infty(B_1)$. Then

$$D\varphi = 2\eta D\eta (\bar{u}_m^\beta u^+) + \eta^2 (\beta \bar{u}_m^{\beta-1} D\bar{u}_m) u^+ + \eta^2 \bar{u}_m^\beta Du^+.$$

Note $D\bar{u}_m \cdot u^+ \bar{u}_m^{\beta-1} = D\bar{u}_m \bar{u}_m^\beta$ because $u^+ = \bar{u}_m$ whenever $D\bar{u}_m \neq 0$ (i.e. $0 \leq u < m$). Hence

$$D\varphi = 2\eta D\eta (\bar{u}_m^\beta u^+) + \eta^2 \bar{u}_m^\beta (\beta D\bar{u}_m + Du^+).$$

Now $\varphi = 0 = D\varphi$ in $\{u \leq 0\}$ (where $u^+ = \bar{u}_m \equiv 0$), so we only integrate in $\{u > 0\}$ (and $D_i u = D_i u^+$ there):

$$\int a^{ij} D_i u D_j \varphi = \int a^{ij} D_i u \left(2\eta D_j \eta \bar{u}_m^\beta u^+ + \eta^2 \bar{u}_m^\beta (\beta D_j \bar{u}_m + D_j u^+) \right).$$

Note

$$D_i u^+ D_j \bar{u}_m = \begin{cases} 0 & \text{if } u \geq m, \\ D_i \bar{u}_m D_j \bar{u}_m & \text{if } u < m. \end{cases}$$

Thus

$$\int a^{ij} D_i \bar{u} \eta^2 \bar{u}_m^\beta \beta D_j \bar{u}_m \geq \lambda \beta \int \eta^2 \bar{u}_m^\beta |D \bar{u}_m|^2,$$

and

$$\int a^{ij} D_i u^+ \eta^2 \bar{u}_m^\beta D_j u^+ \geq \lambda \int \eta^2 \bar{u}_m^\beta |Du^+|^2.$$

For the remaining terms (writing $\bar{u} = u^+$),

$$\begin{aligned} \int a^{ij} D_i u^+ D_j \eta (\bar{u}_m^\beta u^+) \eta &\leq \Lambda \int |Du^+| |D\eta| \bar{u}_m^\beta u^+ \eta \\ &\leq \frac{\lambda}{2} \int \eta^2 \bar{u}_m^\beta |Du^+|^2 + \frac{2\Lambda^2}{\lambda} \int |D\eta|^2 \bar{u}_m^\beta (u^+)^2. \end{aligned}$$

Hence

$$\beta \int \eta^2 \bar{u}_m^\beta |D \bar{u}_m|^2 + \int \eta^2 \bar{u}_m^\beta |Du^+|^2 \leq C \int |D\eta|^2 \bar{u}_m^\beta (u^+)^2.$$

Set $w = \bar{u}_m^{\beta/2} u^+$; then

$$|Dw|^2 \leq (1 + \beta) \{ \beta \bar{u}_m^\beta |D \bar{u}_m|^2 + \bar{u}_m^\beta |Du^+|^2 \},$$

so

$$\int |Dw|^2 \eta^2 \leq C(1 + \beta) \int w^2 |D\eta|^2 \quad \text{or} \quad \int |D(w\eta)|^2 \leq C(1 + \beta) \int w^2 |D\eta|^2.$$

Step 2.

By the Sobolev inequality,

$$\left(\int |\eta w|^{2^*} \right)^{2/2^*} \leq C \int |D(\eta w)|^2 \leq C(1 + \beta) \int w^2 |D\eta|^2.$$

Choose $\eta = 1$ on B_r , $\eta = 0$ on B_R^c , with $|D\eta| \leq \frac{2}{R-r}$. Then

$$\left(\int_{B_r} |w|^{2^*} \right)^{2/2^*} \leq \frac{C(1 + \beta)}{(R-r)^2} \int_{B_R} w^2.$$

By definition $w = \bar{u}_m^{\beta/2} u^+$, so

$$\left(\int_{B_r} (u^+)^{2^*} \bar{u}_m^{\beta 2^*/2} \right)^{2/2^*} \leq \frac{C(1 + \beta)}{(R-r)^2} \int_{B_R} (u^+)^2 \bar{u}_m^\beta.$$

Since $\bar{u}_m \leq u^+$, set $\gamma = 2 + \beta$:

$$\left(\int_{B_r} \bar{u}_m^{(\gamma/2) 2^*} \right)^{2/2^*} \leq \frac{C(\gamma - 1)}{(R-r)^2} \int_{B_R} (u^+)^{\gamma}.$$

Let $m \rightarrow \infty$ and define $\chi = 2^*/2$:

$$\left(\int_{B_r} \bar{u}^{\gamma\chi} \right)^{1/\chi} \leq \frac{C(\gamma - 1)}{(R-r)^2} \int_{B_R} (u^+)^{\gamma}.$$

Therefore

$$\|\bar{u}\|_{L^{\gamma\chi}(B_r)} \leq \left(C \frac{\gamma-1}{(R-r)^2} \right)^{1/\gamma} \|\bar{u}\|_{L^\gamma(B_R)}.$$

Step 3.

Iterate beginning with $\gamma = 2$:

$$\|\bar{u}\|_{L^{2\chi}(B_r)} \leq \left(C \frac{1}{(R-r)^2} \right)^{1/2} \|\bar{u}\|_{L^2(B_R)}.$$

Also iterate the radii:

$$\gamma_i = 2\chi^i \quad (= 2, 2\chi, 2\chi^2, \dots), \quad r_i = \frac{1}{2} + \frac{1}{2^{i+1}} \quad (r_i \rightarrow \frac{1}{2}),$$

with $r_{i-1} - r_i = \frac{1}{2^{i+1}}$. Then

$$\|\bar{u}\|_{L^{\gamma_i}(B_{r_i})} \leq \left(C \frac{\chi^i}{(r_{i-1} - r_i)^2} \right)^{1/\gamma_i} \|\bar{u}\|_{L^{\gamma_{i-1}}(B_{r_{i-1}})},$$

and the prefactor is bounded by a factor of order C^{i/χ^i} . Hence

$$\|\bar{u}\|_{L^{\gamma_i}(B_{r_i})} \leq C^{\sum i/\chi^i} \|\bar{u}\|_{L^2(B_1)},$$

where the sum is bounded since $\chi = 2^*/2 = n/(n-2) > 1$. Letting $i \rightarrow +\infty$,

$$\sup_{B_{1/2}} \bar{u} \leq C \|\bar{u}\|_{L^2(B_1)}, \quad \text{or} \quad \sup_{B_{1/2}} u^+ \leq C \|u^+\|_{L^2(B_1)}.$$

16.2 Comparison principle for elliptic equations

Let Ω be open and bounded, and

$$Lu := a_{ij}D_{ij}u(x) + b_iD_iu(x) + cu(x) \geq 0.$$

- If $c \leq 0$, then L is still a “negative” operator and

$$\max_{\bar{\Omega}} u^+ = \max_{\partial\Omega} u^+.$$

(If $u(x_0) = \max_{\bar{\Omega}} u^+$ with $x_0 \in \text{int}(\Omega)$, then $Lu(x_0) \leq 0$, using $u(x_0) \geq 0$ and $c \leq 0$.)

- If $\|c\|_\infty \leq M$, require $u \leq 0$. Then

$$(L - M)u = a_{ij}D_{ij}u + b_iD_iu + (c - M)u = Lu - Mu \geq -Mu \geq 0$$

(since $Lu \geq 0$, $u \leq 0$, and $c - M \leq 0$). If $u \leq 0$ then $u \equiv 0$ or $u < 0$.

Theorem 16.1 (Narrow domain maximum principle). *Assume $\bar{\Omega} \subset \{0 < x_1 < d\}$ for $d \ll 1$, and*

$$\begin{cases} \Delta u + c(x)u \geq 0 & \text{in } \Omega, \\ u \leq 0 & \text{on } \partial\Omega, \end{cases} \quad (30)$$

where $|c(x)| \leq M$. Then $u < 0$ or $u \equiv 0$ in Ω .

Idea. In this domain, consider $w = \sin(\pi x_1/d)$. Then

$$\begin{cases} \Delta w + (\pi/d)^2 w = 0 & \text{in } \Omega, \\ w > 0 & \text{on } \overline{\Omega}, \end{cases}$$

and we only need $\Delta w + c(x)w \leq 0$, which holds since $(\pi/d)^2 \gg 1$. $\square$

Lemma 16.2. (Ω need not be narrow.) Suppose $Lu \geq 0$ in Ω and

$$\begin{cases} Lw \leq 0 & \text{in } \Omega, \\ w > 0 & \text{on } \overline{\Omega}, \end{cases}$$

where $L := \Delta + c$ with $|c| \leq M$. Then

$$\max_{\overline{\Omega}} \frac{u^+}{w} = \max_{\partial\Omega} \left(\frac{u}{w} \right)^+.$$

Proof. Write $v = u/w$ so $u = vw$. Then

$$\Delta u = \Delta v w + v \Delta w + 2Dv \cdot Dw,$$

hence

$$\Delta v w + v \Delta w + 2Dv \cdot Dw + cvw \geq 0,$$

i.e.

$$\Delta v + \frac{v}{w}(\Delta w + cw) + 2\frac{Dv}{w} \cdot Dw \geq 0.$$

Define

$$\tilde{L}v := \Delta v + 2Dv \cdot \frac{Dw}{w} + \tilde{c}v \geq 0 \quad \text{in } \Omega,$$

where $\tilde{c}(x) := \frac{\Delta w + cw}{w} \leq 0$, and $v^+ = u^+/w$. Assume $v^+(x_0) = \max_{\overline{\Omega}} v^+$ with $x_0 \in \Omega$ and $v(x_0) > 0$. Then (after an “ ε -trick”)

$$\Delta v + 2Dv \cdot \frac{Dw}{w} + \tilde{c}v \leq 0 \quad \text{at } x_0,$$

forcing $x_0 \in \partial\Omega$. $\square$

Back to the narrow-domain case:

$$\max_{\overline{\Omega}} \left(\frac{u}{w} \right)^+ = 0,$$

and by Hopf’s lemma, $u < 0$ in Ω or $u \equiv 0$.

16.3 Moving planes

Theorem 16.3. Suppose $u \in C(\overline{B_1}) \cap C^2(B_1)$ and $u > 0$ in B_1 , with

$$\begin{cases} \Delta u + f(u) = 0 & \text{in } B_1, \\ u = 0 & \text{on } \partial B_1, \end{cases} \quad (31)$$

where f is locally Lipschitz. Then u is radially symmetric in B_1 and $\frac{\partial u}{\partial r}(x) < 0$ for all $x \neq 0$.

Remark 16.4. In fact the symmetry claim only requires Ω bounded and convex in the x_1 -direction and symmetric with respect to $\{x_1 = 0\}$. The proof uses the narrow-domain maximum principle and the small-volume maximum principle.

Exercise 16.5. Show that the conclusion that u is radially symmetric is false without the assumption that u is positive.

Lemma 16.6. Let Ω be a bounded domain that is convex in the x_1 -direction and symmetric with respect to the plane $\{x_1 = 0\}$. Let $u \in C(\bar{\Omega}) \cap C^2(\Omega)$ be a positive solution of

$$\begin{cases} \Delta u + f(u) = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (32)$$

with f locally Lipschitz in $\mathbb{R}$. Then u is symmetric with respect to x_1 and

$$\frac{\partial u}{\partial x_1}(x) < 0 \quad \text{for any } x \in \Omega \text{ with } x_1 > 0.$$

Proof of the ball theorem. The ball theorem follows immediately from the lemma: u is decreasing in any given radial direction (the unit ball is symmetric with respect to any plane through the origin), and u is invariant under reflection across any such hyperplane, hence radially symmetric. $\square$

Proof of the lemma. Use coordinates $x = (x_1, y) \in \Omega$ with $y \in \mathbb{R}^{n-1}$. We prove

$$u(x_1, y) < u(x_1^*, y) \quad \text{for all } x_1 > 0 \text{ and } -x_1 < x_1^* < x_1. \quad (4.3)$$

This implies monotonicity in x_1 for $x_1 > 0$. Letting $x_1^* \rightarrow -x_1$ gives $u(x_1, y) \leq u(-x_1, y)$ for $x_1 > 0$; reversing the direction yields reflection symmetry $u(x_1, y) = u(-x_1, y)$.

To prove (4.3), given any $\lambda \in (0, a)$ with $a = \sup_{\Omega} x_1$, take the moving plane $T_\lambda = \{x_1 = \lambda\}$ and

$$\Sigma_\lambda = \{x \in \Omega : x_1 > \lambda\}.$$

Let $x_\lambda = (2\lambda - x_1, x_2, \dots, x_n)$ be the reflection of x across T_λ , and set

$$w_\lambda(x) = u(x) - u(x_\lambda) \quad \text{for } x \in \Sigma_\lambda.$$

By the mean value theorem,

$$\Delta w_\lambda = f(u(x_\lambda)) - f(u(x)) = -c(x, \lambda) w_\lambda \quad \text{in } \Sigma_\lambda,$$

where $|c(x, \lambda)| \leq \text{Lip}(f)$ (a recurring trick: the difference of two solutions of a semilinear equation satisfies a linear equation with bounded but unknown coefficient). The boundary $\partial\Sigma_\lambda$ consists of a piece of $\partial\Omega$ (where $w_\lambda = -u(x_\lambda) < 0$) and a part of T_λ (where $x = x_\lambda$ and $w_\lambda = 0$). Thus

$$\begin{cases} \Delta w_\lambda + c(x, \lambda) w_\lambda = 0 & \text{in } \Sigma_\lambda, \\ w_\lambda \leq 0 \text{ and } w_\lambda \not\equiv 0 & \text{on } \partial\Sigma_\lambda. \end{cases} \quad (4.5)$$

By the narrow-domain maximum principle, $w_\lambda < 0$ in Σ_λ for λ close to a . Moreover, w_λ attains its maximum 0 over $\bar{\Sigma}_\lambda$ along T_λ , so by Hopf's lemma

$$\left. \frac{\partial w_\lambda}{\partial x_1} \right|_{x_1=\lambda} = 2 \left. \frac{\partial u}{\partial x_1} \right|_{x_1=\lambda} < 0.$$

Hence $\partial u / \partial x_1 < 0$ for $x_1 > 0$, once we know $w_\lambda < 0$ in Σ_λ for all $\lambda \in (0, a)$. Equivalently,

$$u(x_1, x') < u(2\lambda - x_1, x') \quad \text{for all } \lambda \in (0, x_1).$$

Since $c(x, \lambda)$ has no fixed sign, the usual maximum principle fails. There exists $\delta_c > 0$ (depending only on $\|c\|_{L^\infty}$) such that the narrow-domain maximum principle holds for $Lu = \Delta u + c(x)u$ whenever the width in the x_1 -direction is less than δ_c . Thus for $a - \delta_c < \lambda < a$, the domain Σ_λ is narrow enough that $w_\lambda < 0$ in Σ_λ (using $w_\lambda \leq 0$ and $w_\lambda \not\equiv 0$ on $\partial\Sigma_\lambda$).

Now decrease λ (move T_λ left). Let (λ_0, a) be the largest interval such that $w_\lambda < 0$ in Σ_λ for all $\lambda \in (\lambda_0, a)$. If $\lambda_0 = 0$ we are done. Assume for contradiction that $\lambda_0 > 0$. By continuity, $w_{\lambda_0} \leq 0$ in Σ_{λ_0} , and $w_{\lambda_0} \not\equiv 0$ on $\partial\Sigma_{\lambda_0}$. The strong maximum principle gives

$$w_{\lambda_0} < 0 \quad \text{in } \Sigma_{\lambda_0}. \quad (4.7)$$

We claim that for sufficiently small $\varepsilon > 0$,

$$w_{\lambda_0-\varepsilon} < 0 \quad \text{in } \Sigma_{\lambda_0-\varepsilon}, \quad (4.8)$$

contradicting maximality of λ_0 unless $\lambda_0 = 0$.

Choose a simply connected closed set $K \subset \Sigma_{\lambda_0}$ with smooth boundary, nearly all of Σ_{λ_0} in the sense that $|\Sigma_{\lambda_0} \setminus K| < \delta/2$. From (4.7) there exists $\eta > 0$ with $w_{\lambda_0} \leq -\eta$ on K . By continuity,

$$w_{\lambda_0-\varepsilon} < -\frac{\eta}{2} < 0 \quad \text{on } K. \quad (4.9)$$

On the boundary of $\Sigma_{\lambda_0-\varepsilon} \setminus K$ we have $w_{\lambda_0-\varepsilon} \leq 0$ on $\partial\Sigma_{\lambda_0-\varepsilon}$ and $w_{\lambda_0-\varepsilon} < 0$ on ∂K , hence $w_{\lambda_0-\varepsilon} \leq 0$ on $\partial(\Sigma_{\lambda_0-\varepsilon} \setminus K)$. For ε small, $|\Sigma_{\lambda_0-\varepsilon} \setminus K| < \delta$. Choosing δ small enough (depending only on $\|c\|_{L^\infty(\Omega)}$), the maximum principle for domains of small volume yields $w_{\lambda_0-\varepsilon} \leq 0$ in $\Sigma_{\lambda_0-\varepsilon} \setminus K$; the strong maximum principle upgrades this to < 0 . Combined with (4.9) we obtain (4.8), a contradiction. Thus $\lambda_0 = 0$, completing the proof. $\square$

Part V

Numerical methods

17 Fourier spectral methods and foundations

17.1 Course goals and model problem

Homework: once every two weeks, written and/or programming. xycheng@fudan.edu.cn.

Goals for weeks 1–8: solve several classes of PDEs with Fourier spectral methods.

Model problem: convection–diffusion

Convection–diffusion equation

$$\partial_t u + \mathbf{V} \cdot \nabla u = \nu \Delta u + f, \quad \nu > 0, \quad (33)$$

where $\nabla = (\partial_x, \partial_y, \partial_z, \dots)$, $\Delta = \partial_{xx} + \partial_{yy} + \partial_{zz} + \dots$, $u(t, \mathbf{x}) \in \mathbb{R}$. Typical applications include heat conduction, pollutant diffusion, probability densities in statistical mechanics, and option pricing in finance.

Remark 17.1 (Convergence needs analytical information). For the heat equation (parabolic)

$$\begin{cases} \partial_t u - \Delta u = 0 & \text{in } \Omega, \\ u|_{\partial\Omega} = 0, \\ u(0, \mathbf{x}) = u_0(\mathbf{x}), \end{cases}$$

time discretization (implicit Euler)

$$\frac{u^{n+1} - u^n}{\tau} - \Delta u^{n+1} = 0, \quad \tau = \text{time-step size.}$$

On $[t_n, t_{n+1}]$, $t_n = n\tau$, if $u(t_n) = u^n$, integrating yields

$$u(t_{n+1}) - u(t_n) = \int_{t_n}^{t_{n+1}} \Delta u(s) ds.$$

Hence

$$\begin{aligned} u(t_{n+1}) &= u^n + \int_{t_n}^{t_{n+1}} \Delta u^{n+1} ds + \int_{t_n}^{t_{n+1}} (\Delta u(s) - \Delta u^{n+1}) ds \\ &= u^{n+1} + \int_{t_n}^{t_{n+1}} (\Delta u(s) - \Delta u^{n+1}) ds, \end{aligned}$$

so the local error

$$u(t_{n+1}) - u^{n+1} = \int_{t_n}^{t_{n+1}} (\Delta u(s) - \Delta u^{n+1}) ds$$

depends on Δu of the analytic solution u (regularity).

17.2 Hilbert space setup for Fourier spectral methods

Definition 17.2 (Inner product / Hilbert space). A map $(\cdot, \cdot) : H \times H \rightarrow \mathbb{R}$ is an inner product if

- (i) $(u, v) = (v, u)$ for all $u, v \in H$;
- (ii) $u \mapsto (u, v)$ is linear for each fixed v ;
- (iii) $(u, u) \geq 0$ for all $u \in H$;
- (iv) $(u, u) = 0$ iff $u = 0$.

Define $\|u\| := \sqrt{(u, u)}$. A complete inner-product space is a *Hilbert space*.

Lemma 17.3 (Cauchy–Schwarz). $|(u, v)| \leq \|u\| \|v\|$.

Proof. For $\varepsilon > 0$,

$$0 \leq \|u \pm \varepsilon v\|^2 = \|u\|^2 \pm 2\varepsilon(u, v) + \varepsilon^2\|v\|^2,$$

hence

$$\pm(u, v) \leq \frac{1}{2\varepsilon}\|u\|^2 + \frac{\varepsilon}{2}\|v\|^2.$$

Take $\varepsilon = \|u\|/\|v\|$ (the case $\|u\| = 0$ or $\|v\| = 0$ is trivial). □

Example 17.4 ($L^2(\Omega)$). $(u, v) := \int_{\Omega} uv \, dx$ makes $L^2(\Omega)$ a Hilbert space.

Definition 17.5 (Orthogonality / ONB). (i) $u, v \in H$ are orthogonal if $(u, v) = 0$.

- (ii) $\{w_k\}_{k=1}^{\infty} \subset H$ is orthonormal if $(w_k, w_l) = \delta_{kl} = \begin{cases} 0 & k \neq l \\ 1 & k = l \end{cases}$.

- (iii) If $\{w_k\}$ is an orthonormal basis (ONB) and $u \in H$, then

$$u = \sum_{k=1}^{\infty} (u, w_k) w_k, \quad \|u\|^2 = \sum_{k=1}^{\infty} (u, w_k)^2.$$

ONB for $L^2([-\pi, \pi])$

Exercise 17.6. Find an ONB for $L^2([-\pi, \pi])$. (The answer is not unique!)

- **Sol 1.** $\{1, x, x^2, \dots\}$ is a basis but not orthonormal $\implies$ use Gram–Schmidt.

- **Sol 2.** The trigonometric system

$$I = \left\{ \frac{1}{\sqrt{2\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \dots, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \dots \right\}$$

is orthonormal. For $n \neq 0$,

$$\int_{-\pi}^{\pi} \frac{\sin^2 nx}{\pi} dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1 - \cos 2nx}{2} dx = 1,$$

and for $n \neq m$,

$$\int_{-\pi}^{\pi} \sin nx \sin mx dx = \int_{-\pi}^{\pi} (\cos((n-m)x) - \cos((n+m)x)) dx = 0.$$

Similarly for cosines, so I is orthonormal.

Remark 17.7. $\dim L^2(-\pi, \pi) = \infty$. Completeness of I as an ONB is the classical Fourier completeness theorem.

ONB for $L^2(\mathbb{R})$?

Polynomials $1, x, \dots$ and $\sin nx, \cos nx$ are *not* in $L^2(\mathbb{R})$, so the previous constructions fail. One idea is to localize Fourier modes, e.g. $\chi_{[n, n+1]}(x) e^{2\pi i k(x-n)}$ (and analogues).

Linear operators and matrices

Definition 17.8. A map $A : X \rightarrow Y$ between Banach spaces is a *linear operator* if

$$A[\lambda u + \mu v] = \lambda Au + \mu Av \quad \forall u, v \in X, \lambda, \mu \in \mathbb{R}.$$

Range $R(A) := \{Au : u \in X\}$, null space $N(A) := \{u : Au = 0\}$. A is bounded if

$$\|A\| := \sup_{u \neq 0} \frac{\|Au\|}{\|u\|} = \sup_{\|u\|=1} \|Au\| < \infty.$$

Remark 17.9. If X, Y are finite-dimensional, $X \simeq \mathbb{R}^n, Y \simeq \mathbb{R}^m$, then $A \in M_{m \times n}(\mathbb{R})$: discretization turns PDE operators into matrices.

Definition 17.10 (Adjoint / symmetric). If $A : H \rightarrow H$ is bounded linear, its adjoint A^* satisfies $(Au, v) = (u, A^*v)$ for all $u, v \in H$. A is symmetric if $A^* = A$.

Example 17.11. $\frac{d}{dx}$ on $C^\infty([a, b])$ is linear but *not bounded*: $\frac{d}{dx} \sin nx = n \cos nx$, so $\left\| \frac{d}{dx} \sin nx \right\|_C = n \rightarrow \infty$ as $n \rightarrow \infty$.

17.3 Weak convergence (Fourier-relevant examples)

Definition 17.12 (Weak convergence). A sequence $\{u_k\} \subset X$ converges weakly to u , written $u_k \rightharpoonup u$, if

$$\langle u^*, u_k \rangle \rightarrow \langle u^*, u \rangle \quad \forall u^* \in X^*.$$

Example 17.13. In $L^p(\Omega)$, $f_k \rightharpoonup f$ means

$$\int_{\Omega} g f_k dx \rightarrow \int_{\Omega} g f dx \quad \forall g \in L^{p'}(\Omega),$$

where $\frac{1}{p} + \frac{1}{p'} = 1$. (L^p is reflexive for $1 < p < \infty$.)

Example 17.14 ($\sin kx \rightharpoonup 0$). Let $f_k = \sin kx$ in $L^2([-\pi, \pi])$. Then $\sin kx \rightharpoonup 0$ as $k \rightarrow \infty$. Indeed, for all $f \in L^2([-\pi, \pi])$ one needs $\int_{-\pi}^{\pi} \sin kx \cdot f dx \rightarrow 0$. By density assume $f \in C_c^\infty([-\pi, \pi])$; integration by parts gives

$$\begin{aligned} \int_{-\pi}^{\pi} \sin kx \cdot f dx &= -\frac{1}{k} \left(f \cos kx \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \cos kx f'(x) dx \right) \\ &= \frac{1}{k} \int_{-\pi}^{\pi} \cos kx f'(x) dx, \end{aligned}$$

and $|\int \cos kx f'| \leq 2\pi \|f\|_{C^1}$, so the integral tends to 0 as $k \rightarrow \infty$.

Remark 17.15 (Weak $\neq$ strong). (i) **Weak but not strong.** Let $f_k = \chi_{[k, k+1]}$ in $L^2([0, \infty))$. For any $g \in C_c^\infty([0, \infty))$ with $\text{supp } g \subseteq [0, M]$, if $k \geq M$ then $\int_0^\infty f_k g = \int_k^{k+1} g = 0$, so $f_k \rightharpoonup 0$. But

$$\|f_k\|_{L^2([0, \infty))}^2 = \int_k^{k+1} 1 dx = 1 \neq 0,$$

so $\{f_k\}$ does not converge strongly to 0.

(ii) **Strong $\Rightarrow$ weak.** If $\|f_k - f\| \rightarrow 0$, then by Cauchy–Schwarz

$$|\langle u^*, f_k - f \rangle| \leq \|u^*\| \|f_k - f\| \rightarrow 0.$$

17.4 Finite difference methods

17.4.1 Interpolation

Remark 17.16 (Taylor / piecewise linear). $I_1(x) = f(a) + f'(a)(x - a)$ satisfies

$$f(x) - I_1(x) = \frac{f''(\xi)}{2} (x - a)^2 \leq \frac{\|f''\|_\infty}{2} h^2,$$

while the constant interpolant I_0 has error $O(h)$.

On $[a, b]$ with $b = a + Nh$, use values of f at the $N + 1$ nodes $a, a + h, \dots, b$. Advantages of piecewise linear interpolation: derivatives of f are not required; the approximant is continuous; second-order accurate.

On the i -th subinterval $[a + (i - 1)h, a + ih]$,

$$\begin{aligned} I_1(x) &= f(a + (i - 1)h) \frac{a + ih - x}{h} + f(a + ih) \frac{x - (a + (i - 1)h)}{h}, \\ I_0(x) &= f(a + (i - 1)h) \quad (\text{discontinuous}). \end{aligned}$$

Hat-function form:

$$\tilde{I}_h(x) = \sum_j f(x_j) \phi_j(x),$$

where ϕ_j is the standard piecewise-linear “hat” supported on $[x_{j-1}, x_{j+1}]$ (uses information from 3 points, more accurate, but still second-order $O(h^2)$).

17.4.2 Model BVP and L^∞ stability

Consider the 1D Poisson problem

$$\begin{cases} -u_{xx}(x) = f(x), & x \in (0, 1), \\ u(0) = u(1) = 0, \end{cases} \quad f \in C. \quad (34)$$

(Here Δ encodes diffusion/viscosity.) Existence and uniqueness hold; if $f \in C$ then $u \in C^2$; if $f \geq 0$ then $u \geq 0$.

Remark 17.17 (L^∞ estimate). For $\Delta u = f$ in $\Omega \subseteq \mathbb{R}^{2,3}$ with $u|_{\partial\Omega} = 0$,

$$\|u\|_{L^\infty(\Omega)} \leq C\|f\|_{L^\infty(\Omega)}$$

(zero boundary data, so u cannot be a nonzero constant). In the 1D model on $(0, 1)$,

$$\|u\|_{L^\infty(0,1)} \leq \frac{1}{8}\|f\|_{L^\infty(0,1)}.$$

Exercise 17.18 (Homework). For $-u_{xx} = f$ on $(0, 1)$ with $u(0) = u(1) = 0$, prove $\|u\|_{L^\infty} \leq \frac{1}{8}\|f\|_{L^\infty}$. Hint: consider $V = u + \frac{1}{2}(x - \frac{1}{2})^2\|f\|_\infty$.

17.4.3 Central differences and the discrete matrix

Return to (34). Take $h = \frac{1}{N}$, partition the interval into N subintervals, $x_j = jh$, $j = 0, \dots, N$, and write $\hat{u}_j \approx u_j \equiv u(x_j)$.

On $[x_j, x_{j+1}]$, the linear approximation

$$I(x) = u(x_j)\frac{x_{j+1}-x}{h} + u(x_{j+1})\frac{x-x_j}{h}, \quad I'(x) = \frac{u(x_{j+1})-u(x_j)}{h} \approx u'(x).$$

For the second derivative,

$$\begin{aligned} v''(x_j) &\approx \frac{1}{h} \left(v'(x_j + \frac{h}{2}) - v'(x_j - \frac{h}{2}) \right) \\ &\approx \frac{1}{h} \left(\frac{v(x_{j+1}) - v(x_j)}{h} - \frac{v(x_j) - v(x_{j-1}))}{h} \right) = \frac{v(x_{j+1}) - 2v(x_j) + v(x_{j-1}))}{h^2}. \end{aligned}$$

Hence $-\partial_{xx}u = f$ becomes, for $j = 1, \dots, N-1$,

$$-\frac{u(x_{j+1}) - 2u(x_j) + u(x_{j-1}))}{h^2} = f(x_j), \quad (35)$$

with boundary conditions $u(x_0) = u(x_N) = 0$.

In matrix form one may write a discrete system for the full vector $\underline{u} = (u_0, \dots, u_N)^T$, but the natural way to impose Dirichlet data $u_0 = u_N = 0$ is to eliminate the boundary unknowns and solve $B\underline{\tilde{u}} = \underline{f}$ for $\underline{\tilde{u}} = (u_1, \dots, u_{N-1})^T$, where

$$B = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}, \quad \underline{f} = (f(x_1), \dots, f(x_{N-1}))^T.$$

The matrix B is symmetric positive definite: for $v \in \mathbb{R}^{N-1}$,

$$v^T B v = \frac{1}{h^2} \left(v_1^2 + \sum_{i=1}^{N-2} (v_i - v_{i+1})^2 + v_{N-1}^2 \right) \geq 0,$$

with equality iff $v \equiv 0$. Hence B is nonsingular and $\underline{\tilde{u}} = B^{-1}\underline{f}$ exists and is unique.

Remark 17.19. Enlarging to an $(N+1) \times (N+1)$ stencil whose first/last rows are $\frac{1}{h^2}(1, -1, 0, \dots)$ and $\frac{1}{h^2}(\dots, 0, -1, 1)$ produces a *singular* Neumann matrix (constants lie in the kernel). Dirichlet conditions must be enforced by eliminating u_0, u_N as above, not by that enlarged stencil.

17.4.4 Error analysis (qualitative setup)

1. How different are the properties of the discrete solution formed by $\underline{u}$ from those of u ?
2. How accurate as $h \rightarrow 0$?
3. $|u(x_j) - \hat{u}_j| = ?$

(First demo: Matlab code segment 1.)

Model test problem:

$$\begin{cases} -\partial_{xx}u = \sin(\pi x), \\ u(0) = u(1) = 0. \end{cases} \quad (36)$$

18 Finite differences, heat solvers, and Fourier expansions

18.1 Discrete matrix B : non-negativity and stability

Continue the 1D Dirichlet Poisson problem

$$\begin{cases} -\partial_{xx}u = f & \text{in } (0, L), \\ u(0) = u(L) = 0, \end{cases}$$

with $h = L/(n+1)$ and the reduced system $B\underline{u} = \underline{f}$,

$$B = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix} \in \mathbb{R}^{n \times n}, \quad \underline{u} = B^{-1}\underline{f}.$$

The continuous problem satisfies

- (a) non-negativity: $f \geq 0 \Rightarrow u \geq 0$;
- (b) boundedness: $\|u\|_\infty \leq \frac{1}{8}\|f\|_\infty$ (on $(0, 1)$).

Proposition 18.1 (Discrete non-negativity). *Write $B^{-1} = (b_{ij})$. Then $b_{ij} \geq 0$ for all $1 \leq i, j \leq n$. Hence if $f_j \geq 0$ for all j ,*

$$u_i = \sum_{j=1}^n b_{ij} f_j \geq 0.$$

Proposition 18.2 (Discrete stability). *$0 \leq \sum_{j=1}^n b_{ij} \leq \frac{1}{8}$ for every $1 \leq i \leq n$. Consequently*

$$\|\underline{u}\|_\infty = \max_i \left| \sum_j b_{ij} f_j \right| \leq \max_i \left(\sum_j b_{ij} \right) \|f\|_\infty \leq \frac{1}{8} \|f\|_\infty,$$

i.e. $\|B^{-1}\|_\infty \leq \frac{1}{8}$.

Remark 18.3 (Discrete norms). For $v \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$,

$$\|v\|_\infty = \max_{1 \leq j \leq n} |v_j|, \quad \|v\|_2 = \left(\sum_{j=1}^n |v_j|^2 \right)^{1/2},$$

and the induced matrix norms are

$$\|A\|_p = \sup_{v \neq 0} \frac{\|Av\|_p}{\|v\|_p}, \quad \|A\|_\infty = \max_i \sum_j |a_{ij}|, \quad \|A\|_1 = \max_j \sum_i |a_{ij}|, \quad \|A\|_2 = \sqrt{\lambda_{\max}(A^*A)}.$$

All finite-dimensional norms are equivalent: there exist $C_1, C_2 > 0$ with $C_1\|A\|_p \leq \|A\|_q \leq C_2\|A\|_p$. In Matlab, `norm(A,p)`; the discrete L^2 norm of a grid function is $(\sum_i |f(x_i)|^2 \Delta x)^{1/2}$, approximating $(\int_a^b |f|^2 dx)^{1/2}$.

18.2 Truncation error, consistency, and convergence

Assume $v \in C^4$. Taylor expansion yields the central-difference identity

$$\frac{v(x_{j+1}) - 2v(x_j) + v(x_{j-1}))}{h^2} = v''(x_j) + \frac{h^2}{12}v^{(4)}(\xi)$$

for some $\xi \in [x_{j-1}, x_{j+1}]$. Applied to $-\partial_{xx}u = f$,

$$\frac{-u(x_{j+1}) + 2u(x_j) - u(x_{j-1}))}{h^2} = f(x_j) + \tau_j, \quad \tau_j := -\frac{h^2}{12}u^{(4)}(\xi_j)$$

(τ_j = truncation error).

Definition 18.4 (Discretization error). $e_j := u(x_j) - u_j$, where u_j is the numerical solution (discretization error).

Comparing the exact grid values (including truncation) with the numerical scheme gives

$$\frac{-e_{j+1} + 2e_j - e_{j-1}}{h^2} = \tau_j, \quad 1 \leq j \leq n, \quad e_0 = e_{n+1} = 0,$$

or in matrix form

$$B\mathbf{e} = \mathbf{\tau}, \quad \mathbf{\tau} = \frac{h^2}{12}(u^{(4)}(\xi_1), \dots, u^{(4)}(\xi_n))^T.$$

Theorem 18.5 (A priori error estimate). *Using discrete stability* $\|B^{-1}\|_\infty \leq \frac{1}{8}$,

$$\|\mathbf{e}\|_\infty \leq \frac{1}{8}\|\mathbf{\tau}\|_\infty \leq \frac{h^2}{96} \max_{0 \leq x \leq 1} |u^{(4)}(x)|.$$

Thus the scheme is second-order, $O(h^2)$, provided $u \in C^4$.

Definition 18.6 (Consistency). Consistency: $\lim_{h \rightarrow 0} \|\mathbf{\tau}\|_\infty = 0$.

For a general elliptic discretization $B\mathbf{u} = \mathbf{f}$ and the consistency relation $B\hat{\mathbf{u}} = \mathbf{f} + \mathbf{\tau}$, the error $\mathbf{e} = \hat{\mathbf{u}} - \mathbf{u}$ satisfies the *error equation*

$$B\mathbf{e} = \mathbf{\tau} \quad \Rightarrow \quad \mathbf{e} = B^{-1}\mathbf{\tau}.$$

Stability $\|B^{-1}\|_\infty \leq C$ then yields $\|\mathbf{e}\|_\infty \leq C\|\mathbf{\tau}\|_\infty \rightarrow 0$ as $h \rightarrow 0$:

$$\text{stability} + \text{consistency} = \text{convergence}.$$

In the 1D Dirichlet Poisson case, $\|\mathbf{\tau}\|_\infty \leq C|h|^2$ (second-order).

18.3 Neumann and Robin boundary conditions

18.3.1 Pure Neumann is not unique

The continuous problem

$$\begin{cases} -\partial_{xx}u = f & \text{in } (0, 1), \\ u'(0) = u'(1) = 0 \end{cases}$$

is not unique: if u_0 is a solution then so is $u_0 + C$. A well-posed model is the reaction–diffusion BVP with $c > 0$,

$$\begin{cases} -\partial_{xx}u + cu = f, \\ u'(0) = \alpha, \quad u'(1) = \beta. \end{cases}$$

18.3.2 Interior stencil

With $x_i = ih$, $x_0 = 0$, $x_N = 1$, the interior equations are

$$\frac{-u_{i-1} + 2u_i - u_{i+1} + ch^2u_i}{h^2} = f_i, \quad i = 1, \dots, N-1.$$

18.3.3 Method 1: one-sided differences (first-order)

At $x = 0$, $\frac{u_1 - u_0}{h} \approx \alpha$, rewritten as $\frac{1}{h^2}(u_0 - u_1) = -\frac{\alpha}{h}$. At $x = 1$, $\frac{u_N - u_{N-1}}{h} \approx \beta$, rewritten as $\frac{1}{h^2}(u_N - u_{N-1}) = \frac{\beta}{h}$. The resulting $(N+1) \times (N+1)$ system has diagonal entries > 0 , but the boundary approximation is only first-order:

$$\frac{u_1 - u_0}{h} - u'(0) = O(h) \quad (\text{first-order accuracy}).$$

18.3.4 Method 2: ghost points (second-order)

Introduce a ghost value u_{-1} . At x_0 ,

$$\begin{aligned} -u_{-1} + (2 + ch^2)u_0 - u_1 &= f_0h^2, \\ \frac{u_1 - u_{-1}}{2h} &\approx \alpha \quad \Rightarrow \quad u_{-1} = u_1 - 2\alpha h. \end{aligned}$$

Eliminating u_{-1} yields

$$(2 + ch^2)u_0 - 2u_1 = f_0h^2 - 2\alpha h.$$

Similarly at x_N with ghost $u_{N+1} = u_{N-1} + 2\beta h$,

$$-2u_{N-1} + (2 + ch^2)u_N = h^2f_N + 2\beta h.$$

The matrix equation is

$$\frac{1}{h^2} \begin{pmatrix} 2 + ch^2 & -2 & & & \\ -1 & 2 + ch^2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 + ch^2 & -1 \\ & & & -2 & 2 + ch^2 \end{pmatrix} \begin{pmatrix} u_0 \\ \vdots \\ u_N \end{pmatrix} = \begin{pmatrix} f_0 - \frac{2\alpha}{h} \\ f_1 \\ \vdots \\ f_{N-1} \\ f_N + \frac{2\beta}{h} \end{pmatrix}.$$

Taylor expansion confirms second-order accuracy of the centered Neumann formula:

$$\frac{u(h) - u(-h)}{2h} = u'(0) + O(h^2) \quad (\text{second-order accuracy!!}).$$

18.3.5 Robin / mixed BVP

Exercise 18.7 (Robin).

$$\begin{cases} -\partial_{xx}u = f, \\ u(0) = \alpha, \quad u'(1) = \beta \end{cases}$$

combines Dirichlet and Neumann (homework).

18.4 2D Dirichlet Poisson: five-point stencil

Consider

$$\begin{cases} -\Delta u = f & \text{in } \Omega = (0, 1) \times (0, 1), \\ u|_{\partial\Omega} = 0, \end{cases} \quad -\Delta u = -\partial_{xx}u - \partial_{yy}u.$$

18.4.1 Five-point method

Central differences give

$$\begin{aligned} -\partial_{xx}u(x, y) &\approx \frac{-u(x - \Delta x, y) + 2u(x, y) - u(x + \Delta x, y)}{(\Delta x)^2}, \\ -\partial_{yy}u(x, y) &\approx \frac{-u(x, y - \Delta y) + 2u(x, y) - u(x, y + \Delta y)}{(\Delta y)^2}. \end{aligned}$$

At grid point (i, j) , with $u_{ij} \approx u(x_i, y_j)$,

$$-\Delta u_{ij} \approx \frac{-u_{i-1,j} + 2u_{ij} - u_{i+1,j}}{(\Delta x)^2} + \frac{-u_{i,j-1} + 2u_{ij} - u_{i,j+1}}{(\Delta y)^2}.$$

Homogeneous Dirichlet: $u_{0,j} = u_{n+1,j} = 0$ and $u_{i,0} = u_{i,m+1} = 0$.

18.4.2 Matrix form

Lexicographic ordering $\underline{u} = (u_{11}, u_{21}, \dots, u_{n1}, u_{12}, \dots, u_{nm})^T$ produces a block-tridiagonal system $A\underline{u} = \underline{f}$. For $\Delta x = \Delta y = h$ and $n = m$, the five-point stencil reads

$$\frac{-u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1} + 4u_{ij}}{h^2} = f_{ij}.$$

In block form,

$$A = \begin{pmatrix} A_x + 2I_y & -I_y & 0 & \cdots \\ -I_y & A_x + 2I_y & -I_y & \ddots \\ 0 & -I_y & A_x + 2I_y & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix},$$

where

$$A_x = \frac{1}{(\Delta x)^2} \begin{pmatrix} 2 & -1 & \\ -1 & 2 & -1 \\ & \ddots & \ddots \end{pmatrix}, \quad I_y = \frac{1}{(\Delta y)^2} I.$$

The discrete energy identity

$$V^T A V = \sum_{i,j} \left[\frac{1}{(\Delta x)^2} (V_{ij} - V_{i-1,j})^2 + \frac{1}{(\Delta y)^2} (V_{ij} - V_{i,j-1})^2 \right] \geq 0$$

with equality iff $V = 0$ shows that A is SPD, hence the discrete solution exists uniquely. Truncation error for smooth u is

$$\tau_{ij} = -\frac{(\Delta x)^2}{12} \partial_x^4 u(\xi_i, y_j) - \frac{(\Delta y)^2}{12} \partial_y^4 u(x_i, \eta_j) = O((\Delta x)^2, (\Delta y)^2).$$

18.4.3 Jacobi iteration

Assuming $\Delta x = \Delta y = h$, rewrite the stencil as

$$\frac{4u_{ij}^{(k+1)}}{h^2} = f_{ij} + \frac{1}{h^2} (u_{i-1,j}^{(k)} + u_{i+1,j}^{(k)} + u_{i,j-1}^{(k)} + u_{i,j+1}^{(k)}).$$

In vector form $U^{(k+1)} = \frac{h^2}{4} F + \frac{1}{4} C U^{(k)}$. The exact discrete solution satisfies the same fixed-point relation, so

$$\|U^{(k+1)} - U\| \leq \frac{1}{4} \|C\| \|U^{(k)} - U\|.$$

Convergence follows from the contraction condition $\frac{1}{4} \|C\| < 1$ (fixed-point theorem).

Remark 18.8 (Matlab residual loop). Input $u^{(0)}$; update

$$u_{ij}^{(k+1)} = \frac{h^2}{4} f_{ij} + \frac{1}{4} (u_{i-1,j}^{(k)} + u_{i+1,j}^{(k)} + u_{i,j-1}^{(k)} + u_{i,j+1}^{(k)});$$

set residual $\text{res}^{(k+1)} = \|u^{(k+1)} - u^{(k)}\|$ and iterate until the residual is small.

18.5 Heat equation: finite-difference time stepping

Model IBVP (1D heat / parabolic):

$$\begin{cases} \partial_t u = \partial_{xx} u, & t > 0, 0 < x < 1, \\ u(0, x) = f(x), & \text{(IC)} \\ u(t, 0) = u(t, 1) = 0. & \text{(Dirichlet BC)} \end{cases}$$

(Continuous classification, heat kernel / Duhamel, continuous maximum principle and energy estimates: see the heat-equation sections above; omitted here.)

18.5.1 Forward Euler (explicit)

Central difference in space and forward difference in time give truncation $O((\Delta x)^2, \Delta t)$ and the scheme

$$u_i^{k+1} = u_i^k + \frac{\Delta t}{(\Delta x)^2} (u_{i+1}^k + u_{i-1}^k - 2u_i^k),$$

with $x_i = i\Delta x$, $t_k = k\Delta t$, $u_i^k \approx u(t_k, x_i)$. Dirichlet: $u_0^k \equiv u_N^k \equiv 0$; Neumann uses ghost points as in the Poisson case. IC: $u_i^0 = f(i\Delta x)$.

Theorem 18.9 (von Neumann stability for Forward Euler). *Substitute $u_n^k = \phi_k e^{in\Delta x \theta}$. The growth factor satisfies*

$$\phi_{k+1} = \left(1 - \frac{4\Delta t}{(\Delta x)^2} \sin^2\left(\frac{\Delta x \theta}{2}\right) \right) \phi_k.$$

Requiring $|\phi_{k+1}| \leq |\phi_k|$ for all θ yields

$$\frac{\Delta t}{(\Delta x)^2} \leq \frac{1}{2}, \quad \text{i.e.} \quad \Delta t \leq \frac{1}{2}(\Delta x)^2.$$

For $\partial_t u = c\partial_{xx} u$, the condition becomes $\Delta t \leq \frac{1}{2c}(\Delta x)^2$.

Pros: easy to implement. Cons: restrictive CFL-type constraint; regularity of the update is limited (explicit discrete Laplacian maps $H^2 \rightarrow L^2$).

18.5.2 Backward Euler (implicit)

$$\frac{u_n^{k+1} - u_n^k}{\Delta t} = \frac{u_{n+1}^{k+1} + u_{n-1}^{k+1} - 2u_n^{k+1}}{(\Delta x)^2}.$$

With $\lambda = \Delta t / (\Delta x)^2$,

$$AU^{k+1} = U^k, \quad A = \begin{pmatrix} 1+2\lambda & -\lambda & & \\ -\lambda & 1+2\lambda & -\lambda & \\ & \ddots & \ddots & \ddots \\ & & -\lambda & 1+2\lambda \end{pmatrix}.$$

As in the Poisson case, A is invertible and the scheme is **unconditionally stable** (no restriction on λ : large time steps are allowed). Truncation error remains $O(\Delta t, (\Delta x)^2)$.

Remark 18.10 (Direct vs. iterative Backward Euler). **Direct:** unconditional stability and good regularity, but large matrices in higher dimensions. **Iterative** (Jacobi-like on the implicit step):

$$u_j^{(m+1)} = u_j^k + \frac{\Delta t}{(\Delta x)^2} [u_{j+1}^{(m)} + u_{j-1}^{(m)} - 2u_j^{(m)}],$$

with contraction factor $4\Delta t/(\Delta x)^2$; convergence of the inner iteration requires this factor < 1 , so stability/convergence force smaller Δt , though no large matrix solve is needed.

18.5.3 θ -scheme and Crank–Nicolson

For $\theta \in [0, 1]$,

$$\frac{u_n^{k+1} - u_n^k}{\Delta t} = \theta \frac{u_{n+1}^{k+1} - 2u_n^{k+1} + u_{n-1}^{k+1}}{(\Delta x)^2} + (1 - \theta) \frac{u_{n+1}^k - 2u_n^k + u_{n-1}^k}{(\Delta x)^2}.$$

Special cases: $\theta = 0$ Forward Euler; $\theta = 1$ Backward Euler; $\theta = \frac{1}{2}$ Crank–Nicolson.

Expanding about $(x_n, t_{k+1/2})$ shows that Crank–Nicolson cancels the leading $O(\Delta t)$ terms, giving truncation $O((\Delta t)^2, (\Delta x)^2)$. For general θ ,

$$\tau = \left(\frac{1}{2} - \theta\right) \Delta t \partial_{xxt} u + O((\Delta t)^2, (\Delta x)^2).$$

Hence $\tau = O(\Delta t, (\Delta x)^2)$ if $\theta \neq \frac{1}{2}$, and $O((\Delta t)^2, (\Delta x)^2)$ if $\theta = \frac{1}{2}$.

Proposition 18.11 (Stability of the θ -scheme). *Let $\lambda = \Delta t/(\Delta x)^2$. The scheme is*

- *conditionally stable if $0 \leq \theta < \frac{1}{2}$, requiring $2\lambda \leq 1/(1 - 2\theta)$;*
- *unconditionally stable if $\frac{1}{2} \leq \theta \leq 1$.*

Direct Crank–Nicolson: second-order in time, unconditional stability, matrix solve each step. Iterative C–N keeps the same truncation order but needs a smaller Δt for inner convergence.

18.5.4 Higher-order time stepping

BDF methods (with discrete Laplacian $\tilde{\Delta}$):

$$\begin{aligned} \text{BDF1: } \frac{u^{k+1} - u^k}{\Delta t} &= \tilde{\Delta} u^{k+1} && \rightarrow O(\Delta t), \\ \text{BDF2: } u^{k+2} - \frac{4}{3}u^{k+1} + \frac{1}{3}u^k &= \frac{2}{3}\Delta t \tilde{\Delta} u^{k+2} && \rightarrow O((\Delta t)^2), \\ \text{BDF3: } u^{k+3} - \frac{18}{11}u^{k+2} + \frac{9}{11}u^{k+1} - \frac{2}{11}u^k &= \frac{6}{11}\Delta t \tilde{\Delta} u^{k+3} && \rightarrow O((\Delta t)^3). \end{aligned}$$

Classical RK4 for $\dot{y} = f(t, y)$ applied with $f = \tilde{\Delta}$ yields truncation $O((\Delta t)^4)$ using only one past time level, at the cost of weaker regularity requirements and more work per step.

18.6 Fourier sine / cosine expansions for IBVP

18.6.1 Separation of variables (Dirichlet)

For

$$\begin{cases} \partial_t u = \partial_{xx} u, & x \in (0, L), \\ u(0, x) = f(x), \\ u(t, 0) = u(t, L) = 0, \end{cases}$$

set $u = X(x)T(t)$. Then $X''/X = \dot{T}/T = C$. Only $C = -\lambda^2 < 0$ yields physically decaying modes:

$$u = (A \cos \lambda x + B \sin \lambda x) e^{-\lambda^2 t}.$$

Dirichlet forces $A = 0$ and $\lambda = n\pi/L$, hence

$$u(t, x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) e^{-(n\pi/L)^2 t}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx.$$

18.6.2 Non-homogeneous Neumann: particular solution + cosine series

Example 18.12.

$$\begin{cases} \partial_t u = \partial_{xx} u, & 0 < x < 1, t > 0, \\ u_x(t, 0) = 0, \quad u_x(t, 1) = 1, \\ u(0, x) = -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right). \end{cases}$$

Try $w(t, x) = ax^2 + bx + Ct$. Matching the PDE gives $C = 2a$; matching Neumann data gives $a = \frac{1}{2}$, $b = 0$, so $w = \frac{1}{2}x^2 + t$. Set $u = w + v$:

$$\begin{cases} \partial_t v = \partial_{xx} v, \\ \partial_x v(t, 0) = \partial_x v(t, 1) = 0, \\ v(0, x) = -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) - \frac{x^2}{2} =: f(x). \end{cases}$$

Separation with homogeneous Neumann yields the cosine expansion

$$v = b_0 + \sum_{n=1}^{\infty} b_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 t},$$

with $(L = 1)$

$$b_0 = \int_0^1 f(x) dx, \quad b_n = 2 \int_0^1 f(x) \cos(n\pi x) dx \quad (n \geq 1).$$

Remark 18.13 (Summary: BC $\leftrightarrow$ series). • Dirichlet = 0: sine series; $\neq 0$: particular solution + sine.

- Neumann = 0: cosine series; $\neq 0$: particular solution + cosine.
- Odd extension of a Dirichlet datum on $[0, L]$ produces a $2L$ -periodic function; even extension does the same for Neumann (even symmetry).

18.7 Fourier spectral method on a periodic domain

18.7.1 Periodic heat equation

$$\begin{cases} \partial_t u = \partial_{xx} u & \text{in } [0, 2\pi], \\ u(0) = u(2\pi), \quad u^{(n)}(0) = u^{(n)}(2\pi) & \text{(periodic BC)}, \\ u(0, x) = u_0(x). \end{cases}$$

Separation with periodic BC forces eigenvalues $\lambda_n = n \in \mathbb{Z}$,

$$u_n(t, x) = e^{-n^2 t} (A_n \cos nx + B_n \sin nx).$$

Matching u_0 gives

$$A_0 = \frac{1}{2\pi} \int_0^{2\pi} u_0 dx, \quad A_n = \frac{1}{\pi} \int_0^{2\pi} u_0 \cos nx dx, \quad B_n = \frac{1}{\pi} \int_0^{2\pi} u_0 \sin nx dx.$$

18.7.2 Time stepping on the Fourier side

On modes $\hat{u}(k)$, ∂_{xx} becomes multiplication by $-|k|^2$.

- **Backward Euler:**

$$\hat{u}^{m+1}(k) = \frac{\hat{u}^m(k)}{1 + |k|^2 \Delta t}.$$

Since $1 + |k|^2 \Delta t > 1$, the scheme is unconditionally stable; easy to compute; best suited to periodic (or simple Dirichlet/Neumann) BC.

- **Forward Euler:**

$$\hat{u}^{m+1}(k) = (1 - \Delta t |k|^2) \hat{u}^m(k),$$

requiring $\Delta t \lesssim 2/|k|_{\max}^2$ (stability bad). Spatial truncation in N modes is spectrally accurate.

- **Crank–Nicolson:**

$$\hat{u}^{m+1}(k) = \frac{1 - \frac{1}{2}\Delta t |k|^2}{1 + \frac{1}{2}\Delta t |k|^2} \hat{u}^m(k),$$

with amplification factor magnitude < 1 (unconditional stability).

Remark 18.14 (Accuracy checks / spectral BC). Practical accuracy checks: (i) compare against a reference with very small Δt ; (ii) method of manufactured (“forced”) solutions. Spectral methods extend to simple Dirichlet / Neumann conditions via odd/even extensions or particular solutions as above.

19 Phase-field models, exponential integrators, and fluids

19.1 Phase-field models

Classical phase-field models:

- Allen–Cahn (1979): Fe–Mn mixing / interface motion;
- Cahn–Hilliard (1958): solid–liquid interfaces;
- MBE: molecular beam epitaxy (liquid crystals).

Applications: dendritic growth, lithium batteries, crystal precipitation.

On the periodic domain $\mathbb{T}^d$ (repeating microstructure; convenient for spectral methods), following the Li–Tang–Qiao (*SINUM* 2016) framework, the model equations are

$$\partial_t u = \nu \Delta u + u - u^3 \quad (\text{AC}), \quad (37)$$

$$\partial_t u = -\Delta(\nu \Delta u + u - u^3) \quad (\text{CH}). \quad (38)$$

Three leading journals in numerical analysis: *SINUM*, *Math. Comp.*, *Numer. Math.*

Remark 19.1 (AC maximum principle). For smooth solutions,

$$\|u(t)\|_\infty \leq \max\{\|u_0\|_\infty, 1\} \quad \forall 0 \leq t \leq T.$$

The standard proof takes $g = u^2$, $g^\varepsilon = g - \varepsilon t$, and at a maximum point uses the AC equation together with $u \Delta u \leq 0$ to reach a contradiction if $\|u\|_\infty > 1$; then let $\varepsilon \rightarrow 0$.

The 1D stationary equation $\partial_{xx} u + u - u^3 = 0$ has the solution $u = \tanh(x/\sqrt{2})$; the interface-type approximation $\tilde{u} = \tanh(x/\varepsilon)$ has transition-layer width ε .

19.2 Gradient flow and energy dissipation

Define the energy

$$E(u) := \int_{\mathbb{T}^d} \left(\frac{\nu}{2} |\nabla u|^2 + \frac{(u^2 - 1)^2}{4} \right) dx$$

(kinetic / potential). For AC,

$$\frac{dE}{dt} = \int \nu \nabla u \cdot \nabla u_t + (u^3 - u) u_t dx = \int u_t (-\nu \Delta u + u^3 - u) dx = -\|u_t\|_2^2 \leq 0.$$

For CH (mass conservation $\bar{u} = 0$ is needed to define $(-\Delta)^{-1}$),

$$\frac{dE}{dt} = -\| |\nabla|^{-1} u_t \|_2^2 \leq 0.$$

Gradient flow $\Rightarrow H^1$ a-priori control.

Scaling-critical space (AC)

For $\partial_t u = \Delta u + u - u^3$, the scaling $\lambda^s u(\lambda^2 t, \lambda x)$ yields the critical index $s = 1$, hence

$$\|\lambda u(\lambda^2 t, \lambda x)\|_{\dot{H}^k(\mathbb{R}^d)} = \lambda^{1+k-d/2} \|u\|_{\dot{H}^k},$$

and the critical index satisfies $k + 1 = d/2$ (e.g. $d = 2$: L^2 ; $d = 3$: $\dot{H}^{1/2}$). Thus H^1 a-priori control $\Rightarrow$ GWP.

Interface motion

- AC: $\{u = 0\}$ moves by mean curvature; in 2D this is curve shortening $V = -K$; in 3D it is mean curvature flow (non-conservative). References: Alikakos–Bates–Chen (ABC), Pego (89), Chen (94), Fei Lin, Wang–Zhang (23), etc.
- CH: Mullins–Sekerka flow (mass-conserving).

19.3 Energy-stable spectral schemes for CH

For the evolution equation $\partial_t u = Gu + f(u)$, the naive semi-implicit scheme

$$\frac{u^{n+1} - u^n}{\tau} = Gu^{n+1} + f(u^n), \quad G \in \{\Delta, -\Delta^2, -(-\Delta)^{\alpha+1}\} \ (\alpha \in (0, 1)),$$

i.e. $u^{n+1} = (u^n + \tau f(u^n))/(I - G\tau)$. Below we mainly discuss $-\Delta^2$ -type CH (Li–Qiao–Tang 2016): the goal is energy stability $\tilde{E}(u^{n+1}) \leq \tilde{E}(u^n)$, unconditionally for any $\tau > 0$, which requires a stabilization term; $f(u) = u^3 - u$.

Spectral method (frequency truncation $P_N = P_{\leq N}$) with stabilization ($\star$):

$$\frac{u^{n+1} - u^n}{\tau} = -\nu \Delta^2 u^{n+1} + A \Delta(u^{n+1} - u^n) + \Delta P_N f(u^n), \quad u^0 = P_N u_0. \quad (39)$$

Theorem 19.2 (Li–Qiao–Tang; energy decay). *If $A \gg \nu^{-1} |\log \nu|^2$ (note: not sharp), then $E(u^{n+1}) \leq E(u^n)$.*

Sketch. Take the inner product of ($\star$) against $\langle \cdot, (-\Delta)^{-1}(u^{n+1} - u^n) \rangle$, yielding

$$\begin{aligned} \frac{1}{\tau} \|\nabla|^{-1}(u^{n+1} - u^n)\|_2^2 &= \frac{\nu}{2} \left(\|\nabla u^n\|_2^2 - \|\nabla u^{n+1}\|_2^2 - \|\nabla(u^{n+1} - u^n)\|_2^2 \right) \\ &\quad - A \|u^{n+1} - u^n\|_2^2 - (P_{\leq N} f(u^n), u^{n+1} - u^n). \end{aligned}$$

Rearranging and using the integral remainder for $F(u) = \frac{1}{4}u^4 - \frac{1}{2}u^2$ ($F' = f$),

$$F(u^{n+1}) - F(u^n) = f(u^n)(u^{n+1} - u^n) + \int_{u^n}^{u^{n+1}} (3s^2 - 1)(u^{n+1} - s) ds,$$

one obtains

$$\begin{aligned} E(u^{n+1}) - E(u^n) &+ \frac{1}{\tau} \|\nabla|^{-1}(u^{n+1} - u^n)\|_2^2 + \frac{\nu}{2} \|\nabla(u^{n+1} - u^n)\|_2^2 \\ &+ \left(A + \frac{1}{2}\right) \|u^{n+1} - u^n\|_2^2 \\ &\leq \|u^{n+1} - u^n\|_2^2 \left(\|u^n\|_\infty^2 + \frac{1}{2} \|u^{n+1}\|_\infty^2 \right). \end{aligned}$$

By interpolation $\frac{1}{\tau} \|\nabla|^{-1} \delta u\|_2^2 + \frac{\nu}{2} \|\nabla \delta u\|_2^2 \geq \sqrt{2\nu/\tau} \|\delta u\|_2^2$, so whenever

$$A + \frac{1}{2} + \sqrt{\frac{2\nu}{\tau}} \geq \|u^n\|_\infty^2 + \frac{1}{2} \|u^{n+1}\|_\infty^2 \quad (40)$$

one gets energy dissipation. After the inductive hypothesis $E^n \leq \dots \leq E_0$, one must control $\|u^n\|_\infty, \|u^{n+1}\|_\infty$ (below we take $\mathbb{T}^2$). $\square$

The Fourier form is convenient for computation:

$$u^{n+1} = \frac{I - A\tau\Delta}{I + \nu\tau\Delta^2 - A\tau\Delta} u^n + \frac{\tau\Delta P_N}{I + \nu\tau\Delta^2 - A\tau\Delta} f(u^n).$$

19.4 L^∞ bounds via log-Sobolev / Littlewood–Paley

Lemma 19.3 (Brezis–Gallouët type on $\mathbb{T}^2$). *If $\bar{f} = 0$, then for $s > 1$,*

$$\|f\|_\infty \lesssim \|f\|_{\dot{H}^1} \log(1 + \|f\|_{\dot{H}^s}) + 1.$$

Sketch. Littlewood–Paley: $\|f\|_\infty \lesssim \sum_{j \geq -1} \|\Delta_j f\|_\infty$. Splitting at J and using Bernstein,

$$\|f\|_\infty \lesssim J \|f\|_{\dot{H}^1} + 2^{(1-s)J} \|f\|_{\dot{H}^s}.$$

Optimizing $H(J) = J \|f\|_{\dot{H}^1} + 2^{(1-s)J} \|f\|_{\dot{H}^s}$ yields $J \sim \log(1 + \|f\|_{\dot{H}^s})$. $\square$

In the spectral scheme take $s = 3/2$. Fourier-multiplier estimates give

$$\|u^{n+1}\|_{H^1} \leq \|u^n\|_{H^1} + \frac{1}{A} \|(u^3 - u)^n\|_{H^1},$$

and

$$\begin{aligned} \|u^{n+1}\|_{\dot{H}^{3/2}} &\lesssim \left(\frac{1}{A\tau} + \frac{A}{\nu} \right) \|u^n\|_2 + \frac{1}{\nu} \||\nabla|^{-1/2}((u^n)^3 - u^n)\|_2 \\ &\lesssim \left(\frac{A+1}{\nu} + \frac{1}{A\tau} \right) (E^n + 1). \end{aligned}$$

Hence

$$\|u^n\|_\infty \leq 1 + C_1 \sqrt{\frac{2E_0}{\nu}} \log\left(1 + C_2 \left(\frac{A+1}{\nu} + \frac{1}{A\tau}\right) (E_0 + 1)\right) \lesssim_{E_0} \nu^{-1/2} (1 + \log A + |\log \nu| + |\log \tau|) + 1.$$

Let m_0 denote this logarithmic bound; then choosing

$$A \gg_{E_0} m_0^2 \quad (\text{or } A + \frac{1}{A} \sqrt{2E_0/\nu} \gg_{E_0} m_0^2)$$

closes (40).

19.5 L^2 error estimates for the CH scheme

Theorem 19.4 (L^2 error).

$$\|u(m\tau) - u^m\|_2 \leq A e^{Cm\tau} C_2 (N^{-s} + \tau).$$

Tool: discrete Grönwall. If $(y^{n+1} - y^n)/\tau \leq \tilde{\alpha}_n y^n + \tilde{\beta}_n$, then

$$y^m \leq \exp\left(\tau \sum_{n=0}^{m-1} \tilde{\alpha}_n\right) y_0 + \tau \sum_{k=0}^{m-1} \exp\left(\tau \sum_{j=k+1}^{m-1} \tilde{\alpha}_j\right) \tilde{\beta}_k;$$

in particular

$$y^m \leq \exp\left(\tau \sum \tilde{\alpha}_n\right) \left(y_0 + \tau \sum \tilde{\beta}_k\right).$$

Continuous counterpart: $\partial_t y \leq \alpha(t)y + \beta(t)$ yields $y(t) \leq e^{\int_0^t \alpha} y_0 + \int_0^t e^{\int_s^t \alpha} \beta(s) ds$.

Proposition 19.5 (Auxiliary system). *Consider*

$$\begin{cases} \frac{u^{n+1} - u^n}{\tau} - A\Delta(u^{n+1} - u^n) = -\nu\Delta^2 u^{n+1} + \Delta P_{\leq N} f(u^n), \\ \frac{v^{n+1} - v^n}{\tau} - A\Delta(v^{n+1} - v^n) = -\nu\Delta^2 v^{n+1} + \Delta P_{\leq N} f(v^n) + \Delta G^n. \end{cases}$$

If $\|v^n\|_\infty + \|\nabla v^n\|_2 + \|\nabla u^n\|_2 \leq C$, then

$$\begin{aligned} \|u^m - v^m\|_2^2 &\leq \exp\left(\frac{C}{\nu} m\tau\right) \left(\|u_0 - v_0\|_2^2 + A\tau \|\nabla u_0 - \nabla v_0\|_2^2 \right. \\ &\quad \left. + \frac{\tau}{\nu} \sum_{n=0}^{m-1} \|G^n\|_2^2 \right). \end{aligned}$$

Sketch. Set $e^n = u^n - v^n$. Subtract and take the inner product with e^{n+1} , yielding

$$\begin{aligned} & \frac{1}{2\tau} (\|e^{n+1}\|_2^2 - \|e^n\|_2^2 + \|e^{n+1} - e^n\|_2^2) \\ & + \frac{A}{2} (\|\nabla e^{n+1}\|_2^2 - \|\nabla e^n\|_2^2 + \|\nabla(e^{n+1} - e^n)\|_2^2) + \nu \|\Delta e^{n+1}\|_2^2 \\ & = (P_{\leq N}(f(u^n) - f(v^n)), \Delta e^{n+1})_I - (G^n, \Delta e^{n+1})_{II}. \end{aligned}$$

For $f(u) = u^3 - u$,

$$f(u^n) - f(v^n) = \int_0^1 f'(v^n + se^n) ds \cdot e^n = a_1 e^n + a_2 (v^n)^2 e^n + a_3 v^n (e^n)^2 + a_4 (e^n)^3$$

(write I_1, I_2, I_3 for the last three terms). Young / Sobolev:

$$\begin{aligned} I_1 & \leq \frac{\nu}{8} \|\nabla e^{n+1}\|_2^2 + \frac{C}{\nu} \|e^n\|_2^2, \\ I_2 & \leq \|v^n\|_\infty \|e^n\|_4^2 \|\nabla e^{n+1}\|_2 \leq \frac{C}{\nu} \|e^n\|_2^2 + \frac{\nu}{8} \|\nabla e^{n+1}\|_2^2, \\ I_3 & \leq \|e^n\|_6^3 \|\nabla e^{n+1}\|_2 \leq \frac{C}{\nu} \|e^n\|_2^2 + \frac{\nu}{8} \|\nabla e^{n+1}\|_2^2, \end{aligned}$$

and $II \leq \frac{\nu}{8} \|\Delta e^{n+1}\|_2^2 + \frac{C}{\nu} \|G^n\|_2^2$. Set $y^n := \|e^n\|_2^2 + A\tau \|\nabla e^n\|_2^2$; then $(y^{n+1} - y^n)/\tau \leq (C/\nu)y^n + (C/\nu)\|G^n\|_2^2$, and discrete Grönwall concludes the proof. $\square$

Real case / truncation

Substitute the exact solution into the scheme:

$$\begin{aligned} \frac{u((n+1)\tau) - u(n\tau)}{\tau} & = -\nu \Delta^2 u((n+1)\tau) + A \Delta(u((n+1)\tau) - u(n\tau)) \\ & + \Delta P_{\leq N} f(u(n\tau)) + \Delta P_{>N} f(u(n\tau)) + \Delta G^n, \end{aligned}$$

where (schematically)

$$\begin{aligned} G^n & = -\frac{\nu}{\tau} \int_{t_n}^{t_{n+1}} \partial_t \Delta u \cdot (t_n - t) dt + \frac{1}{\tau} \int_{t_n}^{t_{n+1}} \partial_t f(u) \cdot (t_{n+1} - t) dt \\ & - A \int_{t_n}^{t_{n+1}} \partial_t u dt. \end{aligned}$$

Estimate $\tau \sum \|G^n\|_2^2 \lesssim \tau^2 (1+A)^2 \int_0^{t_m} \|\partial_t \Delta u\|_2^2 dt$, and (assuming $\|f(u(t))\|_{H^s} \lesssim 1$) $\tau \sum \|P_{>N} f(u(t_n))\|_2^2 \lesssim m\tau N^{-2s}$. Initial projection: $\|u_0 - P_N u_0\|_2 \lesssim N^{-s}$, $\|\nabla u_0 - \nabla P_N u_0\|_2 \lesssim N^{-(s-1)}$. Altogether,

$$\|u^m - u(t_m)\|_2 \lesssim (1+A) e^{Ct_m} (\tau + N^{-s}), \quad t_m \leq T.$$

Remark 19.6. The exponential contains $e^{Ct_m/\nu}$; usable for fixed ν , but the estimate fails as $\nu \rightarrow 0$.

Smoothing for fourth-order heat kernel

$$\int_0^t \|e^{-(t-s)\nu\Delta^2} \Delta f\|_2 ds \lesssim \int_0^t (t-s)^{-(2-s)/4} \|f\|_{H^s} ds,$$

since on the Fourier side $e^{-(t-s)\nu|\xi|^4} |\xi|^{2-s} |\xi|^s \hat{f}$, and the scaling $\Delta^2 \sim (t-s)^{-1} \Rightarrow |\xi| \sim (t-s)^{-1/4}$.

19.6 Exponential integrators: motivation, IMEX, and BDF2

Exponential integrator methods. For $\partial_t u = \mathcal{L}u + f(u)$:

1st-order semi-implicit

$$\frac{u^{n+1} - u^n}{\tau} = \mathcal{L}u^{n+1} + f(u^n), \quad u^n \approx u(t_n), \quad t_n = n\tau.$$

Taylor: $u(t_{n+1}) - u(t_n) = \tau \partial_t u(t_n) + \frac{\tau^2}{2} \partial_{tt} u(\varsigma)$, truncation error $O(\tau)$; the RHS cancels $\partial_t u(t_n)$.

2nd-order BDF2

$$\frac{u^{n+1} - \frac{4}{3}u^n + \frac{1}{3}u^{n-1}}{\frac{2}{3}\tau} = \mathcal{L}u^{n+1} + f(u^{n+1}).$$

Expanding at t_{n+1} : the constant and τ^2 terms cancel, leaving $\frac{2}{3}\tau \partial_t u$, so after dividing by $\frac{2}{3}\tau$ the truncation error is $O(\tau^2)$ (remainder $\sim \tau^2 \partial_{ttt} u$).

Remark 19.7 (Truncation vs. regularity). Higher-order truncation $\Leftrightarrow$ higher regularity of the exact solution: $O(\tau) \Leftrightarrow \partial_{tt} u \sim \Delta^2 u$; $O(\tau^2) \Leftrightarrow \partial_{ttt} u \sim \Delta^3 u$. Question: can one lower the regularity requirement on the exact solution?

19.7 First-order exponential integrator (Allen–Cahn)

Model $\partial_t u = \Delta u + f(u)$ (WLOG $\varepsilon = 1$). Duhamel / heat kernel:

$$u(t) = e^{t\Delta} u(t_0) + \int_{t_0}^t e^{(t-s)\Delta} f(u(s)) ds,$$

where $e^{t\Delta} = I + t\Delta + \frac{1}{2!}(t\Delta)^2 + \dots$, and $\widehat{e^{t\Delta}g}(k) = e^{-t|k|^2} \hat{g}(k)$.

On $[t_n, t_{n+1}]$ take $f(u(s)) \approx f(u(t_n))$ (independent of s), yielding

$$u^{n+1} = e^{\tau\Delta} u^n + \frac{1 - e^{\tau\Delta}}{-\Delta} f(u^n),$$

or equivalently

$$\frac{u^{n+1} - u^n}{\tau} = \frac{e^{\tau\Delta} - 1}{\tau} u^n + \frac{e^{\tau\Delta} - 1}{\tau\Delta} f(u^n). \quad (41)$$

Expansion: $(e^{\tau\Delta} - 1)/\tau = \Delta + \frac{\tau}{2}\Delta^2 + \dots$. LHS/RHS Taylor expansions match exactly on linear terms: time differentiation produces no truncation error; but in general this alone is still not enough for an L^2 error estimate.

Theorem 19.8 (Stability; unconditional needs stabilizer). *For (41), multiply by $\frac{\tau\Delta}{e^{\tau\Delta} - 1}(u^{n+1} - u^n)$ and integrate. Set $A = \frac{\Delta}{e^{\tau\Delta} - 1}$; then the LHS is $\|\sqrt{A}(u^{n+1} - u^n)\|_2^2$, and*

$$\widehat{\sqrt{A}v}(k) = \sqrt{\frac{|k|^2}{1 - e^{-\tau|k|^2}}} \hat{v}(k).$$

Since $\frac{\tau|k|^2}{1 - e^{-\tau|k|^2}} \geq 1$, one has $\|\sqrt{A}v\|_2^2 \geq \frac{1}{\tau}\|v\|_2^2$.

Two RHS terms:

$$\begin{aligned} RHS_1 &= (u^n, \Delta(u^{n+1} - u^n)) = \frac{1}{2}(\|\nabla u^n\|_2^2 - \|\nabla u^{n+1}\|_2^2 + \|\nabla(u^{n+1} - u^n)\|_2^2), \\ RHS_2 &= (f(u^n), u^{n+1} - u^n), \quad f(u) = u - u^3 \quad (\text{AC}). \end{aligned}$$

Using $F(u^{n+1}) - F(u^n) = f(u^n)\delta u + \int (1 - 3s^2)(u^{n+1} - s) ds$ and combining yields the energy difference and $\|\sqrt{A}\delta u\|_2^2$. Stability requires $\tau \ll 1$ so that

$$\frac{1}{\tau}\|\delta u\|_2^2 \geq \frac{1}{2}\|\nabla \delta u\|_2^2 + \frac{3}{2} \max\{\|u^{n+1}\|_\infty^2, \|u^n\|_\infty^2\} \|\delta u\|_2^2$$

(e.g. $1/\tau \gg N^2 + \dots$), or adding stabilization terms $S_1(u^{n+1} - u^n) + S_2\Delta(u^{n+1} - u^n)$.

19.8 Stabilized ETD: L^2 error

Consider (one may add $S_2(-\Delta)(u^{n+1} - u^n)$)

$$\begin{aligned}\frac{u^{n+1} - u^n}{\tau} + S_1(u^{n+1} - u^n) &= \frac{e^{\tau\Delta} - 1}{\tau} u^n + \frac{e^{\tau\Delta} - 1}{\tau\Delta} \Pi_N f(u^n), \\ \frac{v^{n+1} - v^n}{\tau} + S_1(v^{n+1} - v^n) &= \frac{e^{\tau\Delta} - 1}{\tau} v^n + \frac{e^{\tau\Delta} - 1}{\tau\Delta} \Pi_N f(v^n) + G^n.\end{aligned}$$

$|I - \Pi_N|$ only produces an N^{-s} error tied to regularity s ; we do not expand this further here.

Define the Fourier multiplier

$$\widehat{Lu}(k) = \sqrt{\frac{\tau|k|^2}{1 - e^{-\tau|k|^2}}} \hat{u}(k),$$

satisfying $\|u\|_2 \leq \|Lu\|_2 \leq \|u\|_2 + \sqrt{\tau}\|Du\|_2$.

Proposition 19.9 (L^2 -error). *If $\|Du^n\|_2 + \|Dv^n\|_2 + \|v^n\|_\infty \leq M$, then*

$$\|u^m - v^m\|_L \leq \exp(C_M T) \left(\|u_0 - v_0\|_L + 4\tau \sum_{n=0}^{m-1} \|LG^n\|_2^2 \right)$$

(the norm $\|\cdot\|_L$ is compatible with L ; $T = m\tau$).

Sketch. $\xi^n = u^n - v^n$. Take the inner product with $\frac{\tau Q}{e^{\tau Q} - 1} \xi^{n+1}$ ($Q = \Delta$):

$$LHS = \frac{1}{2\tau} (\|L\xi^{n+1}\|_2^2 - \|L\xi^n\|_2^2 + \|L(\xi^{n+1} - \xi^n)\|_2^2),$$

stabilization contributions

$$\begin{aligned}LHS_2 &= \frac{S_1}{2} (\|L\xi^{n+1}\|_2^2 - \|L\xi^n\|_2^2 + \|L\delta\xi\|_2^2), \\ LHS_3 &= \frac{S_2}{2} (\|L\nabla\xi^{n+1}\|_2^2 - \|L\nabla\xi^n\|_2^2 + \|L\nabla\delta\xi\|_2^2)\end{aligned}$$

(one may omit S_i first in the analysis). Split the RHS as (i) + (ii) + (iii):

$$\begin{aligned}\text{(i)} &= (\xi^n, \Delta\xi^{n+1}) = -\frac{1}{2} (\|\nabla\xi^{n+1}\|_2^2 + \|\nabla\xi^n\|_2^2) + \frac{1}{2} \|\nabla\delta\xi\|_2^2, \\ \text{(ii)} &= (f(u^n) - f(v^n), \xi^{n+1}), \quad f(u) - f(v) = (a_1 + a_2(v^n)^2)\xi + a_3v(\xi)^2 + a_4(\xi)^3\end{aligned}$$

(a_i can be computed exactly). Young estimates absorb into $\frac{1}{8}\|\xi^{n+1}\|_2^2$, etc.; (iii) contains (LG_n, ξ^{n+1}) . After combining,

$$\begin{aligned}\frac{1}{2} (\|L\xi^{n+1}\|_2^2 - \|L\xi^n\|_2^2) &+ (S_1 - \frac{1}{2})\Delta_n \|L\xi\|_2^2 \\ &+ (S_2 - \frac{1}{2})\Delta_n \|L\nabla\xi\|_2^2 \leq C\|L^2G_n\|_2^2 + C\|L\xi^n\|_2^2.\end{aligned}$$

Discrete Grönwall applies; $S_2 > \frac{1}{2}$ suffices. $\square$

Truncation: $Q_n = -\frac{1}{\tau} \int_{t_n}^{t_{n+1}} \partial_t \Delta u \cdot (t_{n+1} - t) dt + \dots$, $\|Q_n\|_2 \lesssim \tau \|\partial_t \Delta u\|_{L_t^\infty L_x^2}$, hence $\sum \tau \|Q_n\|_2^2 \lesssim \tau \cdot t_m$, and thus $\|u^{n+1} - u(t_{n+1})\|_2 \lesssim \tau$ (first-order scheme).

19.9 Stabilized exponential integrator for CH

(See slides on CH.) Exponential scheme

$$u^{n+1} = e^{-\nu\Delta^2\tau} u^n + \frac{1 - e^{-\nu\Delta^2\tau}}{\nu\Delta} \Pi_N f(u^n) - S\tau\Delta^2(u^{n+1} - u^n), \quad (42)$$

where the last term is the stabilizer. Pros: regularity, accuracy, easy to compute; cons: stability depends on S .

19.10 Incompressible NS / Euler on $\mathbb{T}^2$

Incompressible Navier–Stokes / Euler:

$$\begin{cases} \partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \\ u(0, x) = u_0(x), \quad \operatorname{div} u = 0, \end{cases} \quad (\text{NS}; \nu: \text{viscosity}), \quad (43)$$

$$\begin{cases} \partial_t u + (u \cdot \nabla)u + \nabla p = 0, \\ u(0, x) = u_0(x), \quad \operatorname{div} u = 0, \end{cases} \quad (\text{Euler}). \quad (44)$$

Reynolds number $Re = LU/\nu$ (dimensionless): $Re > 4000$ turbulent, $Re < 2000$ laminar. Questions: vanishing viscosity? stability?

Divergence form and vorticity

If $\operatorname{div} u = 0$, then $\operatorname{div}(u \otimes u) = (u \cdot \nabla)u$. Vorticity: in 3D, $\omega = \nabla \times u$; in 2D, the scalar $\omega = \nabla^\perp \cdot u$, $\nabla^\perp = (\partial_2, -\partial_1)$. Applying $\nabla^\perp \cdot$ to 2D NS, since $\nabla^\perp \cdot \nabla p = 0$ and $\nabla^\perp \cdot ((u \cdot \nabla)u) = (u \cdot \nabla)\omega$, one gets

$$\partial_t \omega + (u \cdot \nabla)\omega = \nu \Delta \omega.$$

Viscosity approximation / semi-implicit scheme

Viscous approximation ($\mathbb{T}^d$ periodic; see slides): large viscosity is smooth and stable; small viscosity produces vortices; focus on the vanishing-viscosity rate. Semi-implicit scheme

$$\begin{cases} \frac{u^{n+1} - u^n}{\tau} + u^n \cdot \nabla u^{n+1} + \nabla p^{n+1} = \nu \Delta u^{n+1}, \\ \operatorname{div} u^{n+1} = 0. \end{cases}$$

An L^2 error needs $\partial_{tt}u \in L^2$ (truncation), corresponding to $\Delta^2 u \in L^2$, i.e. $u \in H^4$; the nonlinearity further requires $\partial_t[(u \cdot \nabla)u] \in L^2$, heuristically needing $\|\Delta u\|_2 \|\nabla u\|_\infty$, etc., still pointing to $u \in H^4$.

19.11 Leray projection and exponential NS schemes

Definition 19.10 (Leray projection). $\mathbb{P}f = f - \nabla q$, where $\Delta q = \operatorname{div} f$ ($q = (\operatorname{div} f)/\Delta$). Then $\operatorname{div}(\mathbb{P}f) = 0$ (removing the $\operatorname{div} \neq 0$ part). Properties: $\mathbb{P}u = u$, $\mathbb{P}\nabla p = 0$, $\mathbb{P}\partial_t u = \partial_t u$.

Projected NS: $\partial_t u + \mathbb{P}(u \cdot \nabla u) = \nu \Delta u$. Duhamel (normal exponential method):

$$u(t_{n+1}) = e^{\tau \nu \Delta} u(t_n) - \int_{t_n}^{t_{n+1}} e^{(t_{n+1}-s)\nu \Delta} \mathbb{P}(u \cdot \nabla u)(s) ds.$$

Naive freezing $u(s) \approx u(t_n)$

From NS, $u(s) = u(t_n) + \nu \int_{t_n}^s \Delta u - \int_{t_n}^s \mathbb{P}(u \cdot \nabla u)$, one gets

$$u(t_{n+1}) = e^{\tau \nu \Delta} u(t_n) - \int_{t_n}^{t_{n+1}} e^{(t_{n+1}-s)\nu \Delta} \mathbb{P}(u(t_n) \cdot \nabla u(t_n)) ds + R_n,$$

$$R_n = - \int e^{(t_{n+1}-s)\nu \Delta} \mathbb{P}[u(s) \cdot \nabla u(s) - u(t_n) \cdot \nabla u(t_n)] ds.$$

Substituting $u(s) - u(t_n)$ requires at least $u \in H^3$.

New idea: freeze along heat flow

Approximate $u(s)$ by $e^{(s-t_n)\nu\Delta}u(t_n)$:

$$\begin{aligned} & \int_{t_n}^{t_{n+1}} e^{(t_{n+1}-s)\nu\Delta} \mathbb{P}(u(s) \cdot \nabla u(s)) ds \\ &= \int_{t_n}^{t_{n+1}} e^{(t_{n+1}-s)\nu\Delta} \mathbb{P}(e^{(s-t_n)\nu\Delta}u(t_n) \cdot \nabla e^{(s-t_n)\nu\Delta}u(t_n)) ds - R_{n,1}. \end{aligned}$$

The two terms I_1, I_2 in $R_{n,1}$ give $\|R_{n,1}\|_2 \lesssim \tau^2 \|u\|_{H^2}^3$ (at most needing $u \in H^2$).

Write $V(s) = e^{(s-t_n)\nu\Delta}u(t_n)$, $g(s) = e^{(t_{n+1}-s)\nu\Delta} \mathbb{P}(V(s) \cdot \nabla V(s))$. Use $g(s) = g(t_{n+1}) - \int_s^{t_{n+1}} g'(\sigma) d\sigma$. Since $\Delta \mathbb{P} = \mathbb{P} \Delta$, the Fourier definition (2D)

$$\widehat{\mathbb{P} \begin{pmatrix} f \\ g \end{pmatrix}}(k) = \begin{pmatrix} \hat{f} - \frac{k_1 \hat{f} + k_2 \hat{g}}{|k|^2} k_1 \\ \hat{g} - \frac{k_1 \hat{f} + k_2 \hat{g}}{|k|^2} k_2 \end{pmatrix},$$

verifies $\operatorname{div} \mathbb{P} = 0$. Computation yields

$$g'(s) = \nu e^{(t_{n+1}-s)\nu\Delta} \mathbb{P} \left(- \sum_j \partial_j V \cdot \partial_j \nabla V \right),$$

hence $\|g'(s)\|_2 \lesssim \nu \|V\|_{W^{2,2+\varepsilon}}^2$, so $\|R_{n,2}\|_2 \lesssim \nu \tau^2 \|u\|_{W^{2,2+\varepsilon}}^2$, and

$$\int_{t_n}^{t_{n+1}} g(t_{n+1}) ds = \tau \mathbb{P}(e^{\nu\tau\Delta}u(t_n) \cdot \nabla e^{\nu\tau\Delta}u(t_n)).$$

Implicit gradient for stability

The scheme $u^{n+1} = e^{\tau\Delta}u^n - \tau \mathbb{P}(e^{\tau\Delta}u^n \cdot \nabla e^{\tau\Delta}u^n)$ loses stability as $\tau \rightarrow 0$ (accuracy disappears). Switch to ∇u^{n+1} :

$$u^{n+1} = e^{\tau\Delta}u^n - \tau \mathbb{P}(e^{\tau\Delta}u^n \cdot \nabla u^{n+1}). \quad (45)$$

The exact solution satisfies $u(t_{n+1}) = e^{\tau\Delta}u(t_n) - \tau \mathbb{P}(e^{\tau\Delta}u(t_n) \cdot \nabla u(t_{n+1})) - R_{n,1} - R_{n,2} - \mathbb{P}R_{n,3}$, where

$$R_{n,3} = \tau e^{\tau\Delta}u(t_n) \cdot \nabla (e^{\tau\Delta}u(t_n) - u(t_{n+1}))$$

(written via Duhamel as an integral), $\|R_{n,3}\|_2 \lesssim \tau^2 \|u\|_{H^2}^3$. Summary: local truncation $\|R_{n,1}\|_2 + \|R_{n,2}\|_2 + \|R_{n,3}\|_2 \lesssim \tau^2$, with regularity needs H^2 , $W^{2,2+\varepsilon}$, H^2 respectively.

Theorem 19.11 (Energy stability). *For (45), if $\operatorname{div} u^{n+1} = 0$, then $\|u^{n+1}\|_2 \leq \|u^n\|_2$.*

Proof. Take the inner product with u^{n+1} . Since $(\mathbb{P}f, u^{n+1}) = (f, u^{n+1})$ and $\operatorname{div}(e^{\tau\Delta}u^n) = 0$,

$$\int_{\mathbb{T}^2} e^{\tau\Delta}u^n \cdot \nabla \frac{1}{2} |u^{n+1}|^2 dx = 0.$$

Hence $\|u^{n+1}\|_2^2 = (e^{\tau\Delta}u^n, u^{n+1}) \leq \|u^n\|_2 \|u^{n+1}\|_2$. □

 L^2 error

Set $e^n = u^n - u(t_n)$. Error equation (*):

$$\begin{aligned} e^{n+1} + \tau \mathbb{P}(e^{\tau\Delta}u^n \cdot \nabla e^{n+1}) &= e^{\tau\Delta}e^n - \tau \mathbb{P}(e^{\tau\Delta}e^n \cdot \nabla u(t_{n+1})) \\ &\quad + R_{n,1} + R_{n,2} + \mathbb{P}R_{n,3}. \end{aligned}$$

Test with e^{n+1} : $LHS = \|e^{n+1}\|_2^2$,

$$\begin{aligned} RHS &\leq \frac{1}{2} \|e^n\|_2^2 + \frac{1}{2} \|e^{n+1}\|_2^2 + C\tau \|\nabla u(t_{n+1})\|_\infty \|e^n\|_2 \|e^{n+1}\|_2 + C\tau^2 \|e^{n+1}\|_2 \\ &\leq \frac{1}{2} \|e^n\|_2^2 + \frac{1}{2} \|e^{n+1}\|_2^2 + C\tau \|e^n\|_2^2 + C\tau \|e^{n+1}\|_2^2 + C\tau^3 \end{aligned}$$

(needs $u \in W^{1,\infty}$). Rearranging $(1 - C\tau) \|e^{n+1}\|_2^2 \leq (1 + C\tau) \|e^n\|_2^2 + C\tau^3$, discrete Grönwall $\Rightarrow \|e^n\|_2^2 \lesssim \tau^2$. Overall one still needs $u \in W^{2,2+\varepsilon}$ (from $R_{n,2}$).

20 Selected advanced topics

(Source: slides “Numerical Methods in solving PDEs”, 2024.10.12. Phase-field definitions, sharp-interface limits, Fourier spectral basics, and exponential-integrator Duhamel derivations appear in earlier sections of this part; here we keep only non-overlapping scheme formulas, comparison tables, and fluid/SQG highlights.)

20.1 Energy stability as a design criterion

Gradient-flow structure of Allen–Cahn / Cahn–Hilliard implies continuous energy dissipation, but discretization need not inherit it. The slides treat *energy-dissipative* time schemes as the main figure of merit, and organize methods as: convex splitting; stabilized semi-implicit (IMEX / BDF2); operator splitting; (low-regularity) exponential integrators; fully implicit / backward Euler for interface tracking.

20.2 Stabilized semi-implicit schemes

With time step $\tau > 0$, stabilization $A > 0$, and Fourier truncation Π_N :

$$\frac{u^{n+1} - u^n}{\tau} = \nu \Delta u^{n+1} - A(u^{n+1} - u^n) - \Pi_N f(u^n), \quad (46)$$

$$\frac{3u^{n+1} - 4u^n + u^{n-1}}{2\tau} = \nu \Delta u^{n+1} - A\tau(u^{n+1} - u^n) - \Pi_N(2f(u^n) - f(u^{n-1})). \quad (47)$$

Unconditional energy dissipation holds if $A \geq C \cdot h(\nu)$, with $h(\nu)$ as in Table 1 (BDF2 dissipates a *modified* energy).

Table 1: Stabilization $A \geq C \cdot h(\nu)$ for unconditional energy dissipation.

$h(\nu)$	1st-order IMEX	2nd-order BDF2
2D	$\nu^{-1} \log \nu $	$\nu^{-10} \log \nu ^4$
3D	ν^{-3}	ν^{-8}

Remark 20.1 (Typical numerics on the slides). 2D AC with BDF2: $\nu = 0.01$, $A = 1$, $\tau = 0.01$, $N = 256$, seven-ball u_0 (coarsening of small droplets). 2D CH: $\nu = 0.009$, $A = 10$, same grid/time step.

20.3 Fully implicit radial Allen–Cahn and interface iteration

Radial fully implicit backward Euler:

$$\frac{u_{n+1} - u_n}{\tau} = \partial_{rr} u_{n+1} + \frac{1}{r} \partial_r u_{n+1} - \frac{1}{\varepsilon^2} f(u_{n+1}).$$

Theorem 20.2 (C–Li–Promislow–Wetton 2021). Fix $1 < s \leq 2$ and the distinguished limit $\tau = \varepsilon^s$. For suitable small perturbations of a radial interface of radius $R_0 > 1$, there is an interface sequence $\{R_n\}$ obeying

$$\frac{R_{n+1} - R_n}{\tau} = -\frac{1}{R_{n+1}} + \frac{b_1(R_{n+1} - R_n)^3}{\tau \varepsilon^2} + O\left(\frac{\tau}{\varepsilon}\right), \quad b_1 = \frac{\|g''\|_{L^2}^2}{6\|g'\|_{L^2}^2} > 0,$$

until the interface crosses the threshold $R = 1$.

20.4 Strang operator splitting

For AC, split linear diffusion and the ODE $u_t = u - u^3$:

$$\mathcal{S}_{\mathcal{L}}(\tau)a = \exp(\varepsilon^2\tau\Delta)a, \quad \mathcal{S}_{\mathcal{N}}(\tau)b = ((e^{2\tau} - 1)b^2 + 1)^{-1/2}e^\tau b.$$

Strang composition:

$$u^{n+1} = \mathcal{S}_{\mathcal{L}}(\tau/2) \mathcal{S}_{\mathcal{N}}(\tau) \mathcal{S}_{\mathcal{L}}(\tau/2) u^n,$$

equivalently

$$u^{n+1} = e^{\varepsilon^2\frac{\tau}{2}\Delta} \left(\frac{e^\tau e^{\varepsilon^2\frac{\tau}{2}\Delta} u^n}{[1 + (e^{2\tau} - 1)(e^{\varepsilon^2\frac{\tau}{2}\Delta} u^n)^2]^{1/2}} \right).$$

Energy decrease $E(u^{n+1}) \leq E(u^n)$ is nontrivial (explicit energy map is complicated). For CH, the naive nonlinear step $\partial_t u = -\Delta(u - u^3)$ has no usable explicit solver and is not even well-posed; a modified splitting yields uniform Sobolev bounds $\sup_n \|u^n\|_{H^k} \leq C$ (rather than a direct energy identity).

20.5 SQG, Onsager, and convex integration

Dissipative SQG on $\mathbb{T}^2$:

$$\begin{cases} \partial_t \theta + u \cdot \nabla \theta = -\nu \Lambda^\gamma \theta, \\ u = \nabla^\perp \Lambda^{-1} \theta, \quad \Lambda = (-\Delta)^{1/2}, \quad \nu \geq 0, \quad 0 < \gamma \leq 2, \\ \theta|_{t=0} = \theta_0, \quad \int_{\mathbb{T}^2} \theta = 0. \end{cases} \quad (48)$$

Onsager-type question: lowest regularity for which conservation persists. Convex integration (Gromov; De Lellis–Székelyhidi; ...) constructs rough solutions; for time-dependent SQG the *odd-multiplier obstruction* for plane-wave ansatz is removed by a momentum / potential-velocity reformulation $u = \Lambda v$ (Buckmaster–Shkoller–Vicol; Resnick), yielding solutions with $\Lambda^{-1}\theta \in C^\beta$, $\beta \in (\frac{1}{2}, \frac{4}{5})$ (Onsager range $\beta \in (\frac{1}{2}, 1)$).

Theorem 20.3 (C–Kwon–Li 2021, stationary SQG). *For any $\nu \geq 0$, $\gamma \in (0, \frac{3}{2})$, and $\frac{1}{2} < \alpha < \frac{1}{2} + \min(\frac{1}{6}, \frac{3}{2} - \gamma)$, there exist infinitely many zero-mean stationary weak solutions with $\Lambda^{-1}\theta \in C^\alpha(\mathbb{T}^2)$.*

20.6 Stabilized exponential integrator for CH

(The standard Allen–Cahn ETD from Duhamel appears earlier in this part.) For CH the slides emphasize a stabilized scheme (analytic results previously scarce: no maximum principle):

$$u^{n+1} = e^{-\nu\Delta^2\tau} u^n + \frac{1 - e^{-\nu\Delta^2\tau}}{\nu\Delta} \Pi_N[f(u^n)] - S\tau\Delta^2(u^{n+1} - u^n), \quad S > 0.$$

Numerics ($\nu = 0.01$, $\tau = 0.1$, $N = 256$): $S = 0.01$ produces oscillatory non-monotone energy and background artifacts; $S = 0.1$ restores monotone energy decay and clean Ostwald ripening. Manufactured-solution test $u_e = \frac{1}{2}e^{-t} \sin x \sin y$ confirms first-order temporal rates in L^2 and L^∞ .

20.7 Incompressible NS / Euler schemes

On $\mathbb{T}^2$, Euler and Navier–Stokes are the usual inviscid / viscous systems. The inviscid limit $\nu \rightarrow 0$ holds in Sobolev spaces under suitable assumptions (e.g. Masmoudi: H^{s-2} convergence for H^s data, $s > 2$), but *need not* be mirrored by a given discretization.

Semi-implicit spectral scheme (C.–Luo–Wang 2024):

$$\begin{cases} \frac{u^{n+1} - u^n}{\tau} + \Pi_N(u^n \cdot \nabla u^{n+1}) + \nabla p^n = \nu \Delta u^{n+1}, \quad \operatorname{div} u^{n+1} = 0, \\ u^0 = \Pi_N u_0. \end{cases} \quad (49)$$

Unconditional L^2 dissipation $\|u^{n+1}\|_{L^2} \leq \|u^n\|_{L^2}$ and $\sup_n \|u^n\|_{H^s} \leq 4\|u_0\|_{H^s}$; errors of the form

$$\sup_n \|u^{n+1} - v(t_{n+1})\|_{L^2} \leq C_1(\tau + N^{-s+1}),$$

and, for $u_0 \in H^{s+3}$,

$$\sup_n \|u^{n+1} - v(t_{n+1})\|_{H^s} \leq C_2(\tau + N^{-2} + N^2\nu).$$

An iterative Fourier realization with Leray projector $\mathbb{P}$ converges when $\tau \leq C^* \max\{\nu, N^{-1}\}$.

Remark 20.4 (Experiments highlighted on the slides). Two Gaussian vortices at $T = 50$ ($\tau = 0.001$, $N = 128$): small ν yields filamentation / merger; large ν yields a smooth blob. High-order errors are subtle: the $N^2\nu$ term means refining N at fixed ν can *increase* H^8 error even while lower norms improve.